

Theories of Aging

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Main Premise:

- Complementary health care practitioners and Family Doctors are in a unique situation - able to identify at risk patients before end-stage disease is present, and to provide patients with customized nutrition, exercise, supplementation and other lifestyle recommendations that are best suited to their case.
- Part of the assessment includes understanding the body's aging clock and steps that help prevent cancer, which now accounts for 30% of all deaths each year

Main Topics Covered In These Lectures:

1. Relationship Of Aging Process To Risk of Degenerative Disease and Loss of Functionality – key aspects of the body's aging clock
2. Factors Influencing Cancer Risk Other Than Aging Process
3. Nutrition and Lifestyle Factors That Counter or Slow Aging, Helping To Preserve Functional Living, And Reduce Risk of Cancer and Other Age-Related Conditions
4. Review of Conditionally- Essential Nutrients – Nutrients To Supplement As The Body's Synthesis Declines With Age:
 - Glucosamine
 - Coenzyme Q10
 - Choline
 - Alpha-lipoic Acid
 - L-Carnitine (Acetyl-L-Carnitine)
 - Melatonin
 - Topical Hyaluronic Acid

Why We Age

- After age 40-45 – nature assumes your children will soon be self-sufficient and so, nature has no further use for you – you're taking up space
- Because most people don't volunteer to die, nature's plan facilitates that you die slowly without really noticing – It's called aging
- Aging process is similar on a cellular level as are those related to cancer, cognitive decline, osteoarthritis, congestive heart failure etc.

Theories of Aging

1. Telomeres and Hayflick Theory
2. Damage Accumulation Theories
3. Neuro-Endocrine Clock (Hormonal Decline)

Hayflick Theory

- Leonard Hayflick – 1960 (finite cell division)
- e.g. Fibroblasts (50 replications)
- Frozen, recall remaining cell divisions

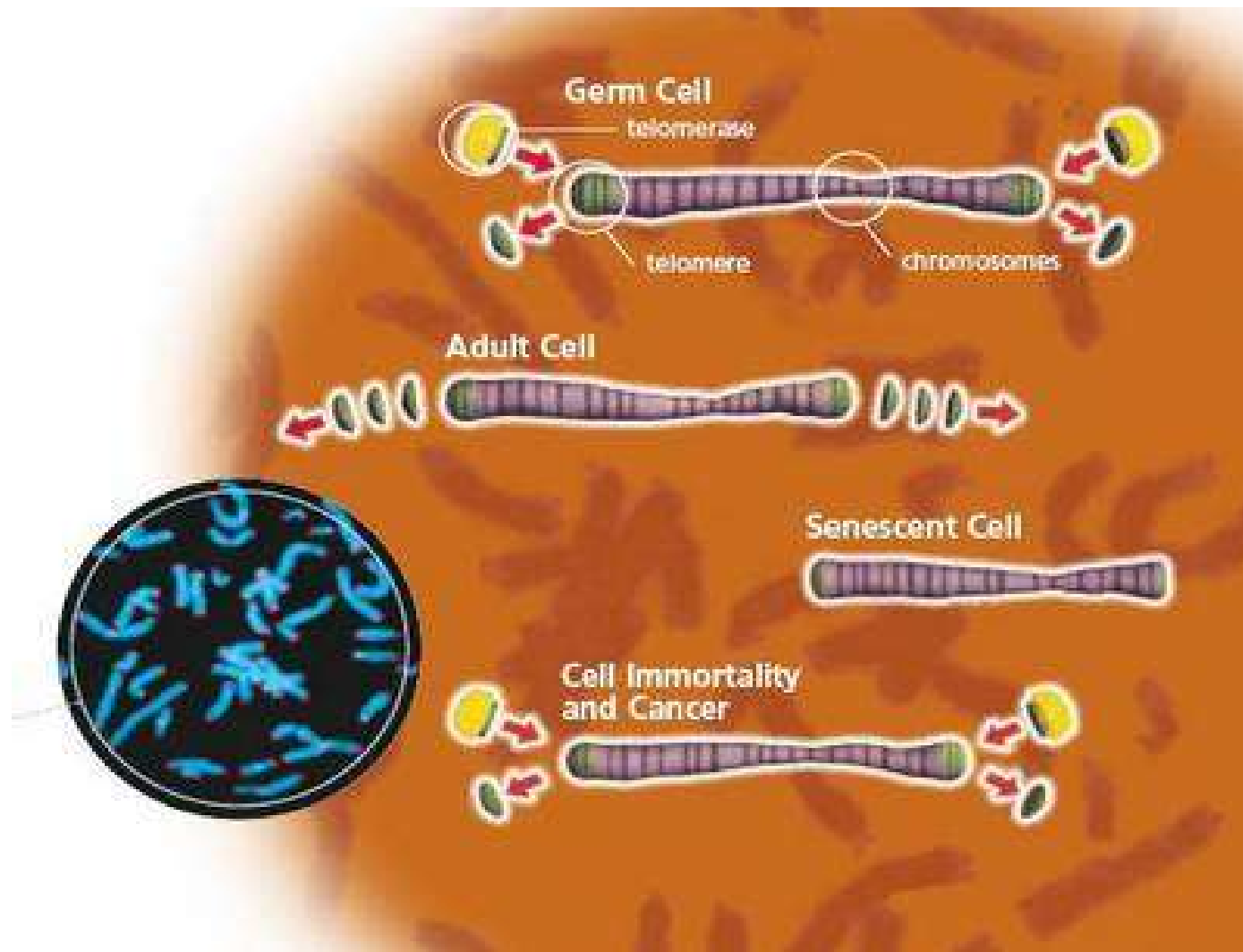
- Long-lived species (humans) – increased number of cell divisions compared to short-lived species (mice)
- Humans live 5X longer than cats, who live 5X longer than mice, who live 25X times longer than fruit flies

- Older peoples' cells in culture peeter out faster than young peoples' cells
- Fetal cells in culture replicate like bunnies and then eventually slow down and stop

Telomere Theory

- Telomeres shorten with each cell division
- At critical level of shortening, cells stop dividing
- Immortal cells: Hemopoietic, spermatazoa, and cancer cells (Telomerase enzyme)

Telomere Shortening



- **Telomerase Research:**

- Anti-aging

- Cancer

- At conception – telomeres: 10,000 base pairs

- At birth – telomeres: 5,000 base pairs

- Versus 13,000,000 base pairs for chromosomes

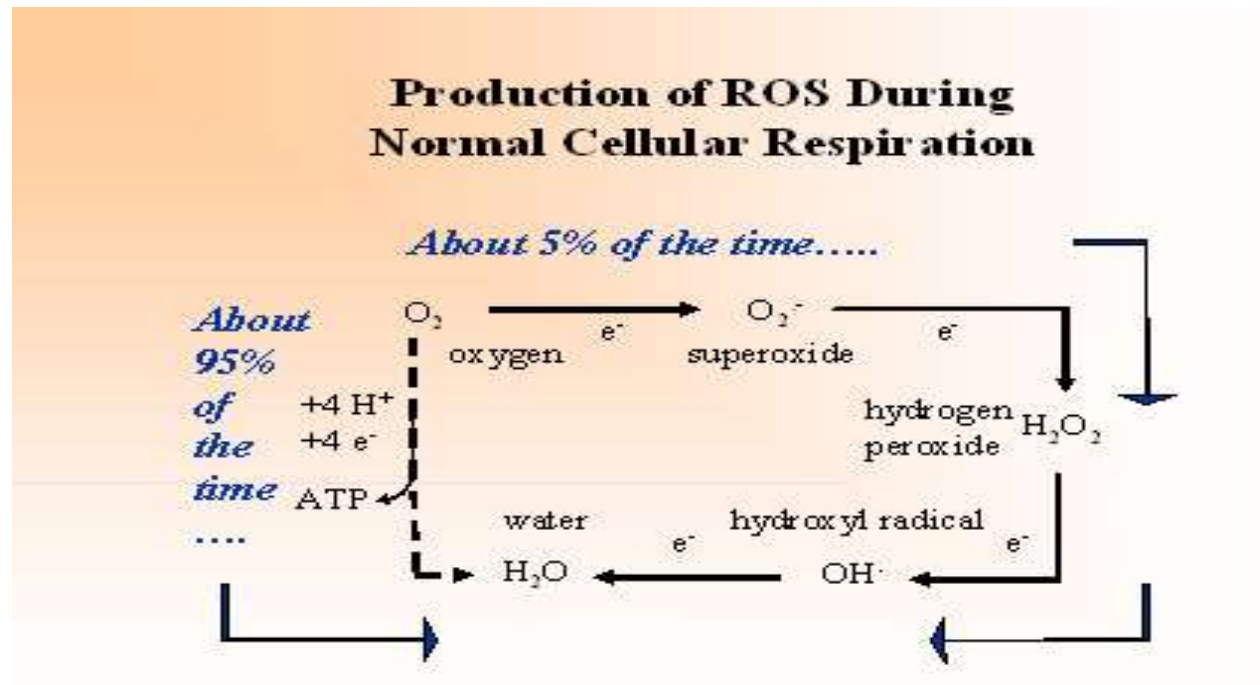
A Major Health Goal= Slow down the rate of cell division (lifestyle has major impact on this).

Damage Accumulation Theories

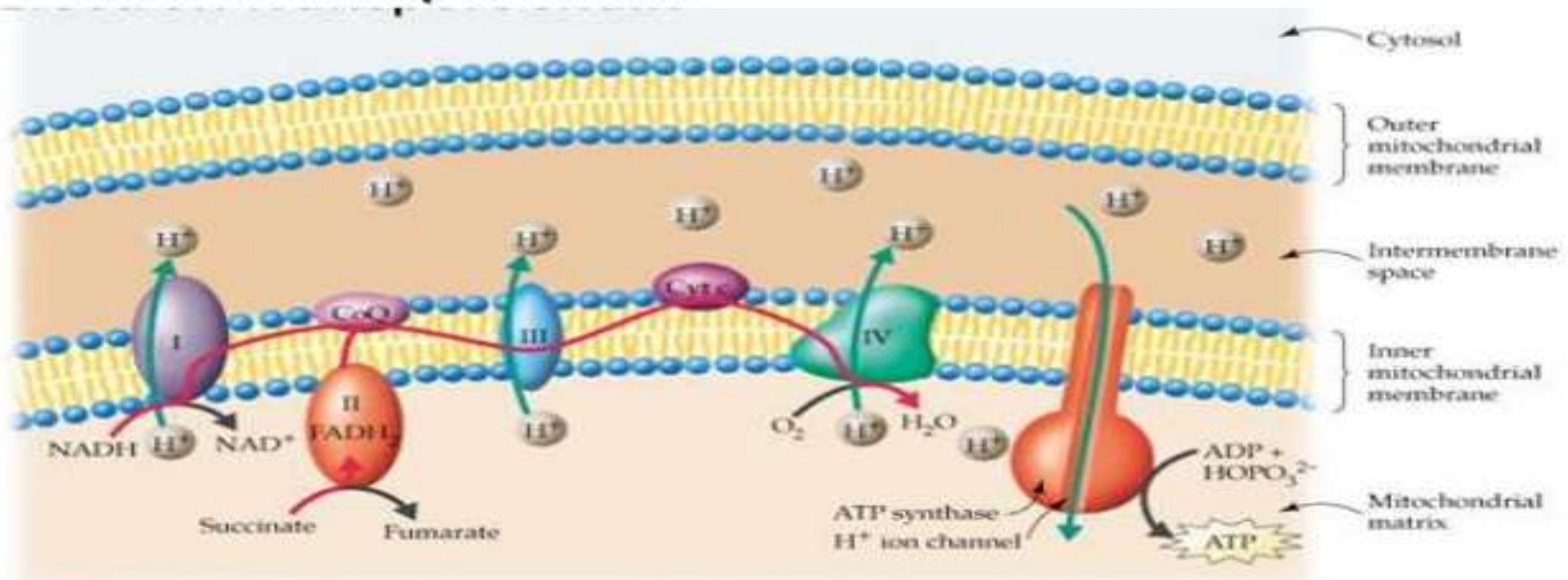
1. Oxygen Free Radicals

- 2-5% of molecular oxygen converted to oxygen free radicals:
 - Superoxide anion
 - Hydrogen Peroxide
 - Hydroxy radicals
 - Singlet oxygen

In about 5 percent of the time, oxygen is reduced to highly reactive superoxide. This is probably due to a “leak” of single electrons at a particular site of the mitochondrial electron transport chain. When superoxide undergoes further reduction, H₂O₂ and the extremely reactive hydroxyl radical is formed. It happens spontaneously or under the influence of various enzymes. The transition from H₂O₂ to hydroxyl radicals is favored in the presence of metals such as **iron or copper**. It also happens during abnormal exposure to **ionizing radiation**.

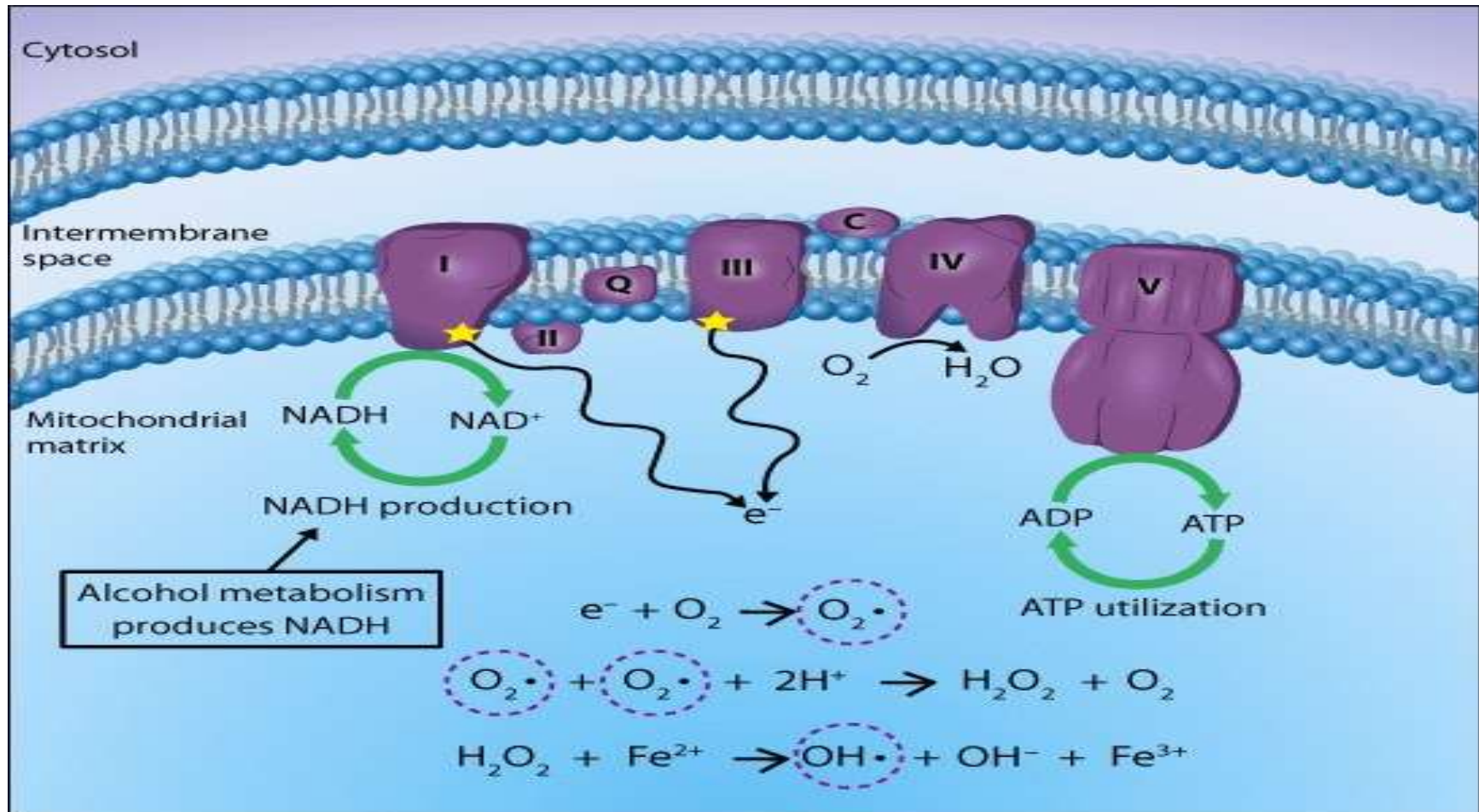


Superoxide Production from Mitochondrial Electron Transport Chain



Leaking' of electron (to oxygen) during electron transport leads to the formation of $O_2^{\bullet -}$ ($O_2 + e^- \longrightarrow O_2^{\bullet -}$)

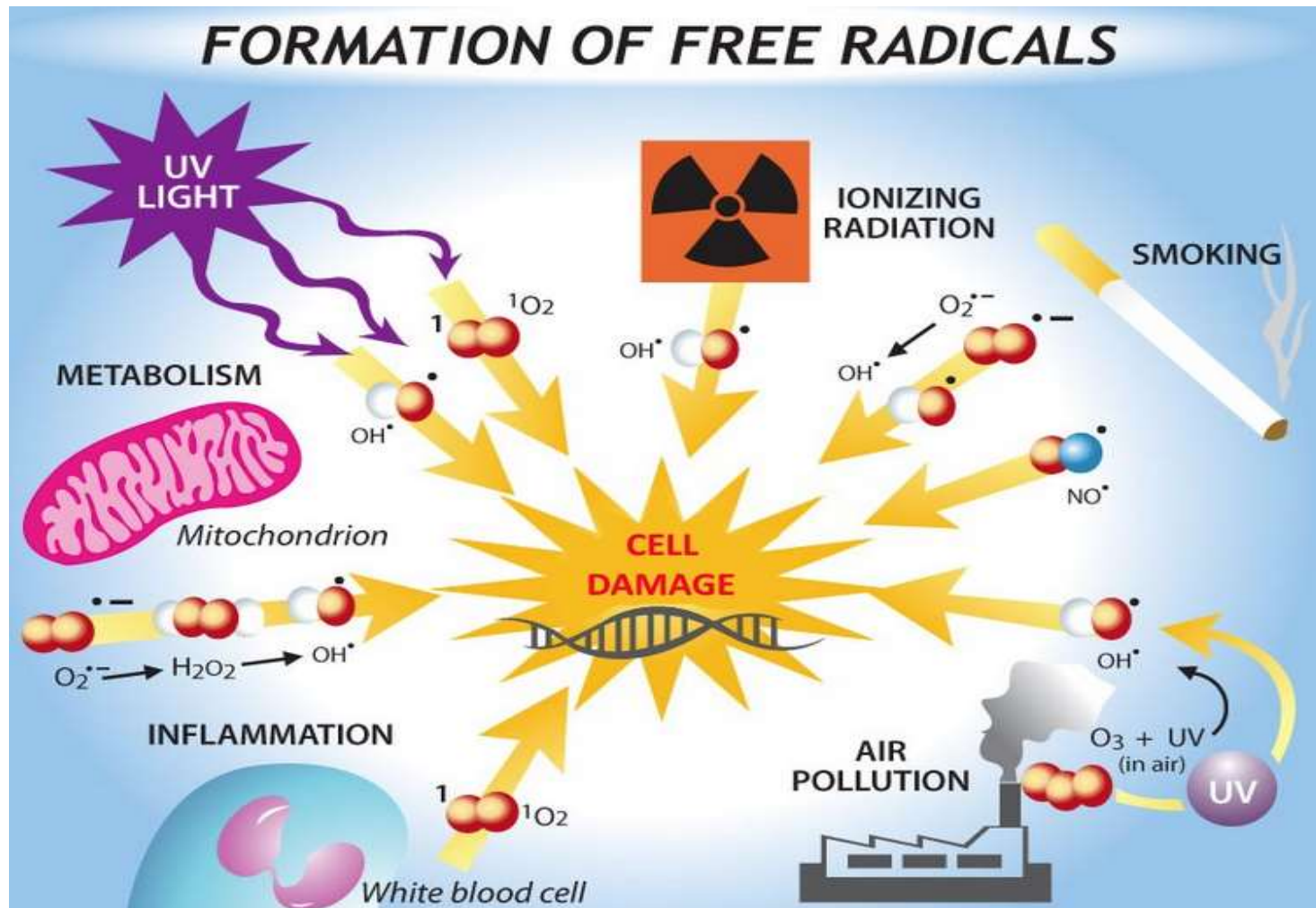
Formation of Superoxide Anion, Hydrogen Peroxide and Hydroxy Radicals

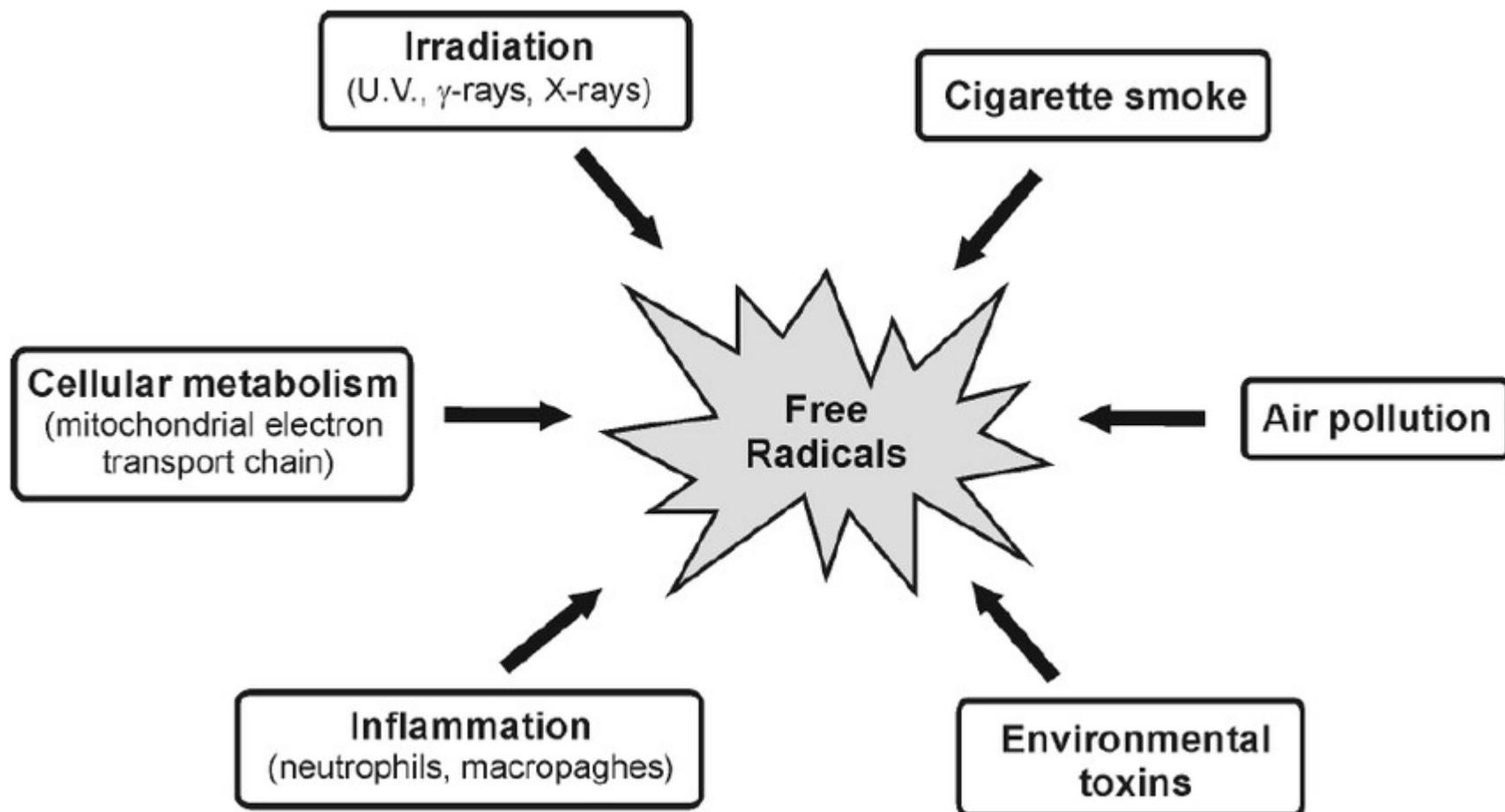


Other Common Free Radicals:

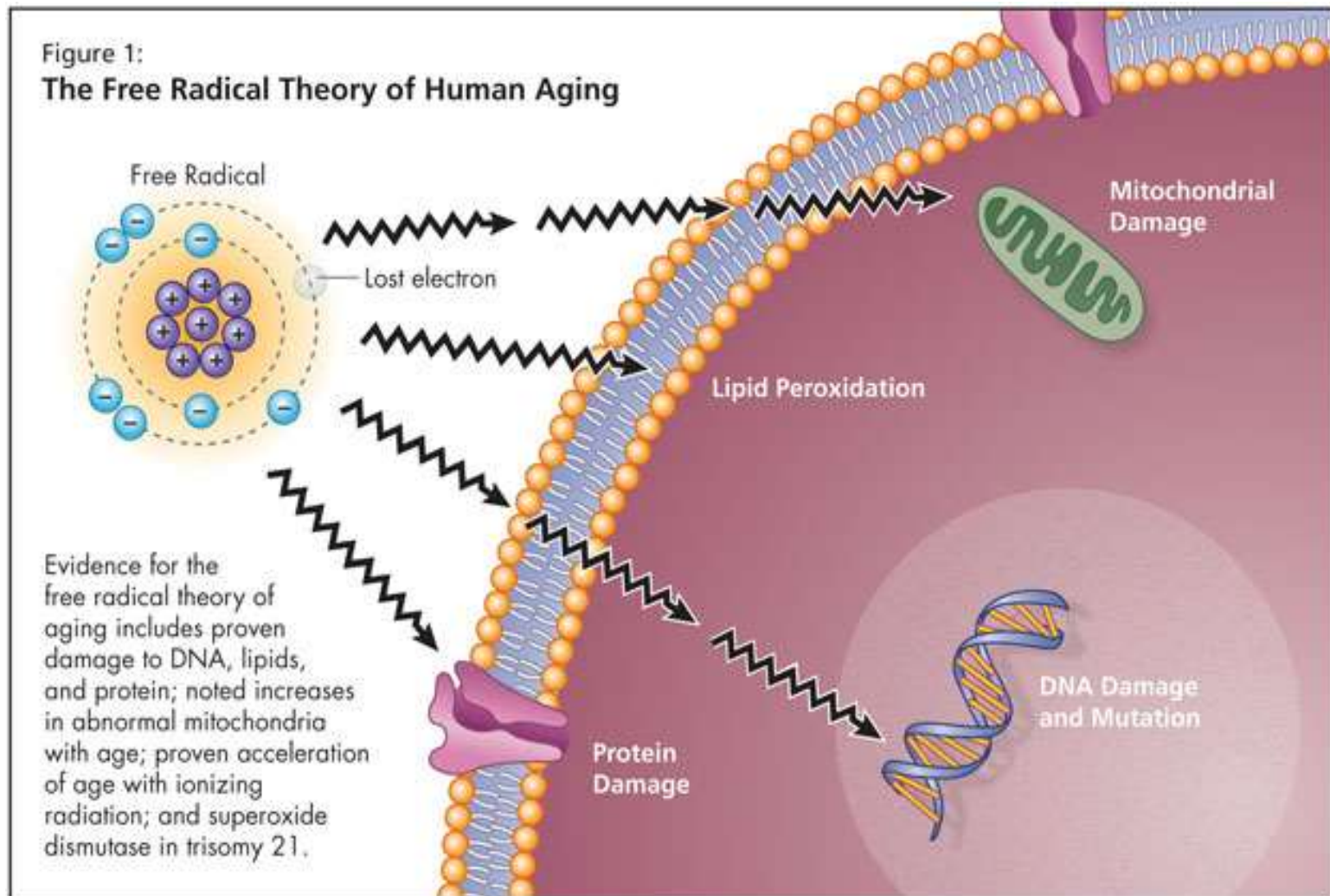
1. Polycyclic Aromatic Hydrocarbons
2. Heterocyclic Amines
3. Nitrosamines
4. Alcohol
5. Cigarette Smoke
6. Air Pollution (ozone and sulfur dioxide, sulfur trioxide, hydrazines)
7. Radiation – sun, tanning beds, X-Rays, CT scan

Note: Inflammation Also Increases Oxidative Stress (Diabetes, Autoimmune Disease, Cancer)





Multi-Modal Damage from Free Radicals on Cellular Level



- Free Radicals Damage:
 - DNA – aging and cancer
 - Mitochondria – decreased cellular energy – Mitochondrial Theory of Aging – (see next slide)
 - Enzymes – compromised function
 - Cell membrane
 - LDL –cholesterol – heart disease
 - Eyes – cataracts and macular degeneration
 - Brain – dementia and neurodegenerative dis
 - Skin – wrinkles, cancer

Mitochondrial Dysfunction in Aging

Mitochondria Have Their Own DNA

A fascinating aspect of mitochondria is the widely held belief that, early in the evolution of life on earth, they were free-living bacteria that became incorporated into the cells of larger (but very primitive) organisms in a symbiotic relationship—the larger organisms providing vital nutrients, and the mitochondria providing ATP.

Scientists base this theory in part on the fact that mitochondria have their own separate DNA molecules (called mitochondrial DNA), which replicate independently of the chromosomal DNA residing in the nuclei of the host cells. Mitochondrial DNA molecules are small, containing a mere 13 genes that code for the synthesis of enzymes involved in cellular respiration.

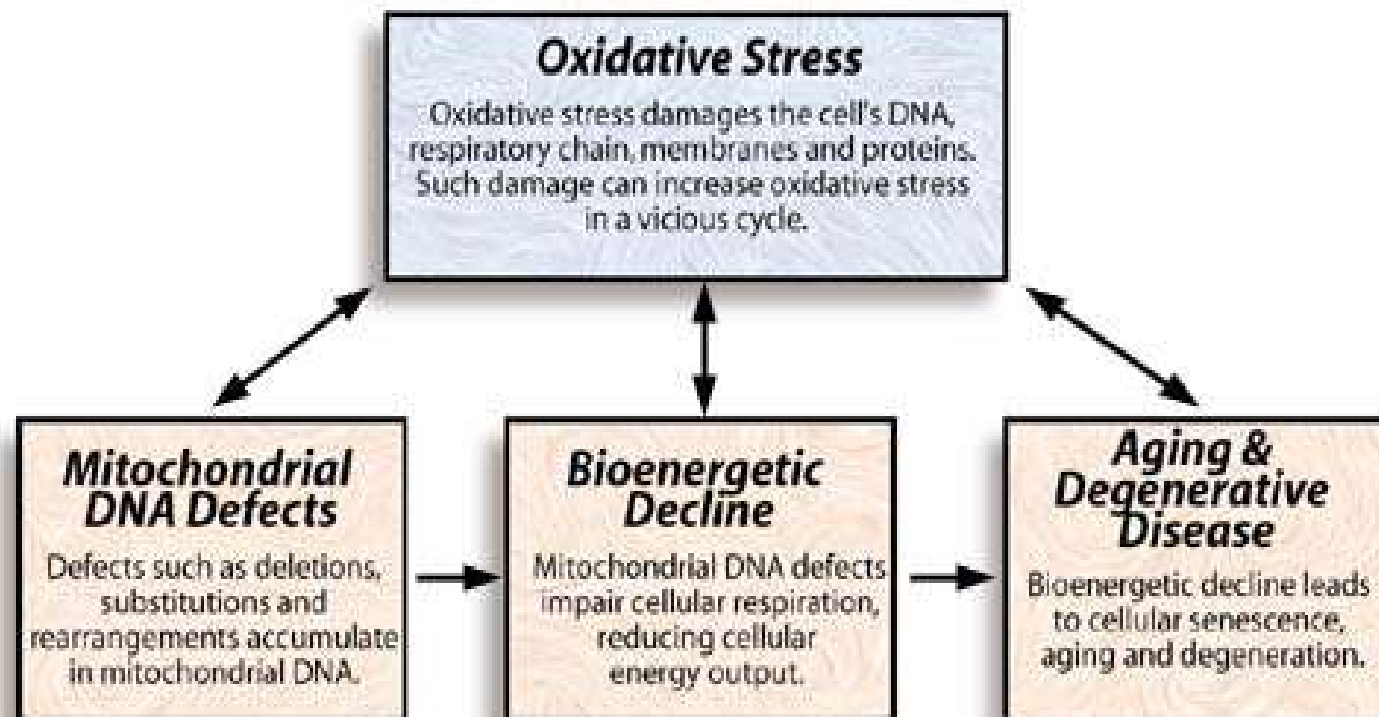
In mice engineered to have more rapid degradation of mitochondrial DNA aging occurred at a younger age and progressed more rapidly

In Mitochondrial Theory of Aging cumulative free radical damage to mtDNA causes mitochondrial dysfunction (decreased energy production for cell) leading to decreased cellular function on many levels. Protecting your mtDNA with CoQ10, Alpha-lipoic acid, nutrients that increase Glutathione, and Vitamin E, is purported to slow mitochondrial dysfunction, especially in diseases where mitochondrial dysfunction is prominent features, such as Diabetes, Parkinson's disease etc.

Reference:

- <http://www.life-enhancement.com/magazine/article/989-can-acetyl-l-carnitine-and-lipoic-acid-slow-the-aging-process>

Cumulative Mitochondrial Damage in Aging



Endogenous Antioxidants

Major Endogenous Antioxidants

1. Superoxide Dismutase (SOD): It quenches superoxide anion. Three forms of superoxide dismutase are present in humans:

SOD1 is located in the cytoplasm – requires **copper and zinc**

SOD2 in the mitochondria – requires **mn (manganese)**

SOD3 is extracellular – requires **copper and zinc**

Mutations in SOD1 can cause Familial Amyotrophic Lateral Sclerosis (ALS) and evidence also shows that wild-type SOD1, under conditions of cellular stress, is implicated in a significant fraction of sporadic ALS cases, which represent 90% of ALS patients.

Animal studies show SOD1 inactivation causes wide range of pathologies, including hepatocellular carcinoma

2. Catalase – It breaks down hydrogen peroxide into oxygen and water. It requires **iron**

3. Glutathione Peroxidase (8 variations of enzyme in body) It quenches fatty acid lipid peroxides- converting to stable alcohol forms and also quenches hydrogen peroxide. It requires **selenium**

4. Uric acid – end product of purine catabolism

5. Melatonin – fat and water soluble antioxidant in brain (made in pineal gland, but declines with age)

6. CoQ10 – fat soluble antioxidant (declines with age)

7. Alpha-lipoic acid- water and fat soluble antioxidant (decline with age)

References for SOD

- Elchuri S, Oberley TD, Qi W, Eisenstein RS, Jackson Roberts L, Van Remmen H, Epstein CJ, Huang TT (Jan 2005). "CuZnSOD deficiency leads to persistent and widespread oxidative damage and hepatocarcinogenesis later in life". *Oncogene*. **24** (3): 367–80.
- Muller FL, Song W, Liu Y, Chaudhuri A, Pieke-Dahl S, Strong R, Huang TT, Epstein CJ, Roberts LJ, Csete M, Faulkner JA, Van Remmen H (Jun 2006). "Absence of CuZn superoxide dismutase leads to elevated oxidative stress and acceleration of age-dependent skeletal muscle atrophy". *Free Radical Biology & Medicine*. **40** (11): 1993–2004.
- Sentman ML, Granström M, Jakobson H, Reaume A, Basu S, Marklund SL (Mar 2006). "Phenotypes of mice lacking extracellular superoxide dismutase and copper- and zinc-containing superoxide dismutase". *The Journal of Biological Chemistry*. **281** (11): 6904–9.
- Milani P, Gagliardi S, Cova E, Cereda C (2011). "SOD1 Transcriptional and Posttranscriptional Regulation and Its Potential Implications in ALS". *Neurology Research International*. **2011**: 458427.
- Deng HX, Hentati A, Tainer JA, Iqbal Z, Cayabyab A, Hung WY, Getzoff ED, Hu P, Herzfeldt B, Roos RP (Aug 1993). "Amyotrophic lateral sclerosis and structural defects in Cu,Zn superoxide dismutase". *Science*. **261** (5124): 1047–51.
- Conwit RA (Dec 2006). "Preventing familial ALS: a clinical trial may be feasible but is an efficacy trial warranted?". *Journal of the Neurological Sciences*. **251** (1–2): 1–2.
- Al-Chalabi A, Leigh PN (Aug 2000). "Recent advances in amyotrophic lateral sclerosis". *Current Opinion in Neurology*. **13** (4): 397–405.
- Gagliardi S, Cova E, Davin A, Guareschi S, Abel K, Alvisi E, Laforenza U, Ghidoni R, Cashman JR, Ceroni M, Cereda C (Aug 2010). "SOD1 mRNA expression in sporadic amyotrophic lateral sclerosis". *Neurobiology of Disease*. **39** (2): 198–203.
- Groner Y, Elroy-Stein O, Avraham KB, Schickler M, Knobler H, Minc-Golomb D, Bar-Peled O, Yarom R, Rotshenker S (1994). "Cell damage by excess CuZnSOD and Down's syndrome". *Biomedicine & Pharmacotherapy = Biomédecine & Pharmacothérapie*. **48** (5–6): 231–40.

Nutritional Antioxidants:

- Vits C, E, Beta-carotene (possibly Vitamin A-
<http://pubs.rsc.org/en/content/articlehtml/2015/ra/c4ra13315c>)
- Lycopene, Lutein (Lycopene – most powerful quencher of singlet oxygen)
- Selenium, zinc, manganese, copper
- CoQ10
- Green tea (catechins)

- Alpha-lipoic acid – fat and water soluble antioxidant
- Silymarin (milk thistle – concentrated in liver)
- Procyanidins (blue and bilberries, hawthorn)
- Polyphenolic compounds (red wine)
- Melatonin
- Grape seed extract (PCO's)
- Baicalein (chinese scullcap)

Bioflavonoids

Flavonol (*e.g.* quercetin , myricetin)

Flavone (*e.g.* apigenin , luteoline)

Flavonolols (*e.g.* taxifolin)

Flavan-3-ols (*e.g.* catechin, epigallocatechin)

Flavonone (*e.g.* hesperetin, naringenin)

Anthocyanidin (*e.g.* cyanidin, delphinidin)

Isoflavone (*e.g.* genistein, daidzein)

Bioflavonoids

- Bioflavonoids widely distributed in fruits and vegetables, are reported to exert multiple biological effects including free radical-scavenging activity. It has been reported that the bioflavonoids have a protective effect on the DNA damage induced by the hydroxyl radicals.
- One of the mechanisms that explains the protective effect of the flavonoids on the DNA is the involvement of the chelating metal ions, such as copper or iron. The flavonoids complexed with the copper or iron prevent the generation of the RO

Recommended Reading on Free Radicals and Antioxidants (see course notes for full PDF article)

Nimse SB, Pal D. Free radicals, natural antioxidants, and their reaction mechanisms. Royal Society of Chemistry. (Review Article) *RSC Adv.*, 2015, **5**, 27986-28006

<http://pubs.rsc.org/en/content/articlehtml/2015/ra/c4ra13315c>

See Course Notes For PDF article

- 2. Cross Linking Of Collagen Due To Free Radicals:

- Rigidity of tissues

- Decrease vital capacity of lungs

Advanced Glycosylated End-Products

- AGE-protein Damage:
 - 1. Clog arteries and increase arterial inflammation linked to heart attack and other vascular events
 - 2. Damage kidneys (kidney failure-dialysis)
 - 3. Cataracts

Target Blood Levels:

- Blood Sugar (Glucose) – 90 mg/dL or 5.0 mmol/L
- Fructosamine fasting levels – good evaluation (205-285 umol/L (1.2 – 2.0 mmol/L)
- Hemoglobin A1c - 20–40 mmol/mol (**4–5.7 %**)

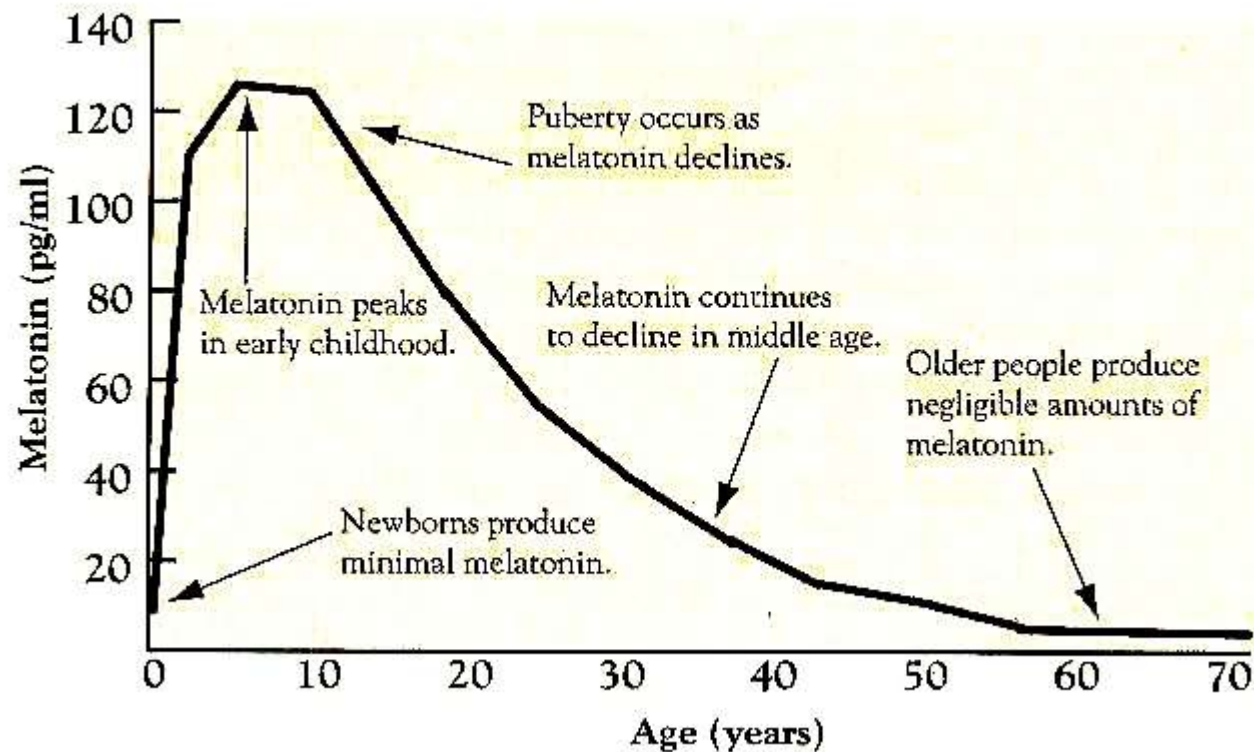
The following are the results when the HbA1c is being used to diagnose diabetes:

- Normal: Less than 5.7%
- Pre-diabetes: 5.7% to 6.4%
- Diabetes: 6.5% or higher

Neuro-Endocrine Clock

- Age-Related Decline In key Hormones Accelerates Aging:
 1. GH and IGF-1
 2. Melatonin (brain, immune, cancer, sleep quality)
 3. Reproductive and Anabolic Hormones - DHEA, testosterone, estrogen, progesterone
 4. Thymus Hormones- weaken immune system

Age-Related Decline In Melatonin – decline triggers puberty



Melatonin required for sleep, brain antioxidant and immune system modulator (low melatonin associated with breast and prostate cancers)

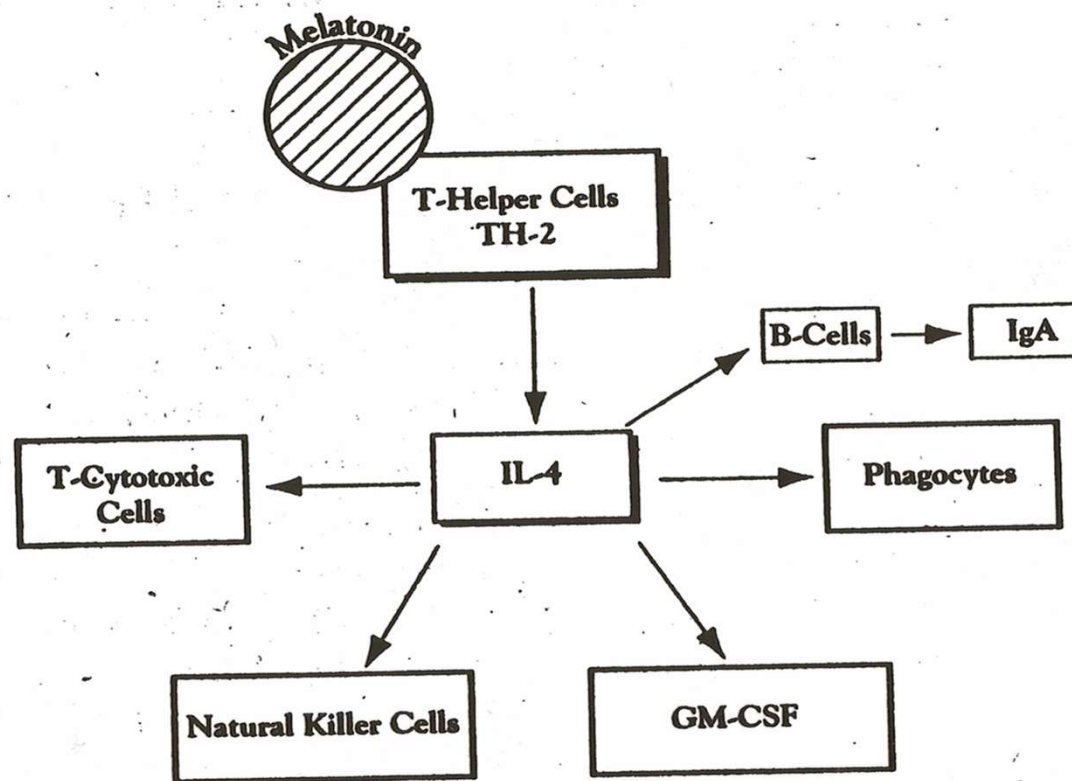
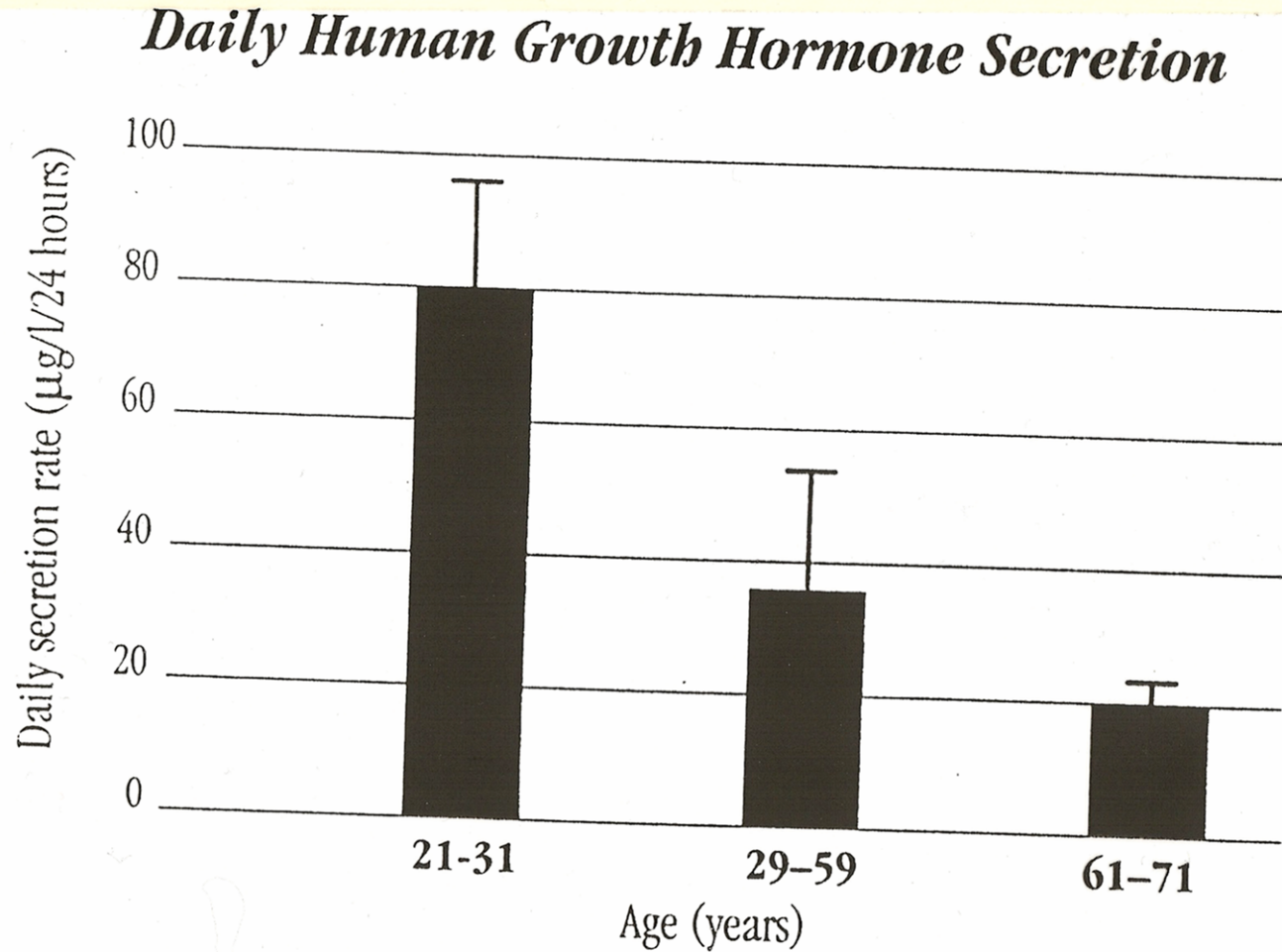


Figure 7. Melatonin Stimulates T-Helper Cells

Age-Related Decline In Growth Hormone: A 20 year old averages 20 ng/ml of HGH in their blood,
a 60 year old averages 2 ng/ml



Oral GH Secretagogues Can Return IGF-1 To Age 35 Levels – 275 ng/mL



Daily HGH secretion rate is reduced progressively with increasing age.

Growth Hormone Released With Exercise (also Testosterone)

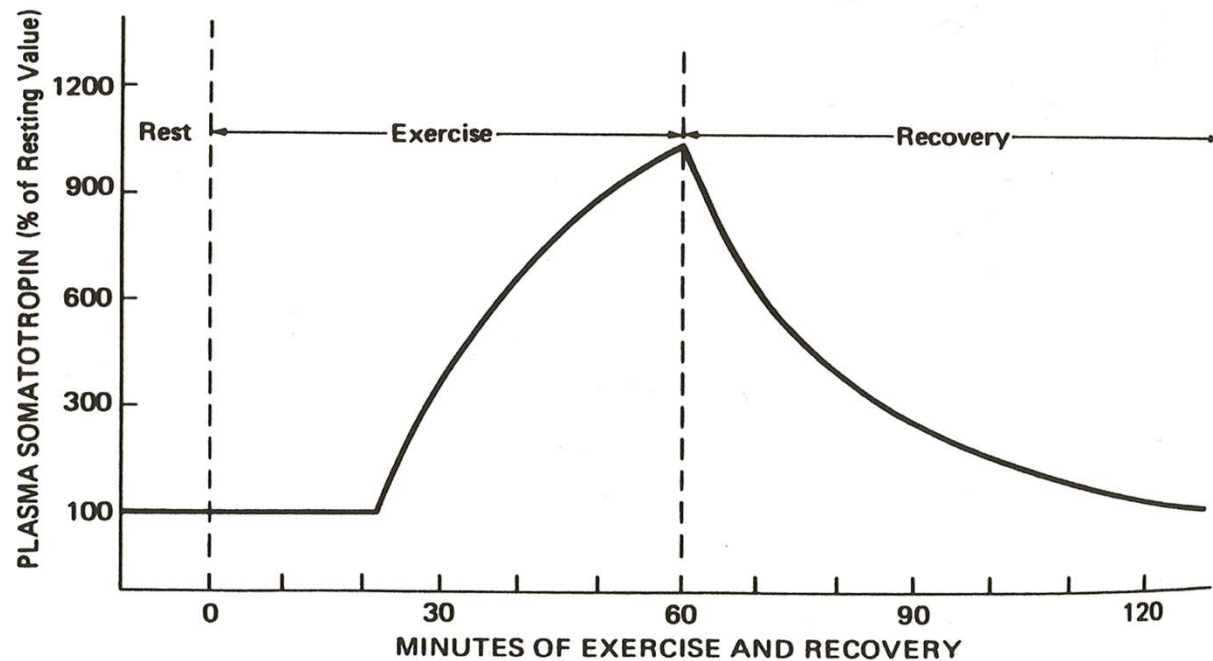
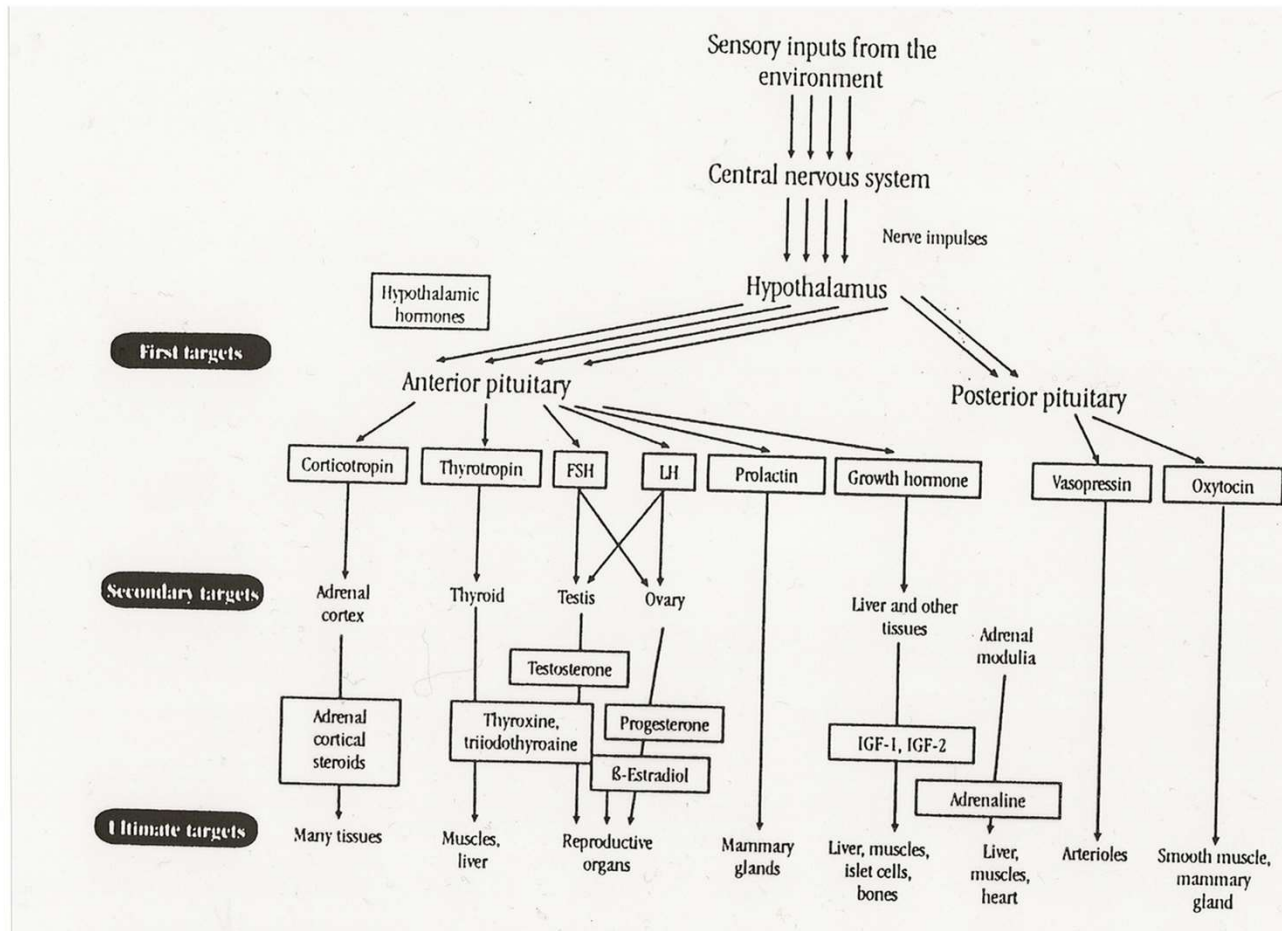
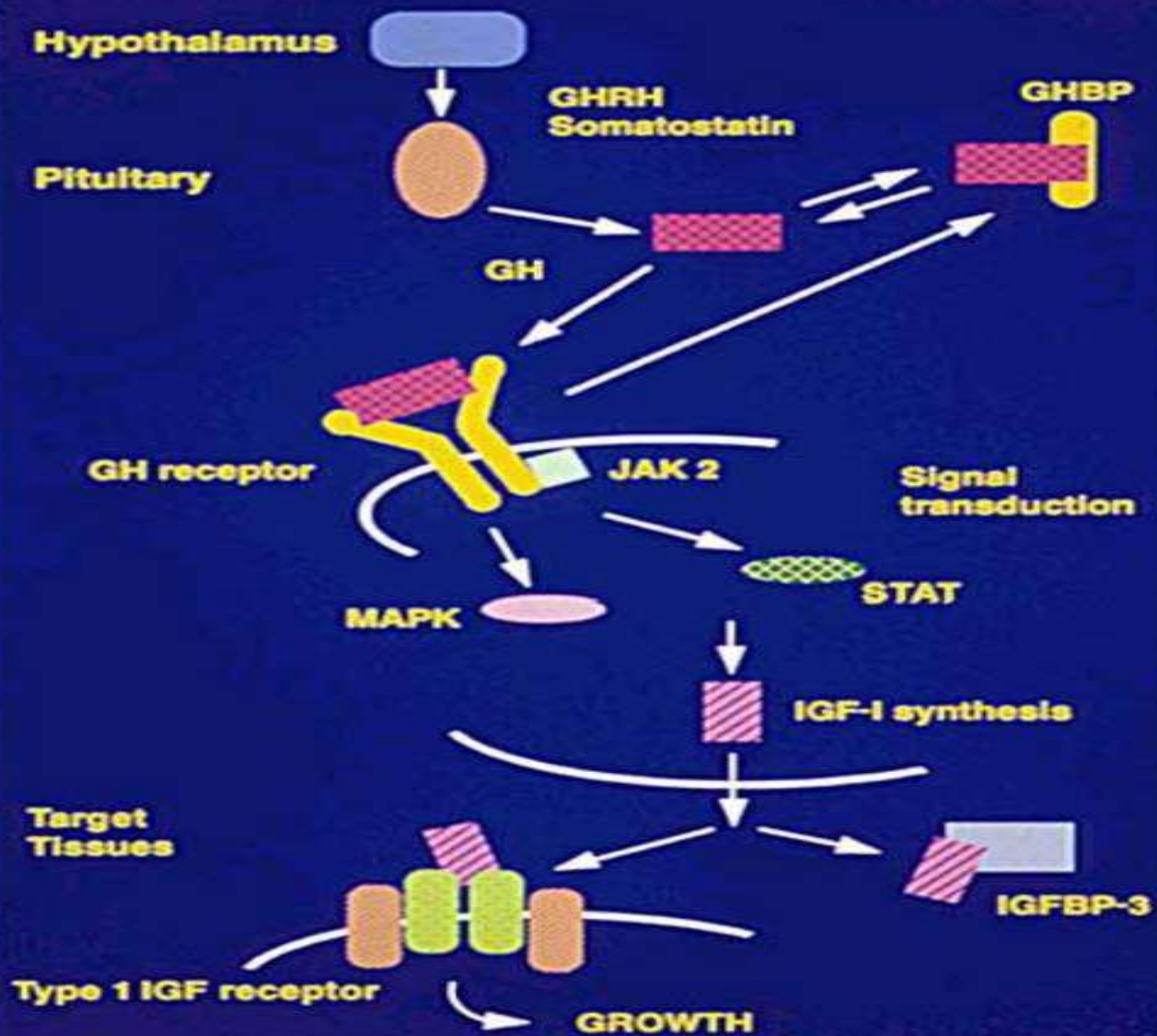


FIGURE 17.2. Diagram of growth hormone concentration in blood plasma during exercise and recovery.

Neuro-Endocrine Signaling



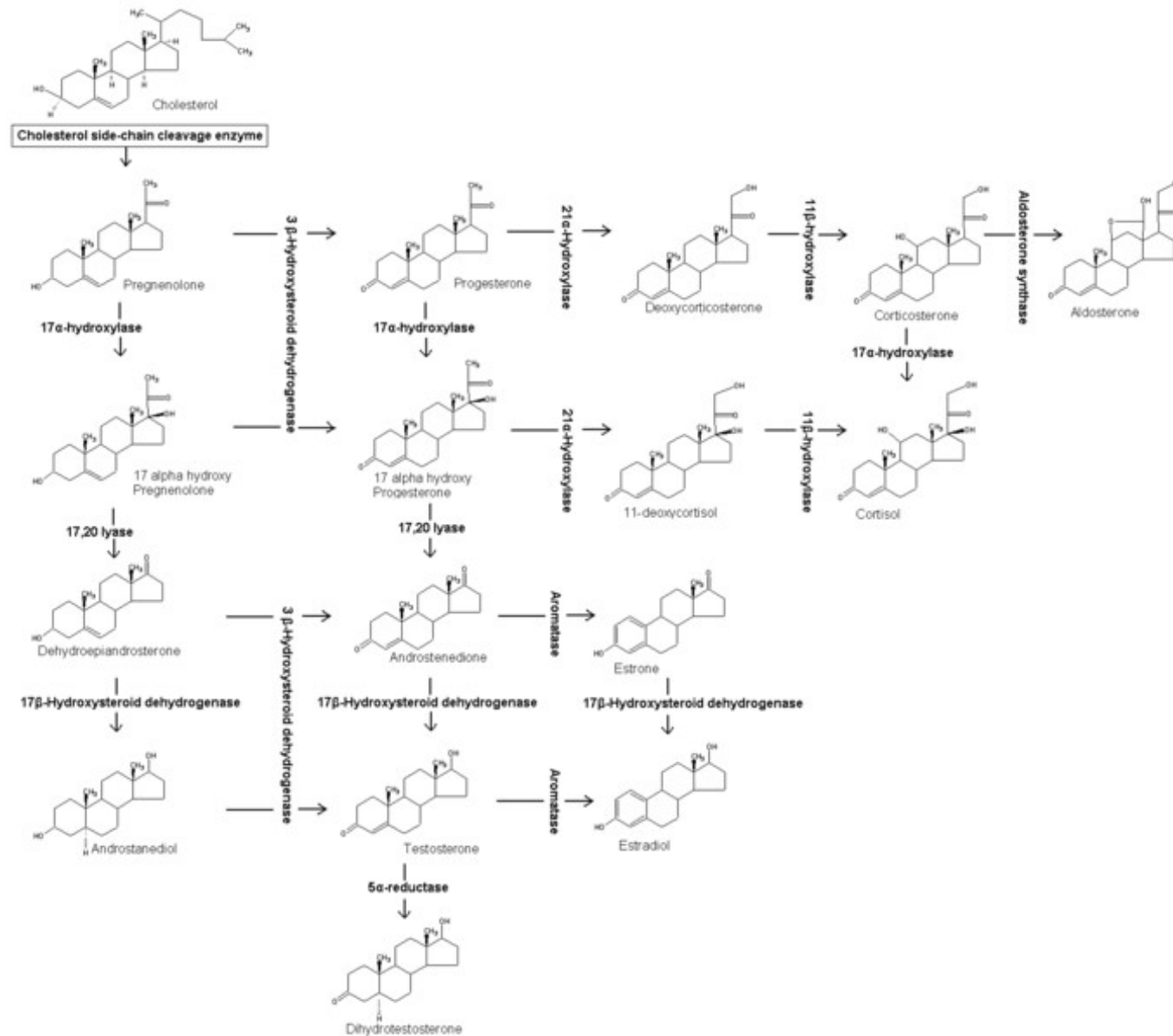
Growth hormone - IGF-I Axis



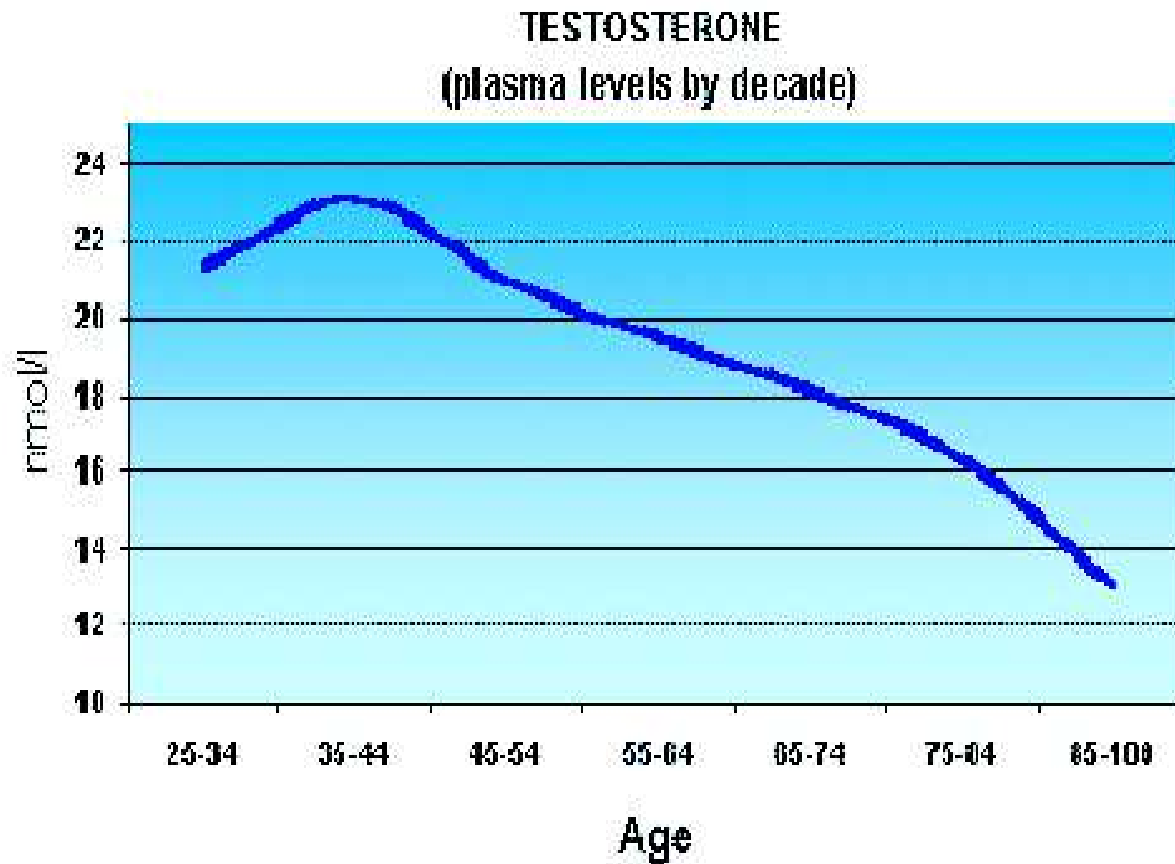
Steroid Hormone Synthesis:

- Cholesterol – Pregnenolone – DHEA – Androstenedione – Testosterone – Estrogen
- Androstenedione – Estrone (fat cells)
- Testosterone – DHT (prostate and liver)

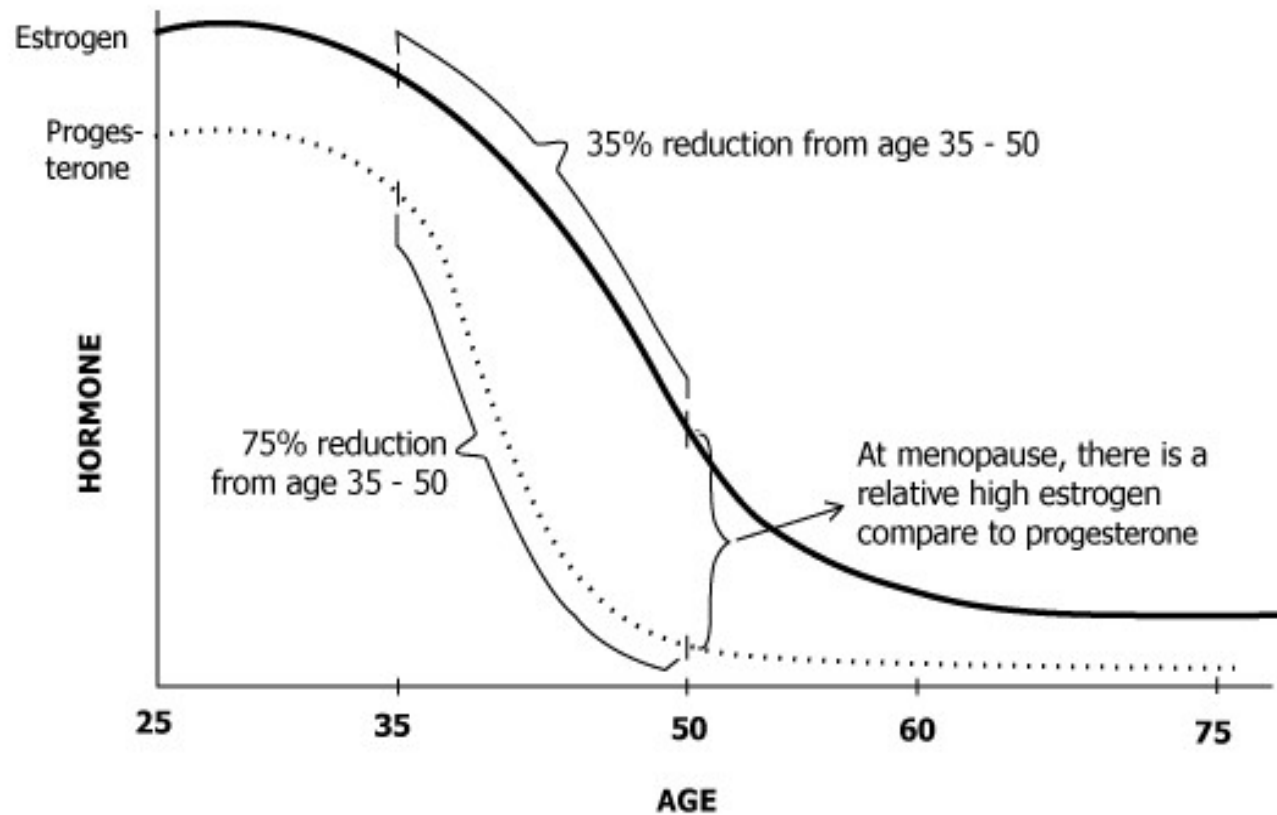
- Steroidogenesis



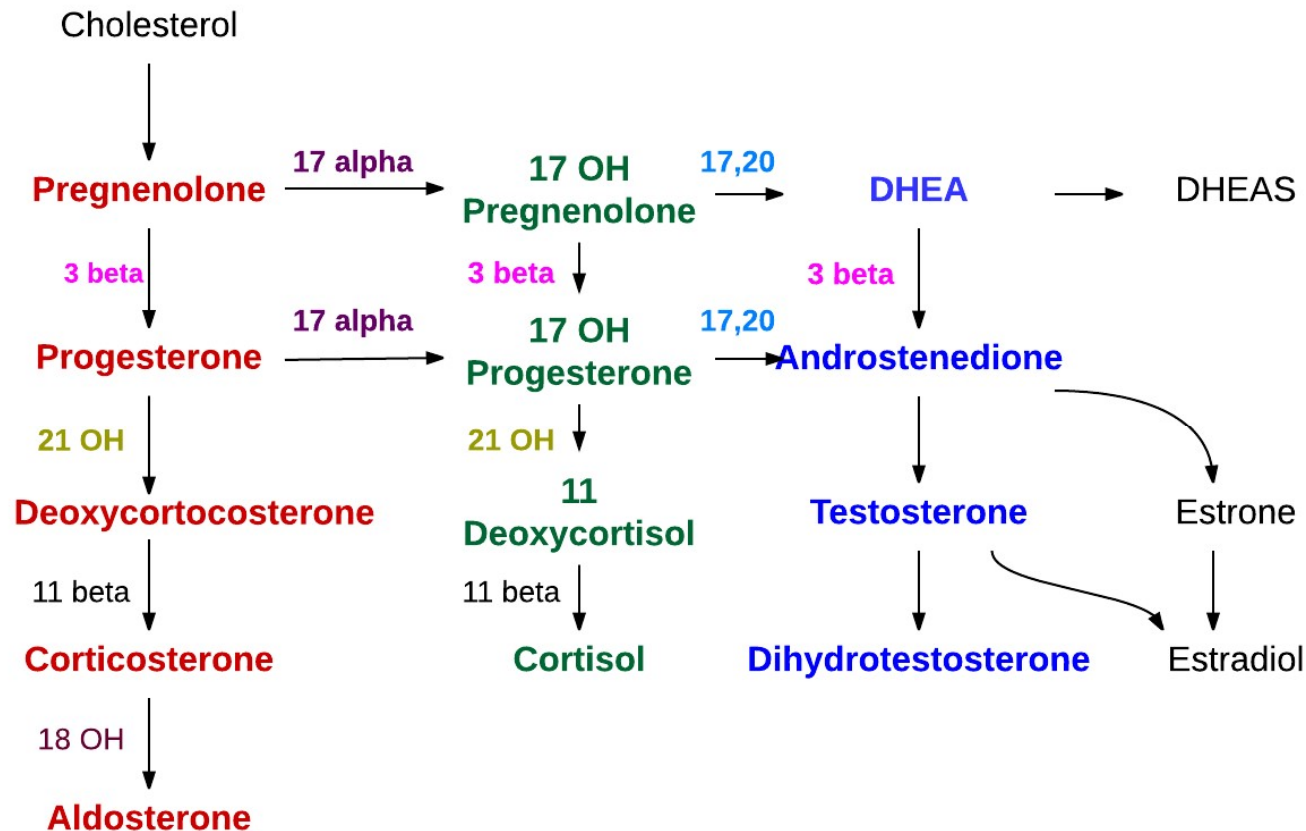
Age-Related Decline In Testosterone: 1% decline annually after 40



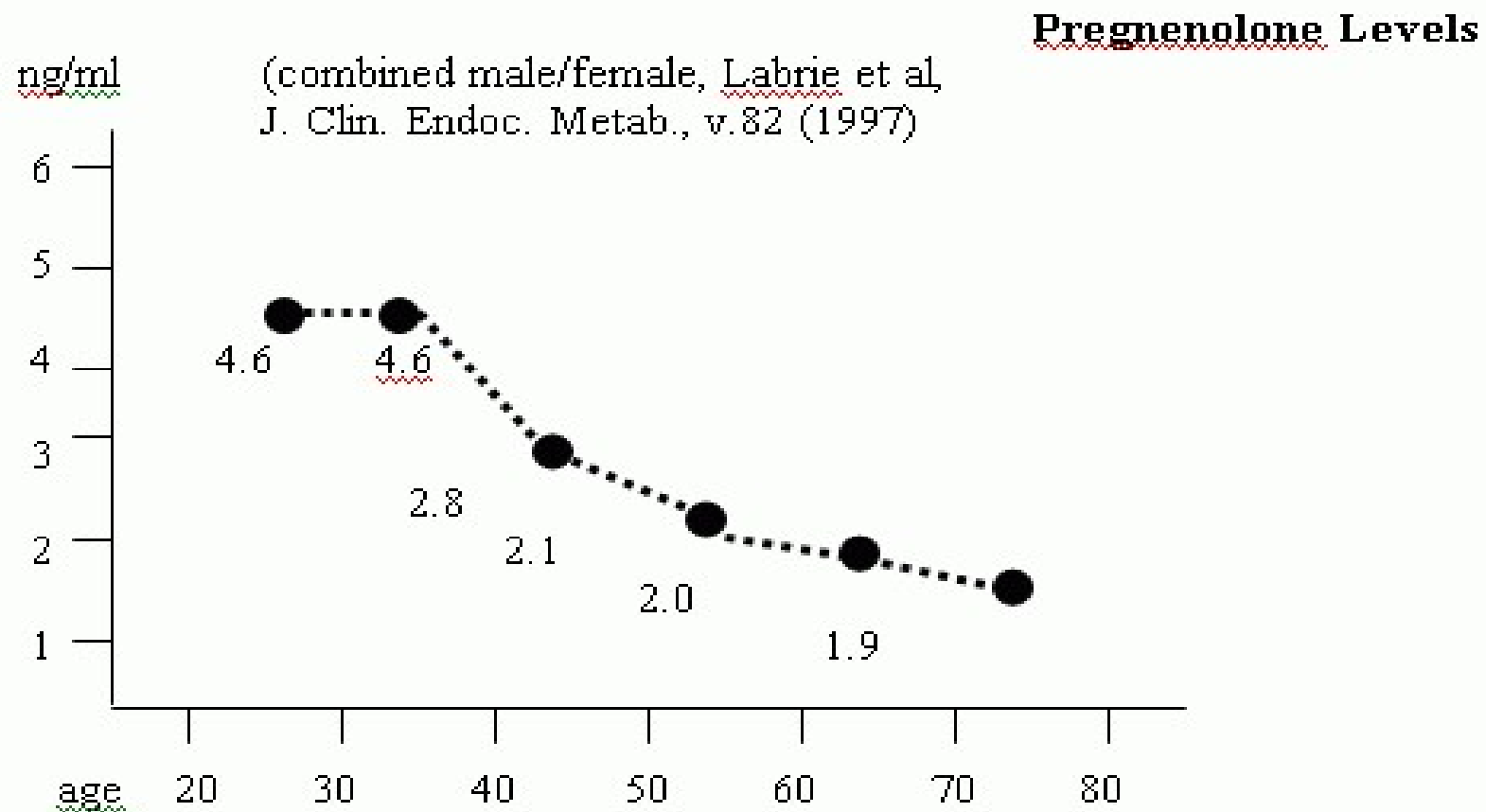
Age-Related Decline In Estrogen and Progesterone In Women



With Aging - Decreased Synthesis Pregnenolone, DHEA, Androstenedione



Decline in Pregnenolone with Age



Decline in DHEA with Age

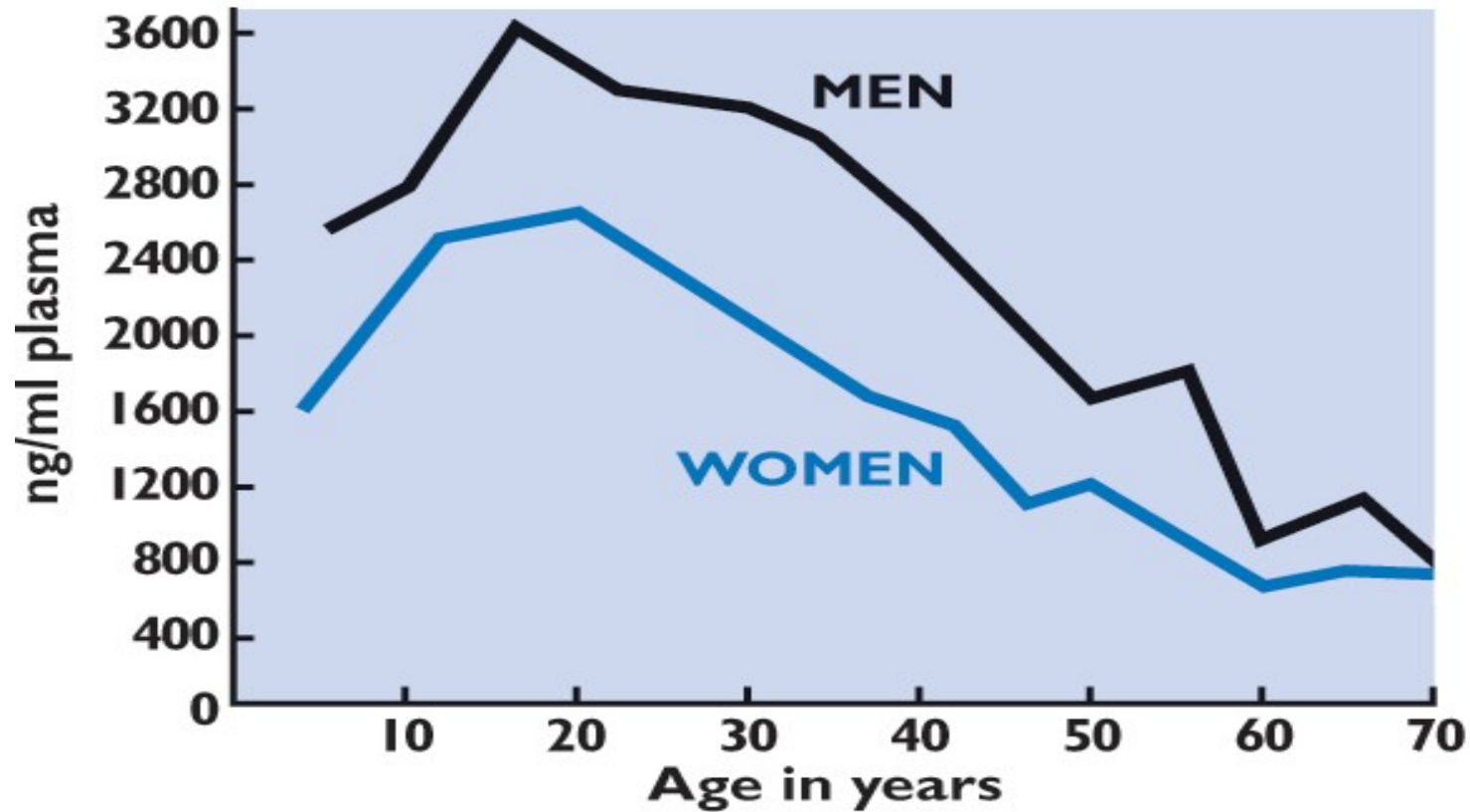


Fig 1. Decline of DHEA-S with age.³

Combating Decline in Anabolic Hormone with Age: Using Nutritional and Lifestyle Medicine

- **Strength Training** – increases muscle protein synthesis (MPS) and Preserves and Maintains Bone Mass - muscles crave amino acids
- **Protein** – 1.0 – 1.25 gm/kg body weight per day – to build and preserve lean mass in aging
- **Creatine Monohydrate** – stimulates MPS
- **Growth Hormone Secretagogues** – if IGF-1 level is below 150 ng/ml – very weak at 100 ng/ml
- **Endurance Fitness** – for cardiorespiratory fitness and maintain insulin sensitivity
- See Other Details in Course Notes (i.e. GH, IGF-1, Melatonin)

More details in Sports Nutrition and Exercise Physiology, and Male and Female Reproductive Health, and Osteoporosis Courses

Nutrition, Environment And Cancer

Factors In Cancer Development:

Many Overlap with Anti-aging

1. Free Radicals - Damage Genes Creating Mutations:

Cigarette smoke

Alcohol

Polycyclic Aromatic Hydrocarbons

Heterocyclic Amines

Air pollutants such as asbestos, benzene, carbon monoxide, chlorine, formaldehyde, ozone, and toluene

Chemical solvents - cleaning products, glue, paints, paint thinners

Pesticides and Herbicides

Water pollutants such as chloroform and other trihalomethanes from chlorination

Radiation

Ultra-violet light

Oxygen - unavoidable

2. Insufficient Antioxidant Protection

5-7 fruit and vegetable servings assoc with 50% reduced risk of cancer in some studies

Antioxidant Supplementation assoc with decreased risk of cancer risk – 48% (EPIC Study) – 500,000 individuals followed for 11 years, from 10 European countries)

A basic multiple vitamin offered no protection (EPIC Study, 2011)

3. Rapid Cell Replication: more genetic mistakes, less time for DNA repair enzymes to fix mistakes, and shorter telomeres

HRT – estrogen, testosterone, growth hormone

High Animal Fat Diet due in part to arachidonic acid endotoxins from gut bacteria (except fish)

Linoleic Acid-rich in some Vegetable Oils (corn oil)

Lack of omega-3 fats - too much PG-2 synthesis, not enough PG-1 & PG-3 (lack of omega-3 fats)

Transfats

Overweight – increase insulin, estrogen, decrease SHBG (diabetics have increased cancer risk and poorer prognosis)

Alcohol - increases estrogen and other hyperproliferation effects

Low Vitamin D blood levels

Insufficient calcium intake – colon cancer (every 300 mg per day associated with 8-9% decrease in colo-rectal cancer)

4. Weakened Immune Function: Decreased cancer killing abilities of immune cells and increased risk of viruses causing cancer (e.g. HPV, various Herpetic viruses):

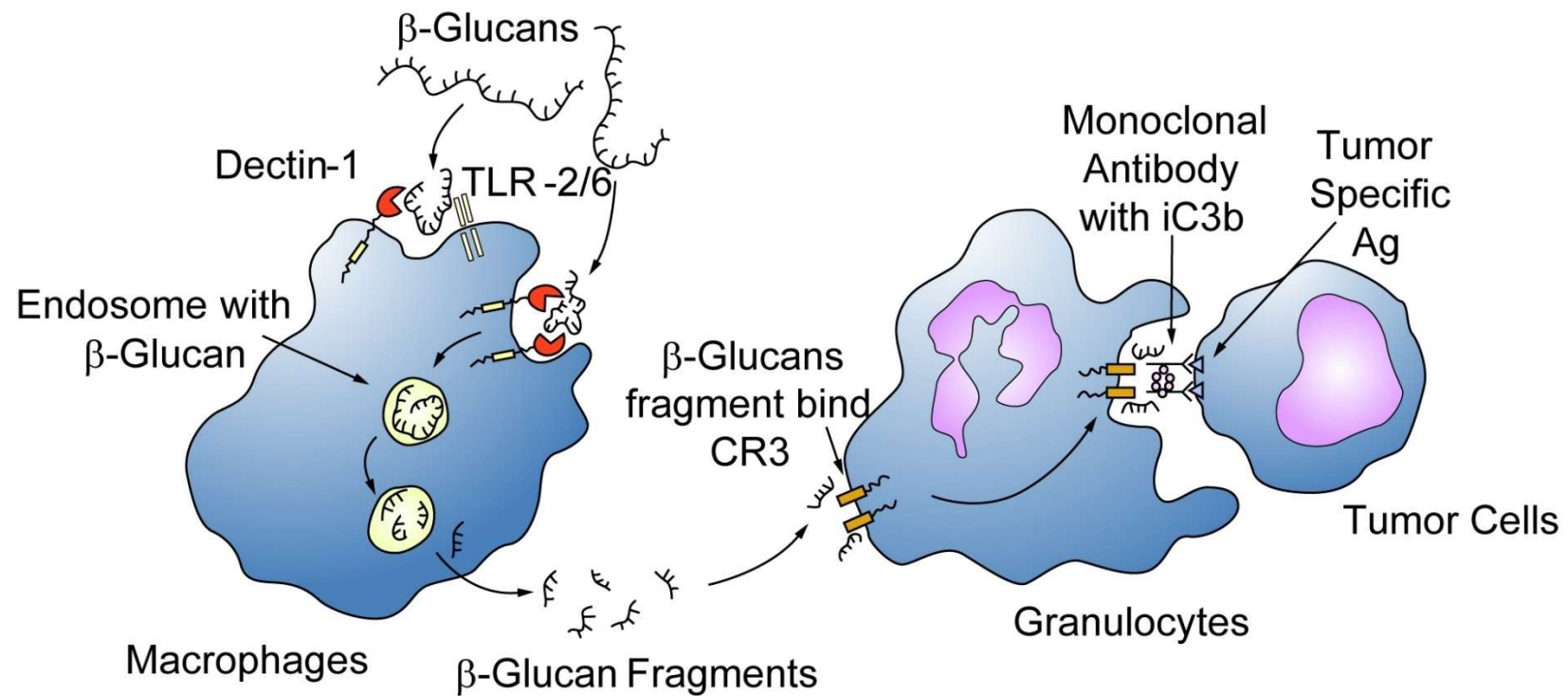
Causes:

- Thymus gland involution with aging – T-cell modulation
- Free radical damage accumulation
- Decreased respiratory bursts (HOOH) neutrophils by age 60
- Drugs: corticosteroids (Prednisone), TNF-inhibitors for autoimmune disease (Humira), anti-rejection drugs (Cyclosporin)
- Nutritional Deficiencies –zinc, vitamin A, vitamin C, vitamin D etc.
- Compromised Immune States: e.g. HIV-infection, Over training
- Depression and Stress: feeling hopeless, high cortisol etc.

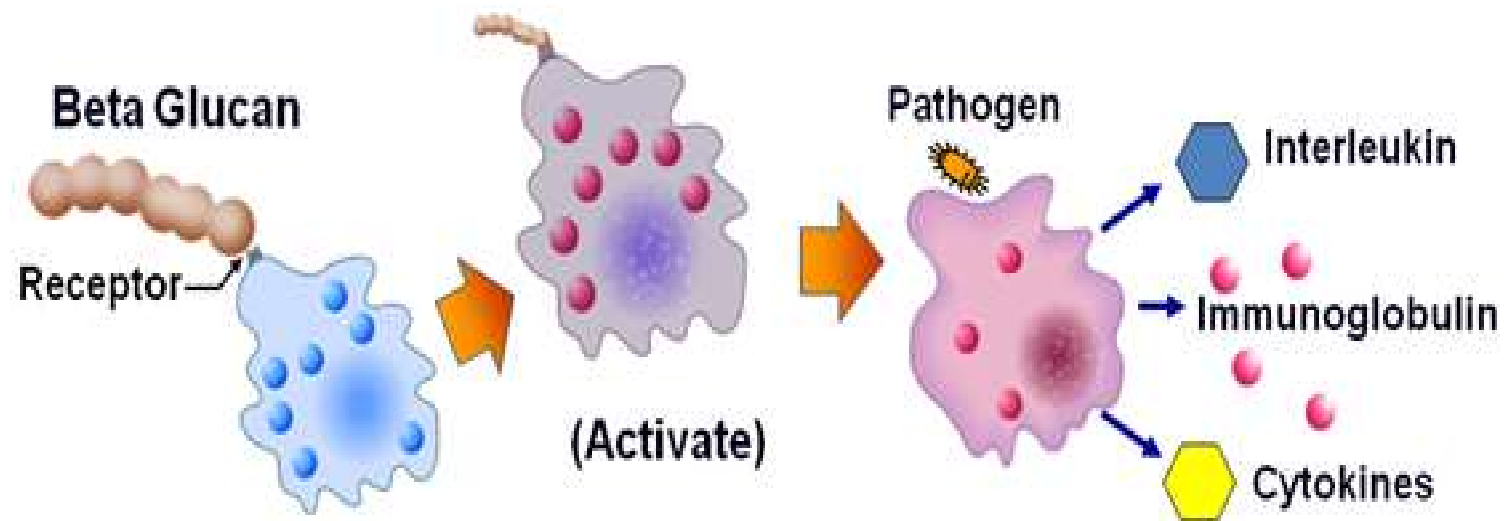
Nutrient Support For Immune System:

- Vitamin C
- Vitamin E
- Beta-carotene
- Vitamin A
- Vitamin D
- Selenium
- Zinc
- Herbs: medicinal mushrooms, astragalus, phytosterols, ginseng

Medicinal Mushrooms – beta-glucans and macrophage receptors



How Does Beta Glucan Work?



Macrophage activation via cell surface receptor

5. Decreased Cellular Differentiation: cells more prone to carcinogens:

Vitamin D – blood level above 85 nmol/L

Vitamin A – metaplasia (precancerous)

Carotenes (beta-carotene, lycopene)

6. Weak Chromosomal Linkages: From hypomethylation

Folic Acid – low blood levels increase cancer risk (hypomethylation of DNA and base pair mutations, as uracil substitutes for thymine in DNA synthesis)

Vitamin B12 – low blood levels associated with increased of certain cancers (decreased DNA synthesis from decreased methylation)

7. Tone down Mitogen Receptors (EDGF, IGF-1, Insulin) – see next slide

Vitamin E Succinate

Vitamin D

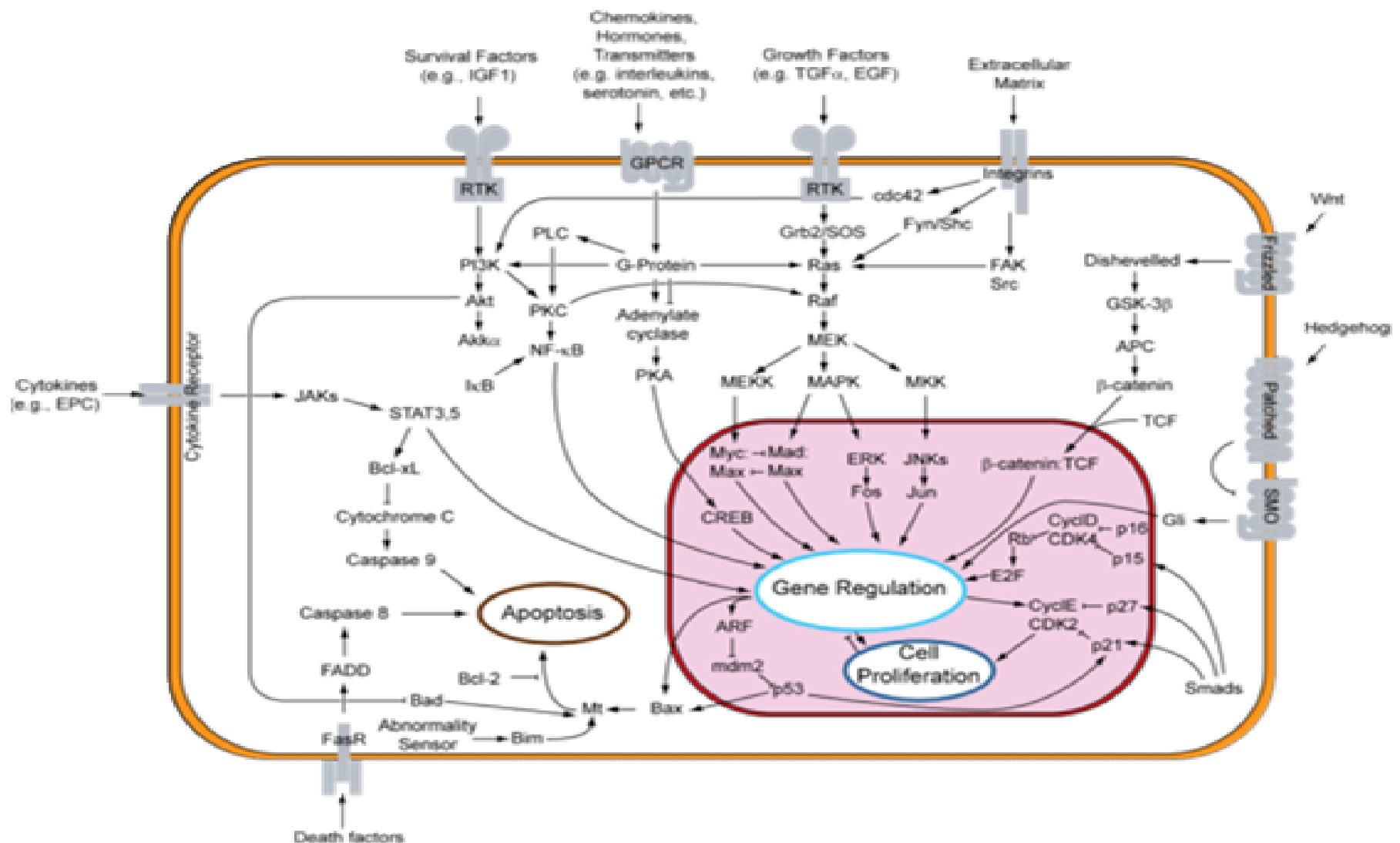
Curcumin

Omega-3 fats (fish and fish oil)

Lycopene (IGF-1 – keep it under 275-300 ng/ml)

Fasting glucose and insulin

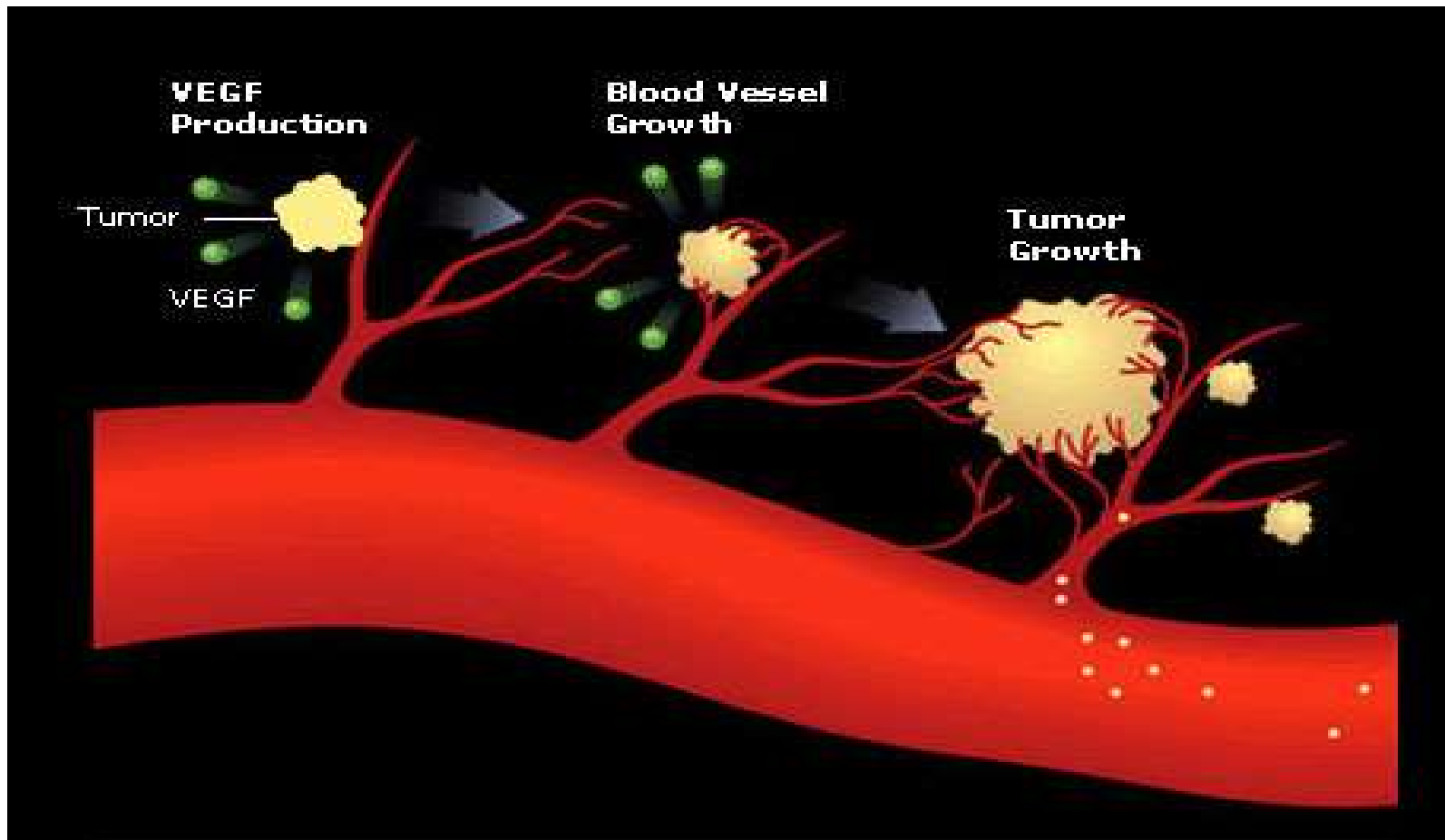
Note: IGF-1, Insulin and EGFR Receptors



8. Nutrients That Prevent Cancer Cells From Gaining Forming New Blood Vessels – anti-angiogenesis (see next slide)

1. Vitamin E Succinate
2. Milk Thistle
3. Soy Isoflavones – up to 80% fewer cancers when people consume 35-50 mg of soy isoflavones per day
4. Curcumin
5. Green Tea (EGCG)
6. Strawberries and Raspberries (ellagic acid)

Angiogenesis from Vascular Endothelial Growth Factor- once tumor mass is 1-2 mm in size

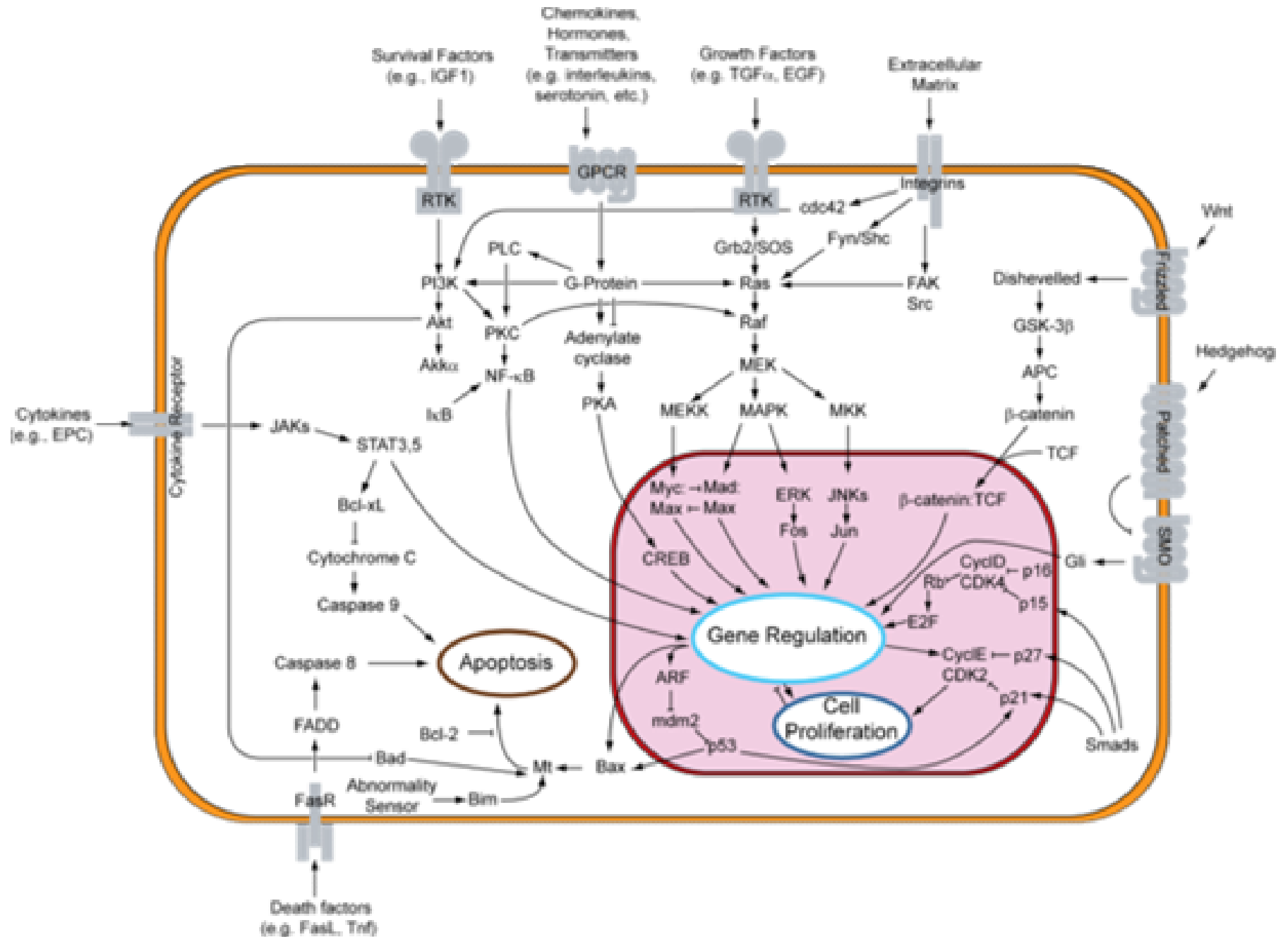


9. Nutrients That Encourage Cancer Cells To Commit Suicide (apoptosis) – see next slide

1. Vitamin E Succinate
2. Selenium
3. Vitamin C (via immune function)
4. Reishi Mushroom Extract
5. Soy Isoflavones
6. Garlic and onions (*disulfide compounds*)
7. Green tea Catechins
8. Melatonin -

At physiological concentrations, melatonin suppresses cell growth and multiplication and inhibits cancer cell proliferation in vitro through specific cell-cycle effects. At pharmacological concentrations, melatonin suppresses cancer cell growth and multiplication. At physiological and pharmacological concentrations, melatonin acts as a differentiating agent in some cancer cells and lowers their invasive and metastatic status by altering adhesion molecules and maintaining gap junction intercellular communication. In other cancer cell types, melatonin, alone or with other agents, induces programmed cell death. (http://physics.fau.edu/observatory/lightpol-melatonin.html#Melat_Summ_NCI)

David Blask, M.D., Ph.D.
Laboratory of Experimental Neuroendocrinology/Oncology
Bassett Research Institute
Cooperstown, NY.



10. Inherited Mutations Can Increase Risk Of Cancer (egfr, brca,1brca2) – Genomic Screening Now Available

Most inherited cancer genes do not guarantee cancer will develop, but increase risk.

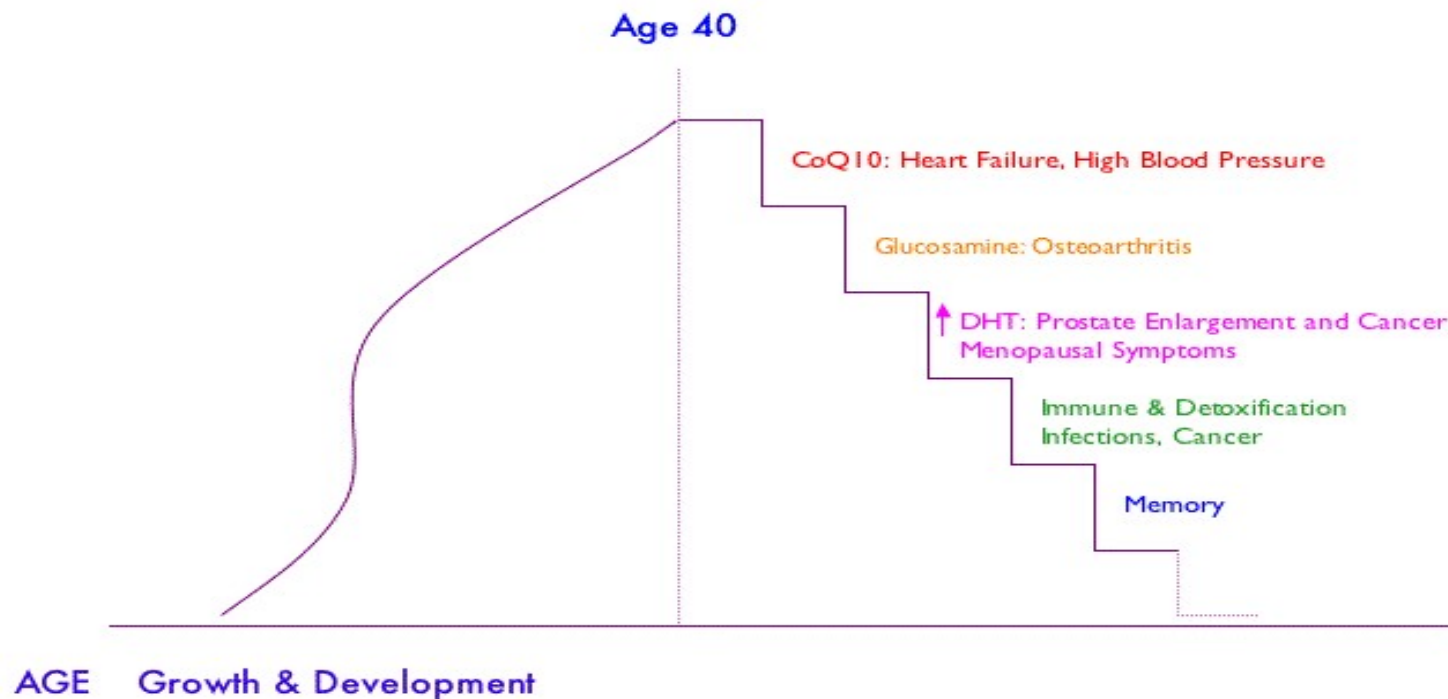
People beat cancer genes by paying attention to preceding 9 strategies

We'll look at specific cancer gene mutations in Cancer Course

JNCI – Willett W. 1996

Estimates of cancer deaths avoidable by dietary change			
Types of cancer	Percent avoidable		
	Doll-Peto (1981)	Willett (1994)	Range (1994)
Lung	20	20	10-30
Colon/rectum	90	70	50-80
Breast	50	50	20-80
Prostate	(with other)	75	20-80
Pancreas	50	50	10-50
Stomach	35	35	30-70
Endometrium	50	50	50-80
Gall bladder	50	50	50-80
Larynx, bladder, cervix, mouth, pharynx, esophagus	20	20	10-30
Other types	10	10	—
Overall estimate	35	32	20-42

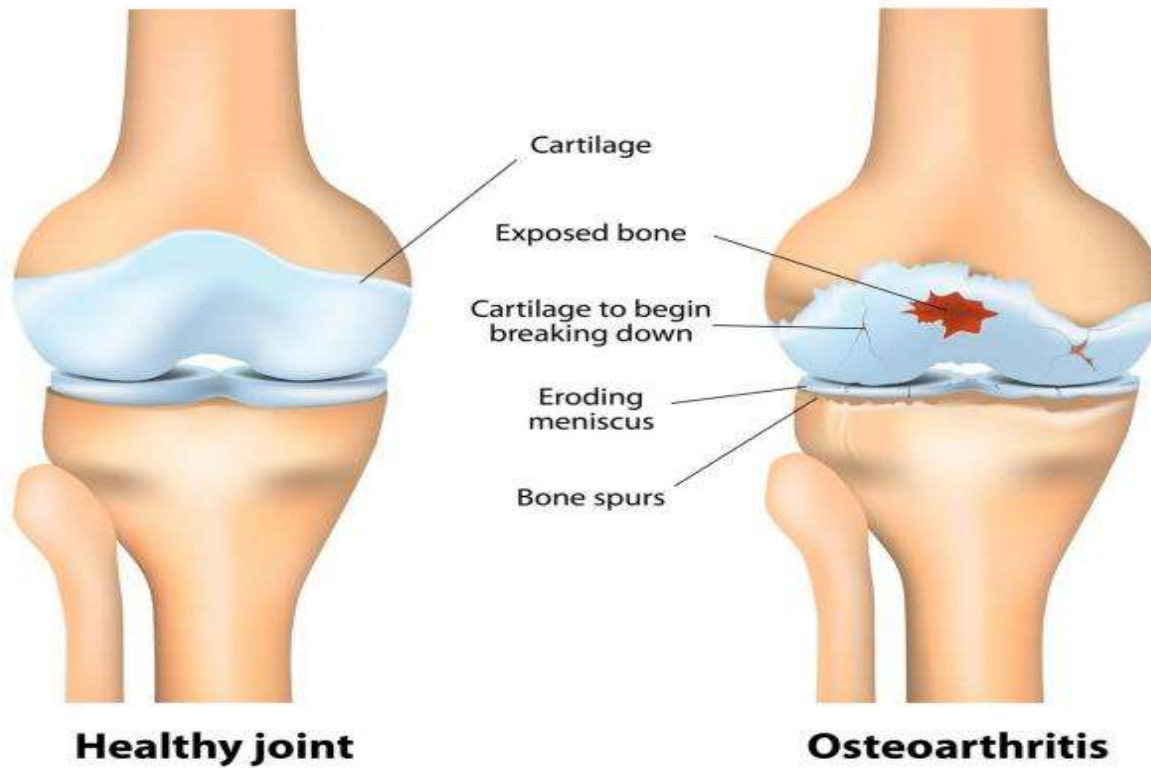
Anti-Aging And Supplements: After Age 40 Genetic Time Bombs Go Off That Can Be Countered To Significant Degree With Targeted Nutritional Supplements



At Age 40

**Joint Cartilage Erosion
(osteoarthritis)**

OSTEOARTHRITIS



Decrease Production Of Glucosamine Due To Decreased Activity of Fructose-6 amide Transferase Enzymen (note glutamime becomes glutamate)

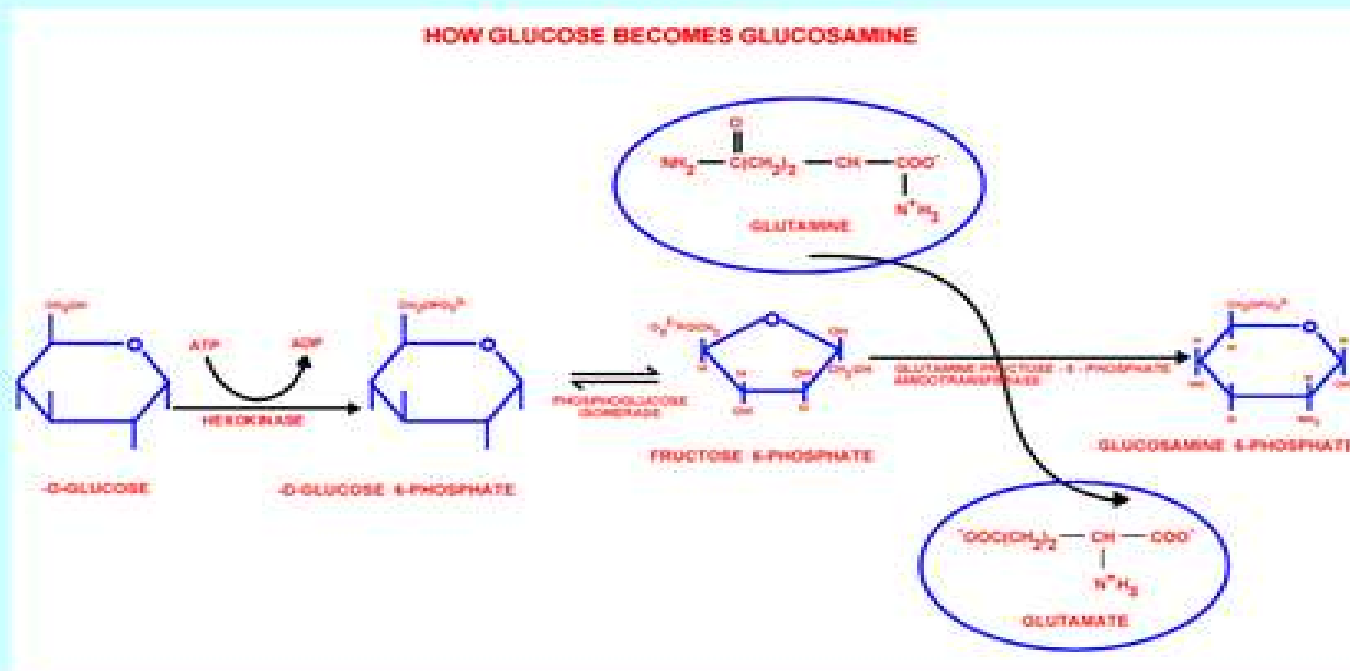


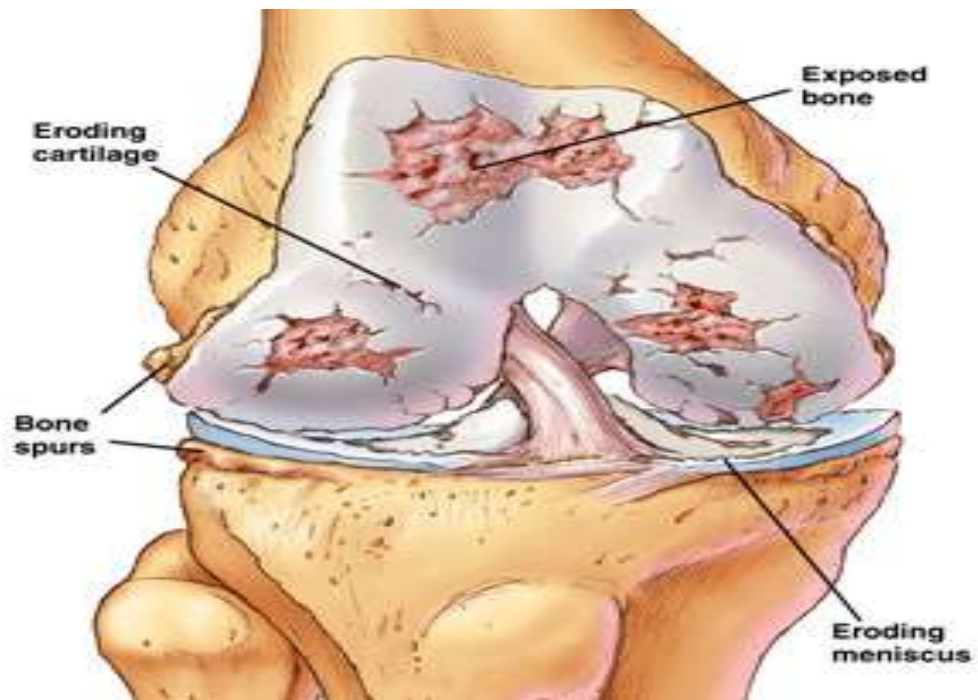


Figure 1



Figure 2

Cartilage Erosion



Osteoarthritis from Decreased Synthesis of Glucosamine at Age 40

Osteoarthritis



Healthy knee joint



Hypertrophy and spurring
of bone and erosion of cartilage

How To Decrease Risk Of Arthritis

Age 40 - Decline in glucosamine synthesis results in:

1. Thinning of joint cartilage
2. Arthritis (80% of people will develop if no strategy in place).

Solution: Supplement with **Glucosamine** and natural anti-inflammatories (**e.g. quercetin, bromelain and MSM**) – begin at age 40

Does Glucosamine Supplementation Work?

- A 2012 review paper in the journal **Therapeutic Advances in Musculoskeletal Disease** instructs medical doctors to use **glucosamine sulfate** in the management of osteoarthritis.
- Medical researchers point out, clinical studies prove that **glucosamine sulfate** (not other forms of glucosamine) **block the release of inflammatory chemicals (NF-kb, interleukin-1) found in osteoarthritis, and Glucosamine sulfate is also the only known substance that blocks the destruction of joint cartilage** – the hallmark feature in osteoarthritis.
- **Lead Researcher: Lucio C. Rovati, MD**
Chief Scientific Officer, Rottapharm | Madaus, Via Valosa di Sopra, 9, 20900 Monza (MB), Italy

At Age 45

**Power Shortage from Decreased
CoQ10 Synthesis**

CoQ10 & Heart Disease

At age 45: Body decreases CoQ10 synthesis

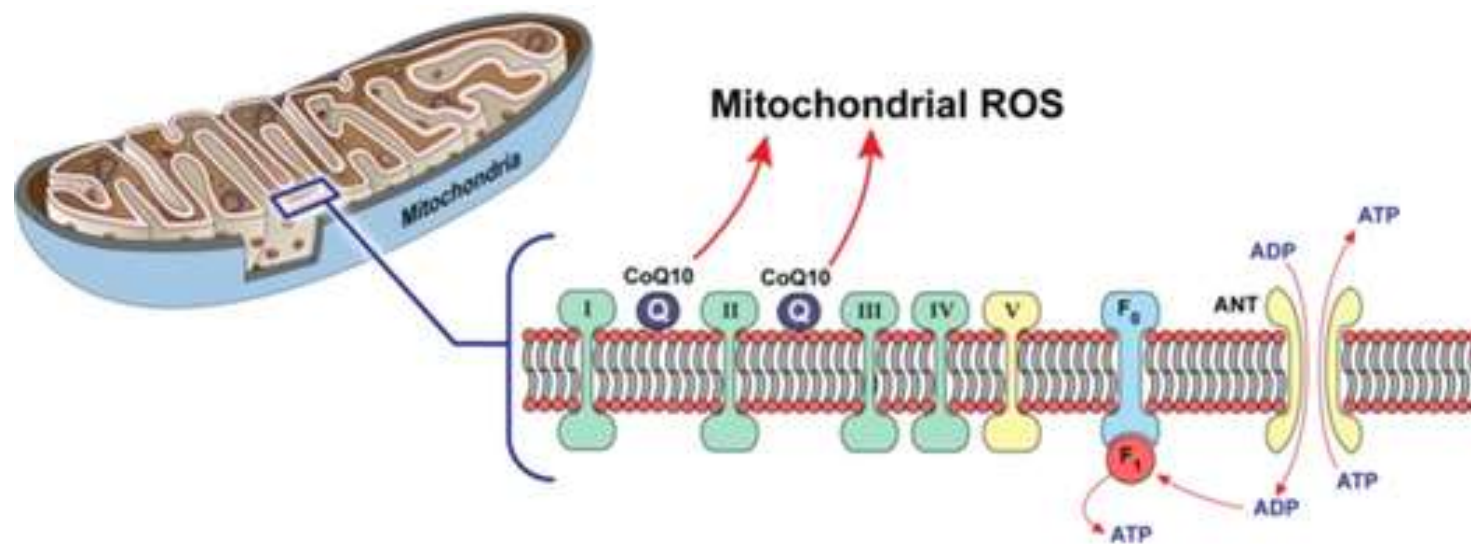
CoQ10 required to convert food into energy virtually all body cells (not in RBC to appreciable degree)

Decreased energy contributes to:

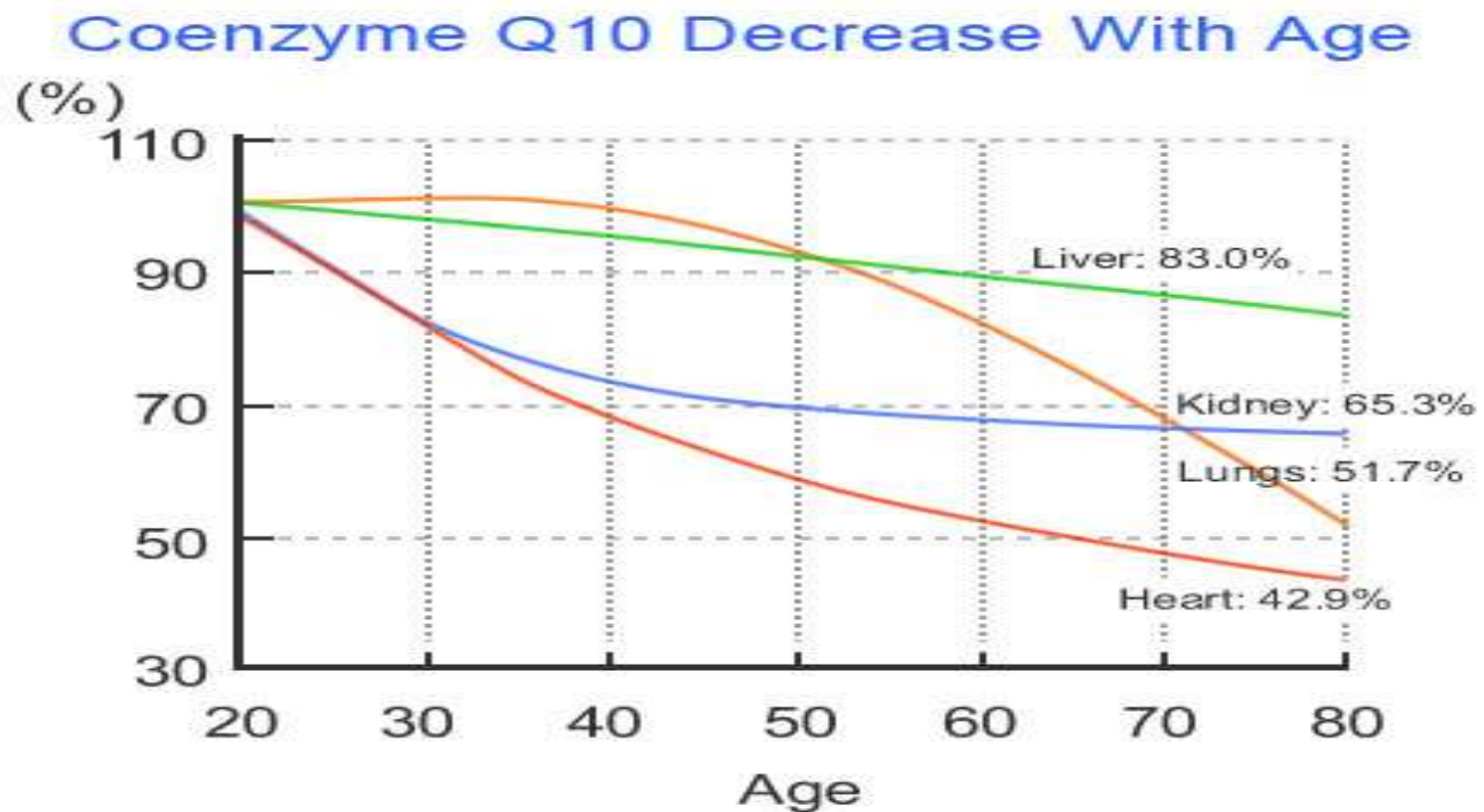
1. **Congestive Heart Failure** (Power Shortage)
2. **High Blood Pressure**

75% of CHF patients show decreased CoQ10 concentrations in heart muscle (*Folkers K*)

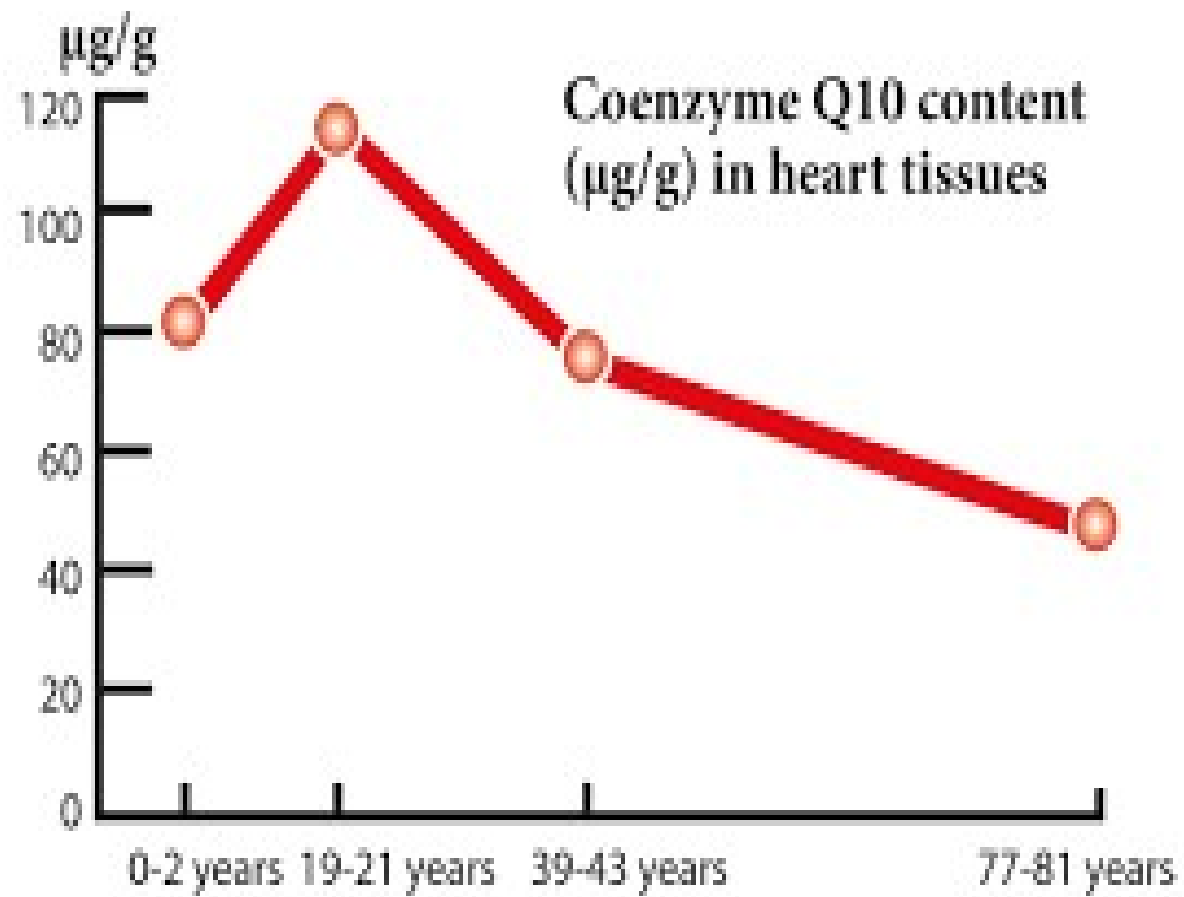
CoQ10 and Mitochondria - 95% of energy production in the body



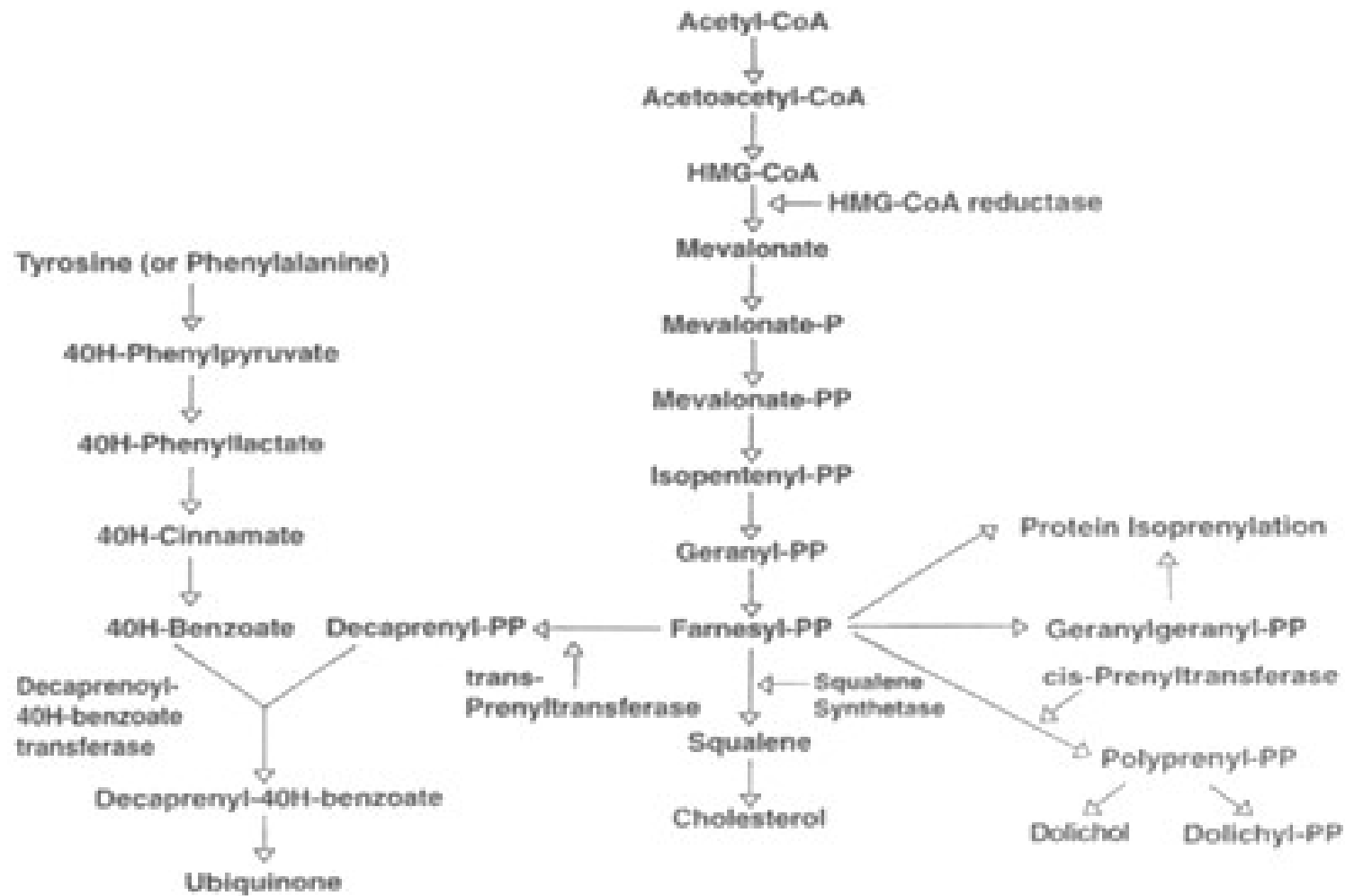
CoQ10 Decline With Age: Note Heart Muscle



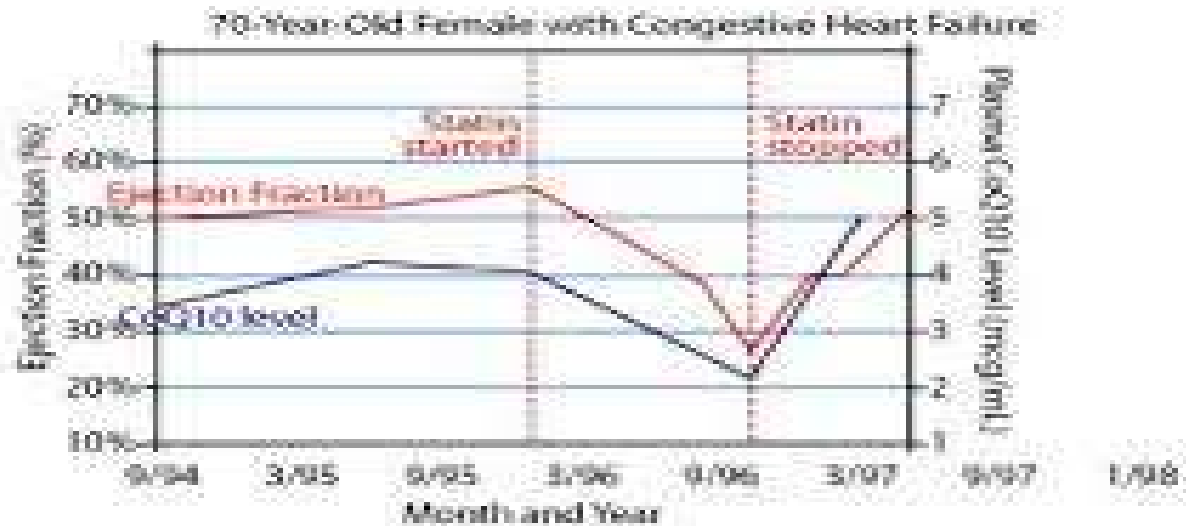
Amendment from A.Kalen et al. (1989) Lipids, 24, 579



CoQ10 Synthesis: Statin Drugs (i.e. Lipitor) block CoQ10 synthesis. If taking statin drug, it is recommended that you supplement with at least 90 mg per day of CoQ10

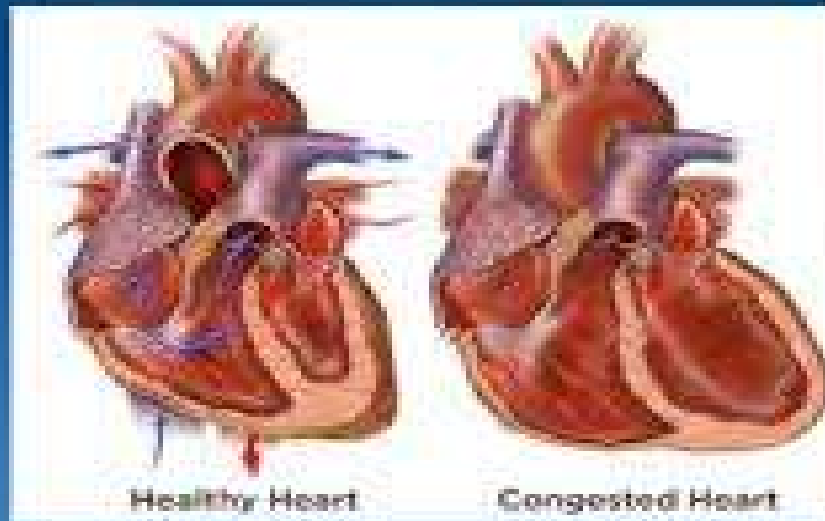


CoQ10 and Heart Ejection Fraction Decline With Statin Drug Use



Congestive Heart Failure: Leading Cause of Hospital Admissions after 65 yrs of Age

Congestive Heart Failure



Most common signs and symptoms

- Fatigue
- Edema
- Shortness of breath
- Chest congestion

Fluid Build Up In Chest With Congestive Heart Failure

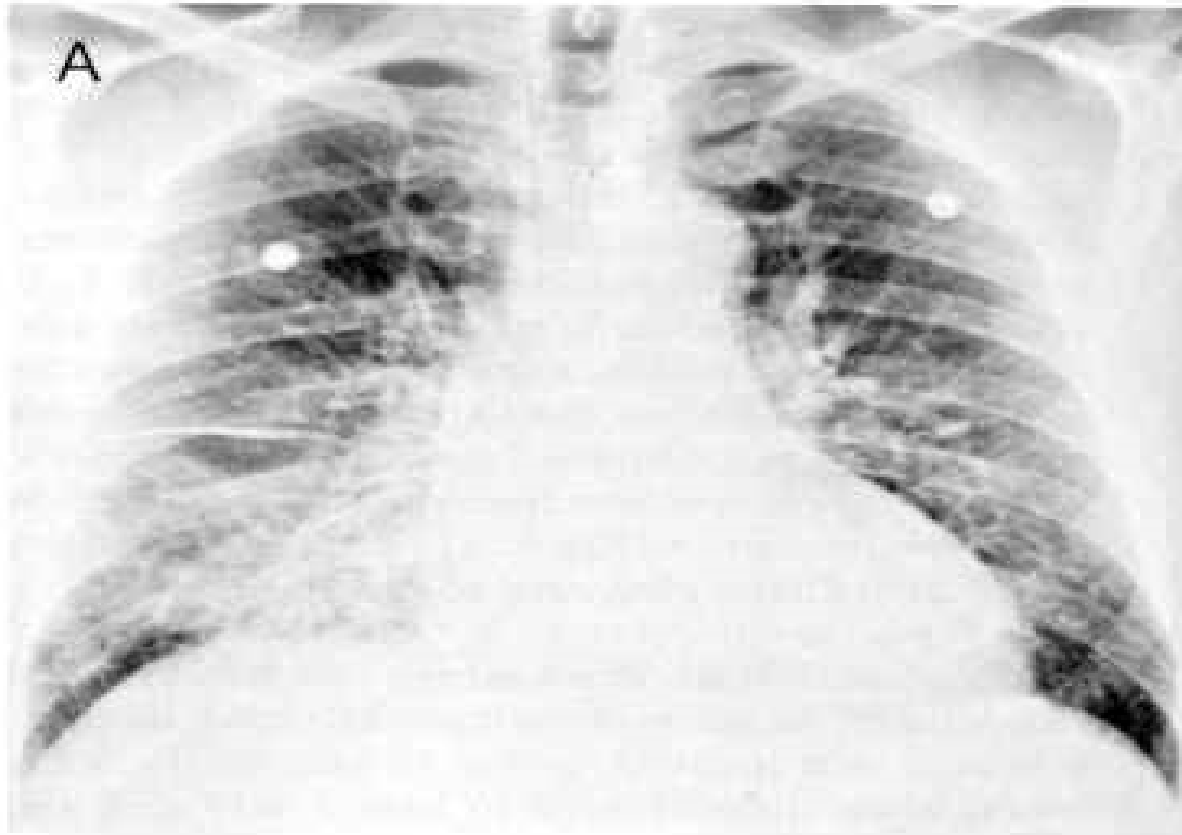
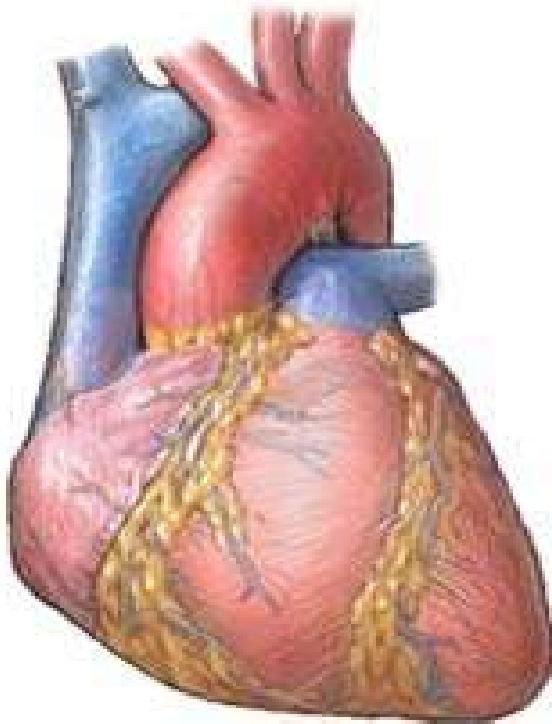


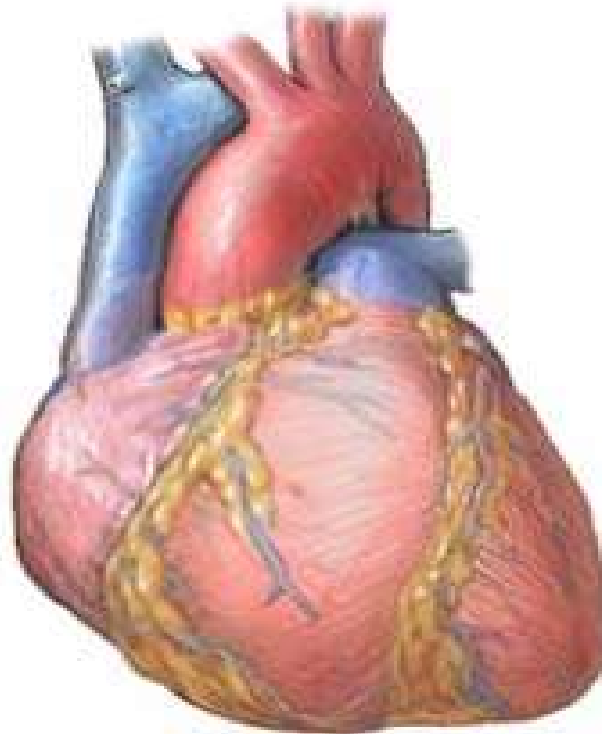
Figure 10-10-10

Congestive Heart Failure

Normal heart



Enlarged heart due to cardiomyopathy



CoQ10 And Hawthorn Work Together

Solution:

By Age 45 – consider supplementing with CoQ10 and Hawthorn (hawthorn increases effectiveness of CoQ10 in energy production. Hawthorn increases cAMP required for ADP then ATP.

Hawthorn (and CoQ10) also **improves blood flow to heart muscle** (endothelial function) –increase nitric oxide from endothelial cells lining arteries

I like Combination Supplement (CoQ10, Hawthorn, Quercetin)

Therapeutic Doses Have Been Helpful in the Following Conditions:

1. High blood pressure
2. Congestive Heart Failure
3. Atrial Fibrillation
4. Dilated Cardiomyopathy
5. Parkinson's Disease

Glucosamine with Natural Anti-inflammatory Agents Supplement Example

Each Capsule Contains:

Glucosamine Sulfate – 500 mg

MSM – 134 mg

Quercetin – 100 mg

Bromelain Enzymes – 100 mg

Dosage:

1. Prevention After 40 - 1-2 capsules per day
2. Arthritis or Cartilage Damage – 3-4 capsules per day
– Fast Results Due Anti-inflammatories

Quercetin Reduces CVD

Adv Nutr January 2012 Adv Nutr vol. 3: 39-46, 2012

Therapeutic Potential of Quercetin to Decrease Blood Pressure: Review of Efficacy and Mechanisms^{1,2}

Abigail J. Larson³,

J. David Symons⁴, and

Thunder Jalili^{4,*}

Abstract

Epidemiological studies beginning in the 1990s have reported that intake of quercetin, a polyphenolic flavonoid found in a wide variety of plant-based foods, such as apples, onions, berries, and red wine, is inversely related to cardiovascular disease. More recent work using hypertensive animals and humans (>140 mm Hg systolic and >90 mm Hg diastolic) indicates a decrease in blood pressure after quercetin supplementation. A number of proposed mechanisms may be responsible for the observed blood pressure decrease such as antioxidant effects, inhibition of angiotensin-converting enzyme activity, and improved endothelium-dependent and -independent function. The majority of these mechanisms have been identified using animal models treated with quercetin, and relatively few have been corroborated in human studies. The purpose of this review is to examine the evidence supporting the role of quercetin as a potential therapeutic agent and the mechanisms by which quercetin might exert its blood pressure-lowering effect.

CoQ10-Hawthorn-Quercetin Supplement Example

Each Capsule Contains:

1. CoQ10 – 30 mg
2. Hawthorn – 37.5 mg (5% flavonoid content) –unique anthocyanidins and proanthocyanidins in berries, leaves and stem.
3. Quercetin – 50 mg

To Keep Heart Strong As You Age:

- By Age 45 – 1 capsule per day
- By Age 60 – 2-3 capsules per day
- Statin Drugs – 3 capsules per day
- High Blood Pressure – 2-3 capsules, twice daily
- Dilated Cardiomyopathy – 3 capsules, twice daily (1/3 of all congestive ht failure – males 20-60yrs)

Age 50

**Immune And Detoxification
Systems Weaken**

Weakened Immune and Detoxification Systems After 50 Can Increase Cancer And Infection Risk

By age 50 the immune system and the detoxification system become sluggish:

Results: 1. Life-threatening infections can more easily occur (pneumonia in older people)

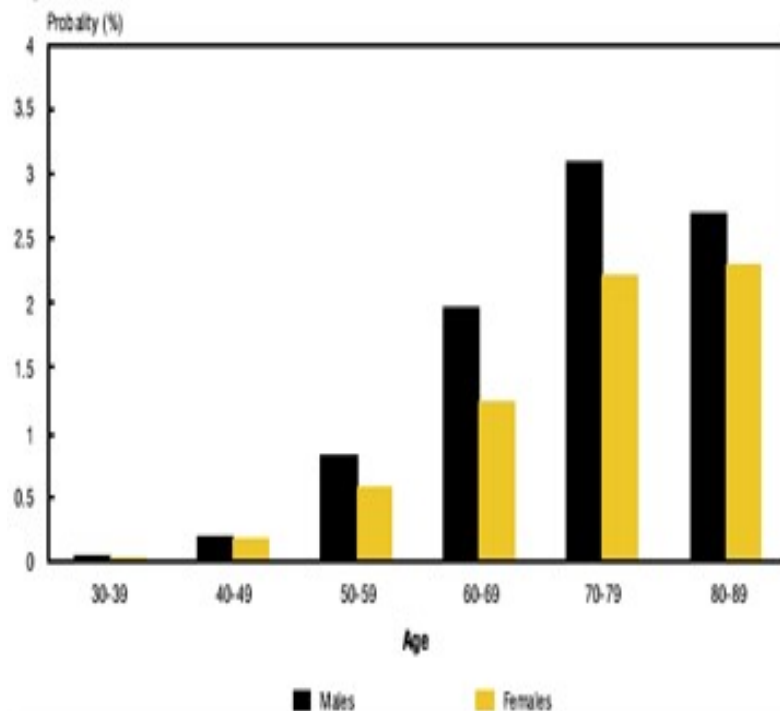
2. Immune cells less able to identify and kill emerging cancer cells

3. Cancer-causing substances more easily accumulate in body with escalation in cancer incidence after 50

Colon and Breast Cancer: Note the abrupt rise in incidence after 50

Figure 10.6

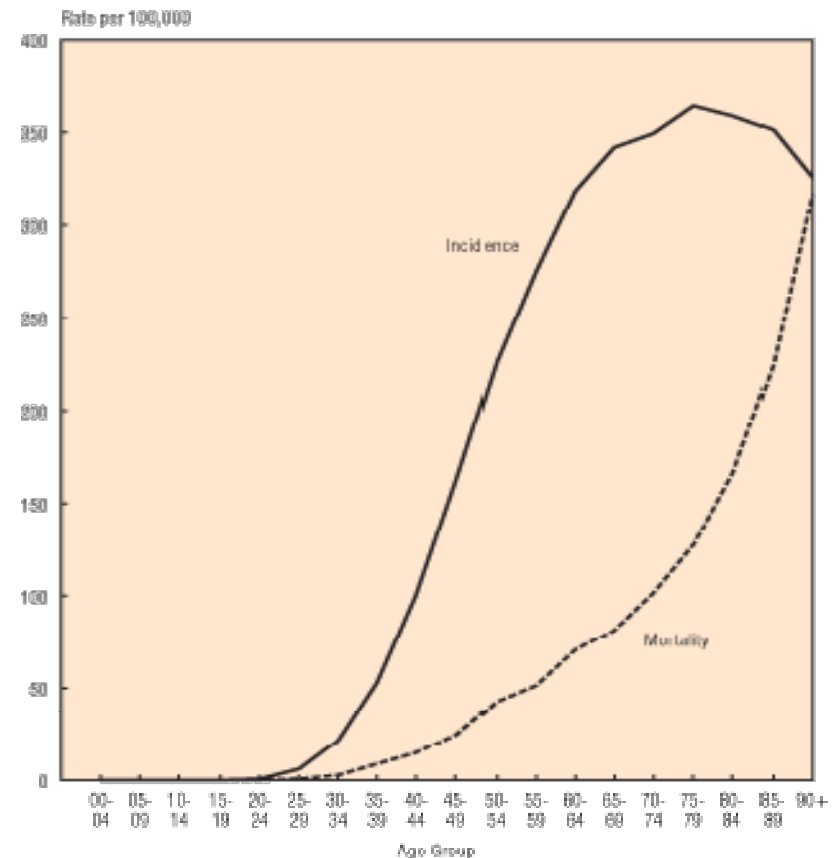
Probability (%) of Developing Colorectal Cancer in the Next 10 Years by Age, Canada



Source: Cancer Bureau, CCDC, Health Canada

Figure 13.1

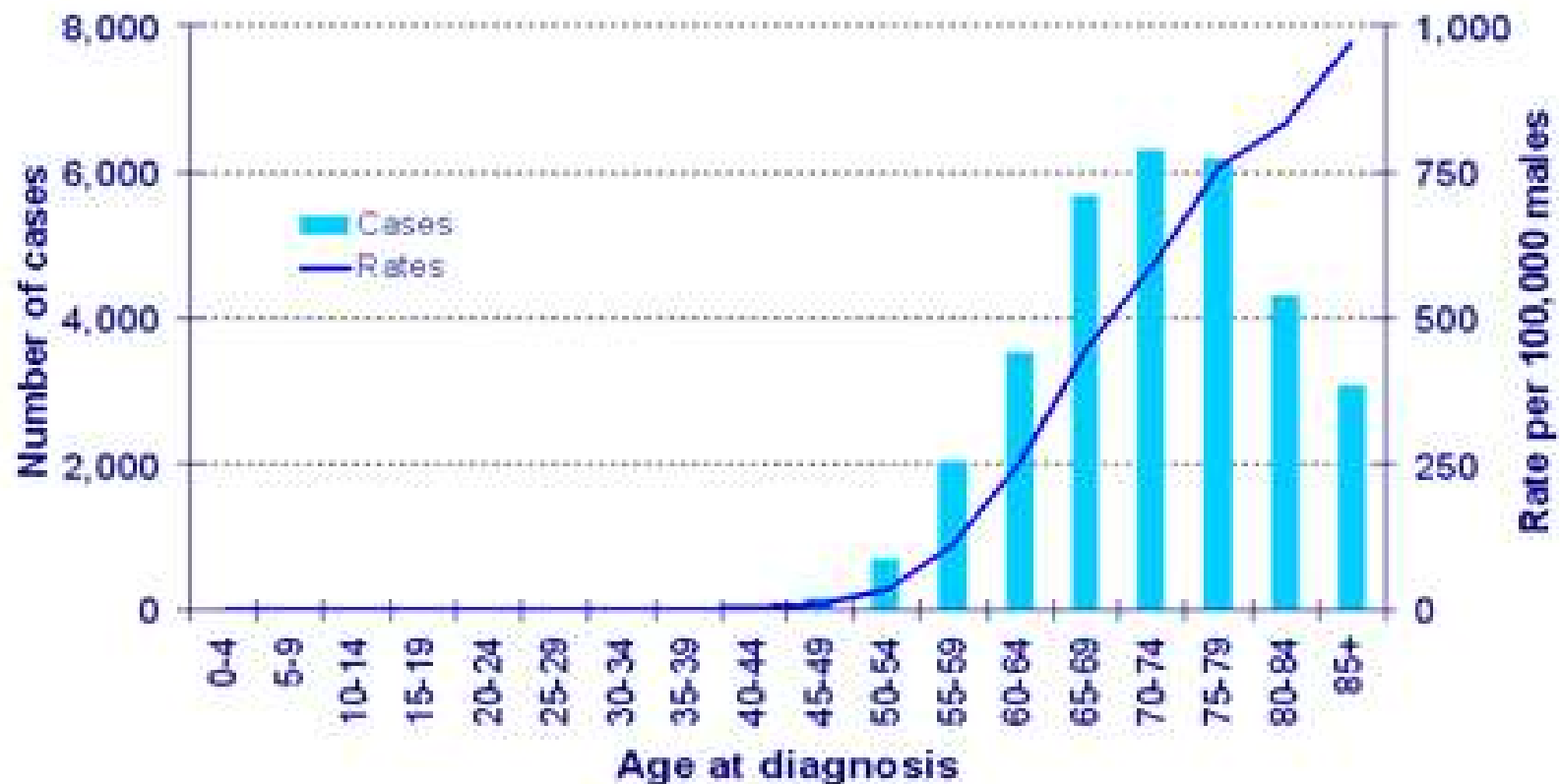
Age-Specific Incidence and Mortality Rates for Breast Cancer, Females, Canada, 1999-2003



Source: Surveillance Division, CCDC, Public Health Agency of Canada

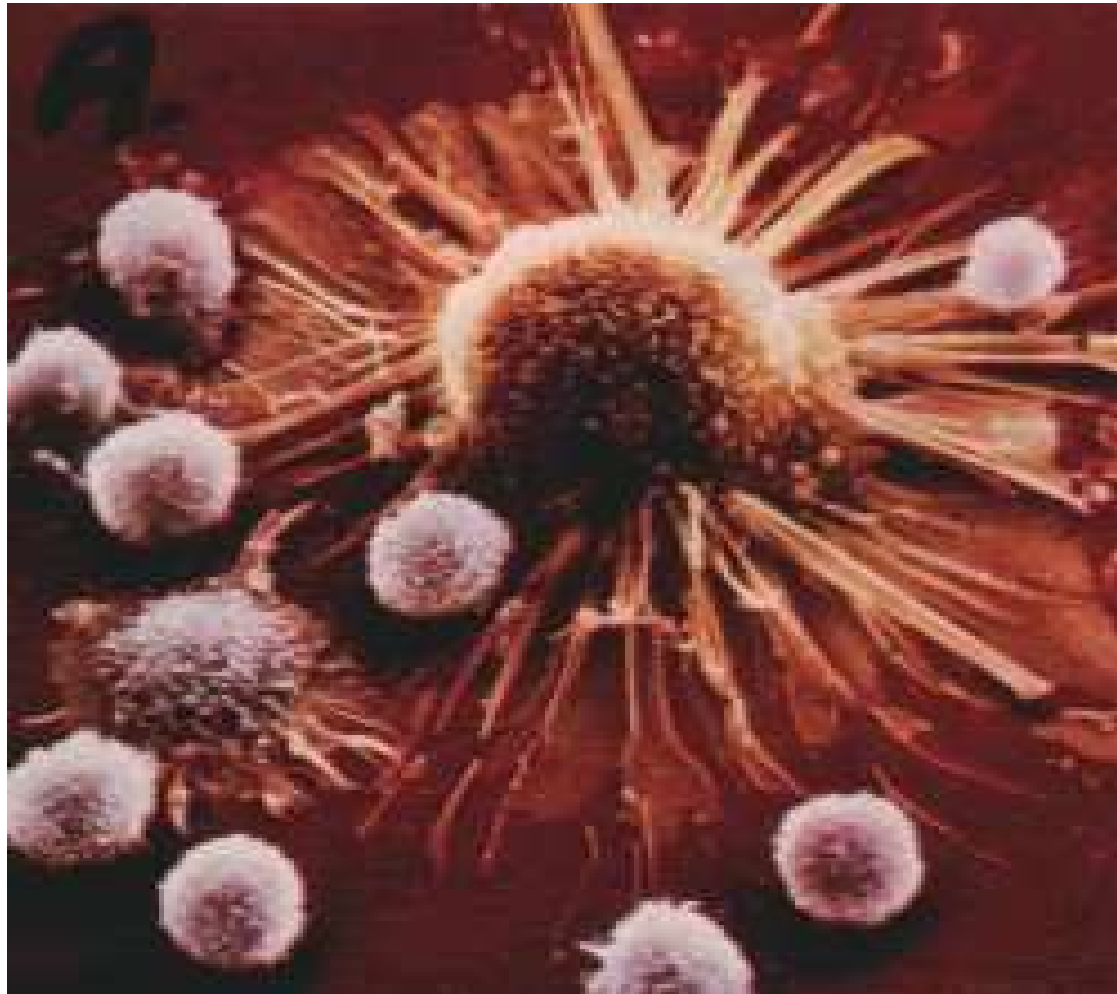
Prostate Cancer Relative To Age

Figure 4.1: Numbers of new cases and age-specific incidence rates, prostate cancer, UK 2002



Before

A fully intact cancer cell surrounded by the immune systems killer t-cells. The cancer cell is surrounded and attacked by the killer T-Cells of the immune system. **Scanning electron microscope** pictures shows killer t-cells attacking the cancer cell. **Cancer cells secrete chemicals that weaken immune cells, but various nutrients can boost immune function and overcome resistance.**



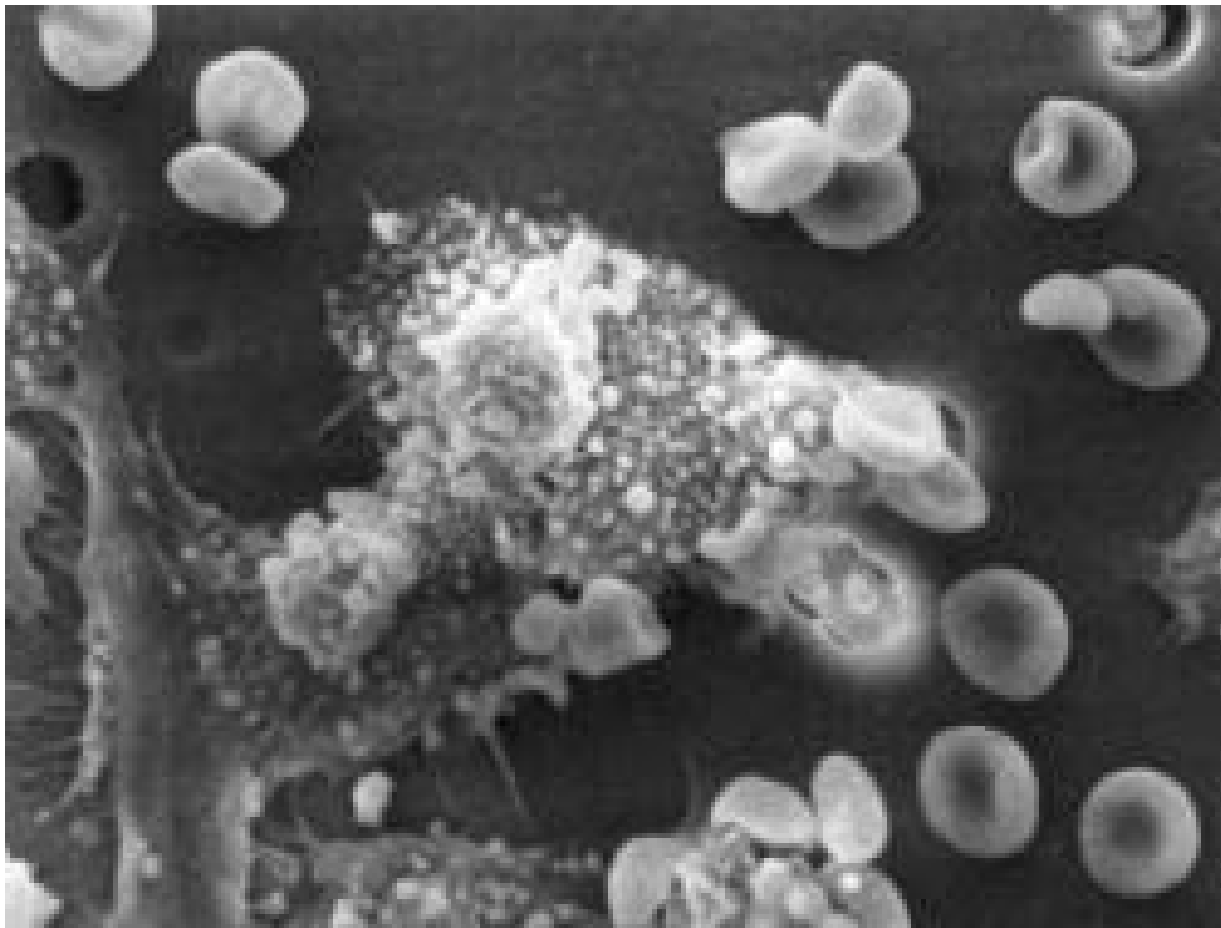
After

Cancer cell is completely flattened and destroyed.

During the killing process, granules in a T-Cell fuse with the cell membrane and release units of the protein PERFORIN. These combine to form pores in the target cell membrane. Thereafter fluid and salts enter so that the target cell eventually bursts.



Macrophages have identified a cancer cell (the large, spiky mass). Upon fusing with the cancer cell, the macrophages (smaller white cells) will inject toxins that kill the tumor cell.



Specific Nutrients Boost Immune And Detoxification Function To Youthful Activity

Antioxidants – Vit C, Vit E, selenium, zinc, beta-carotene (High Potency Multiple Vit and Min)

**At age 50 consider a combination supplement
containing:**

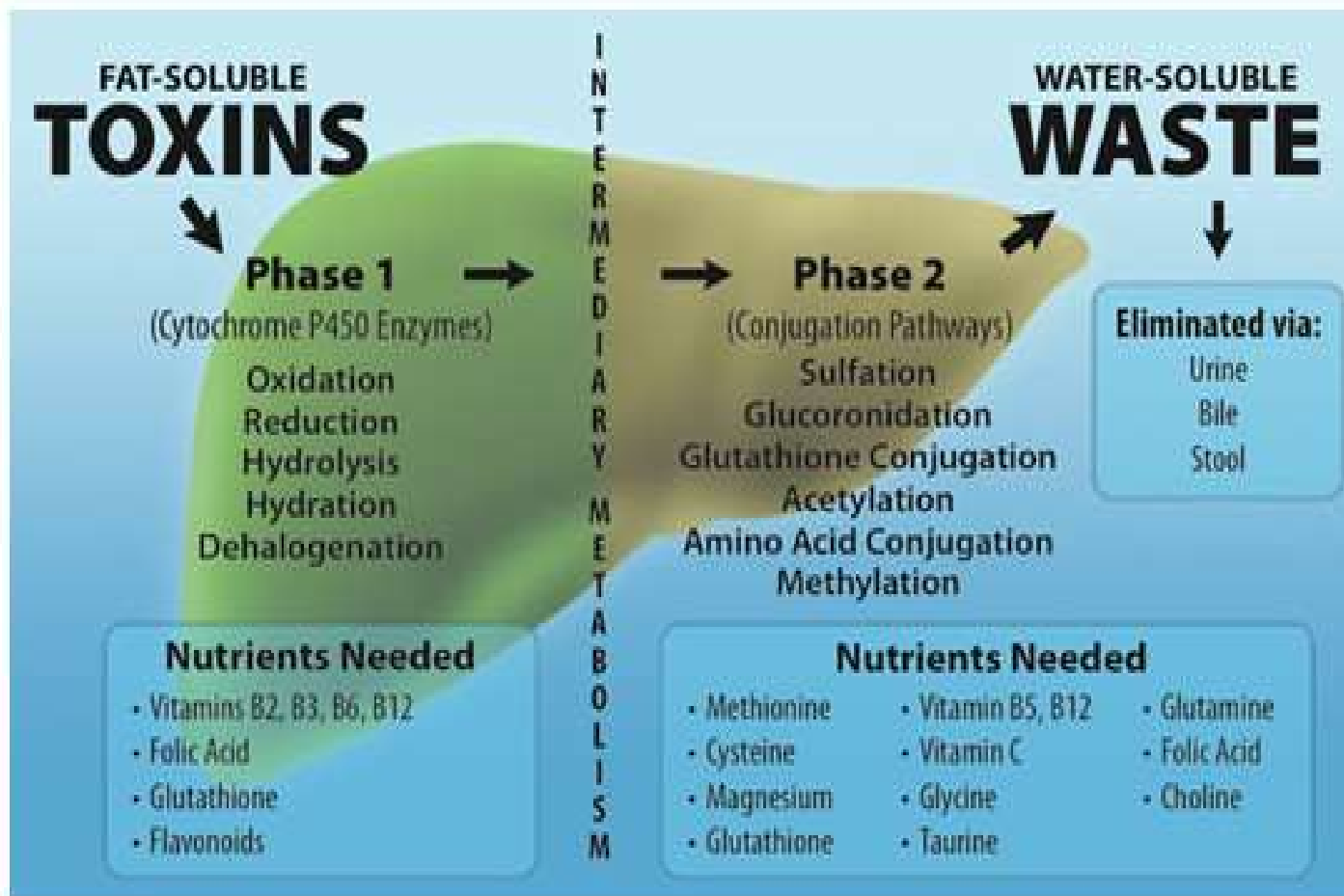
A. Immune Modulators – i.e Reishi mushroom
Extract and Astragalus

B. Detoxifying and Liver Protective Agents i.e. - Milk
Thistle (silymarin) and Indole-3 Carbinol

THE LIVER DETOX PATHWAYS AND ESSENTIAL NUTRIENTS

Detoxification Pathways





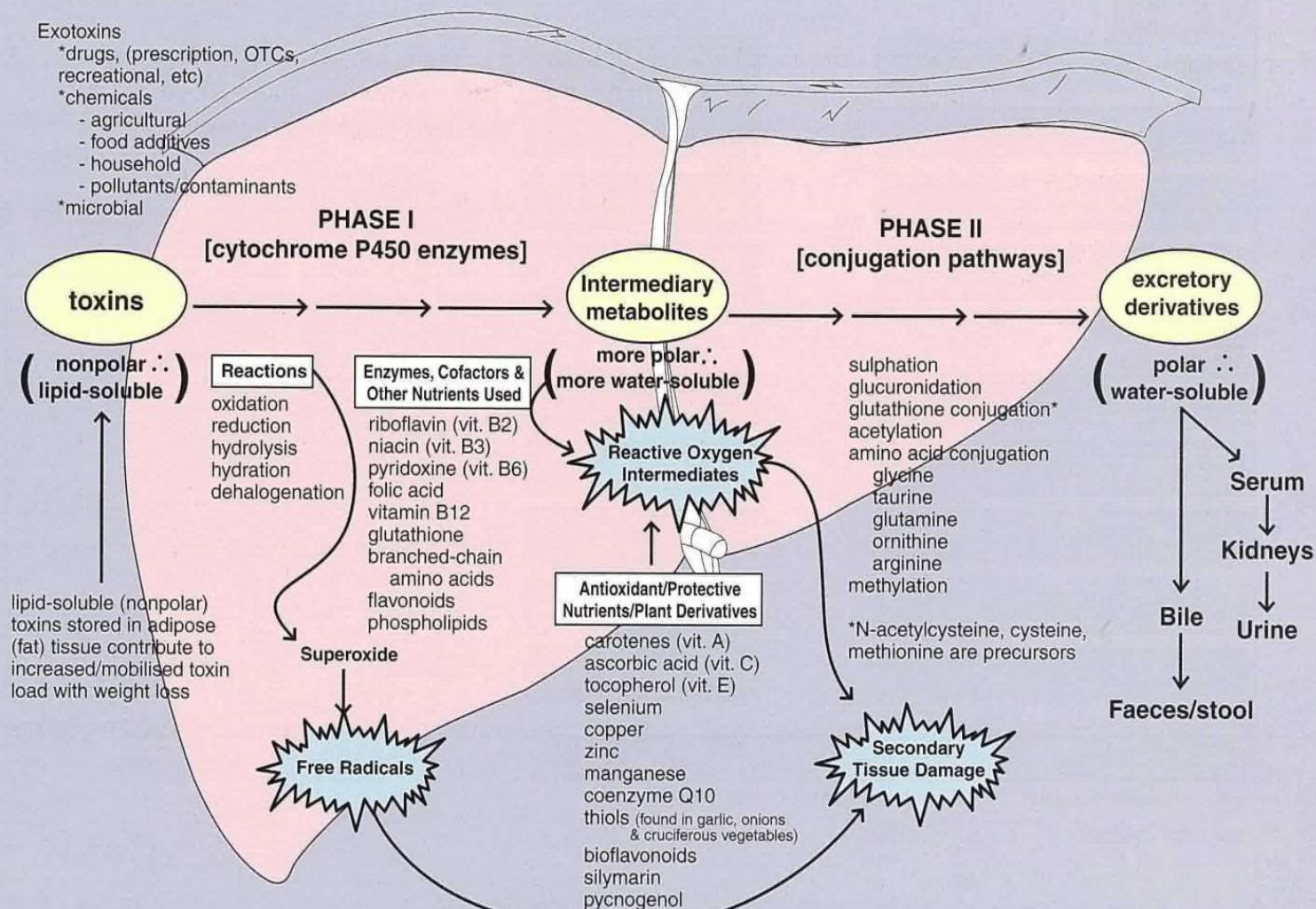
Liver Detoxification Pathways & Supportive Nutrients

Endotoxins

- *end products of metabolism
- *bacterial endotoxins

Exotoxins

- *drugs, (prescription, OTCs, recreational, etc)
- *chemicals
 - agricultural
 - food additives
 - household
 - pollutants/contaminants
- *microbial



Dietary Considerations

1. Cruciferous vegetables – proven to reduce cancer risk (indole-3 carbinol and sulfurophane)
2. Garlic and onions (*Diallyl sulfide*) – *sulfur is major chelator and detoxifying agent – makes things water soluble for excretion. And shows other anti-cancer and immune modulating properties.*
3. Be careful with alcohol (weakens immune system and acts as carcinogen)

Immune and Detox Supplement Example

Ingredients:

1. Reishi Mushroom Extract – 60 mg (std to 10% polysaccharide and 4% terpene content)
2. Astragalus – 200 mg (2:1 extract)
3. Milk Thistle – 300 mg (std. to 80% silymarin)
4. Indole-3-Carbinol – 50 mg

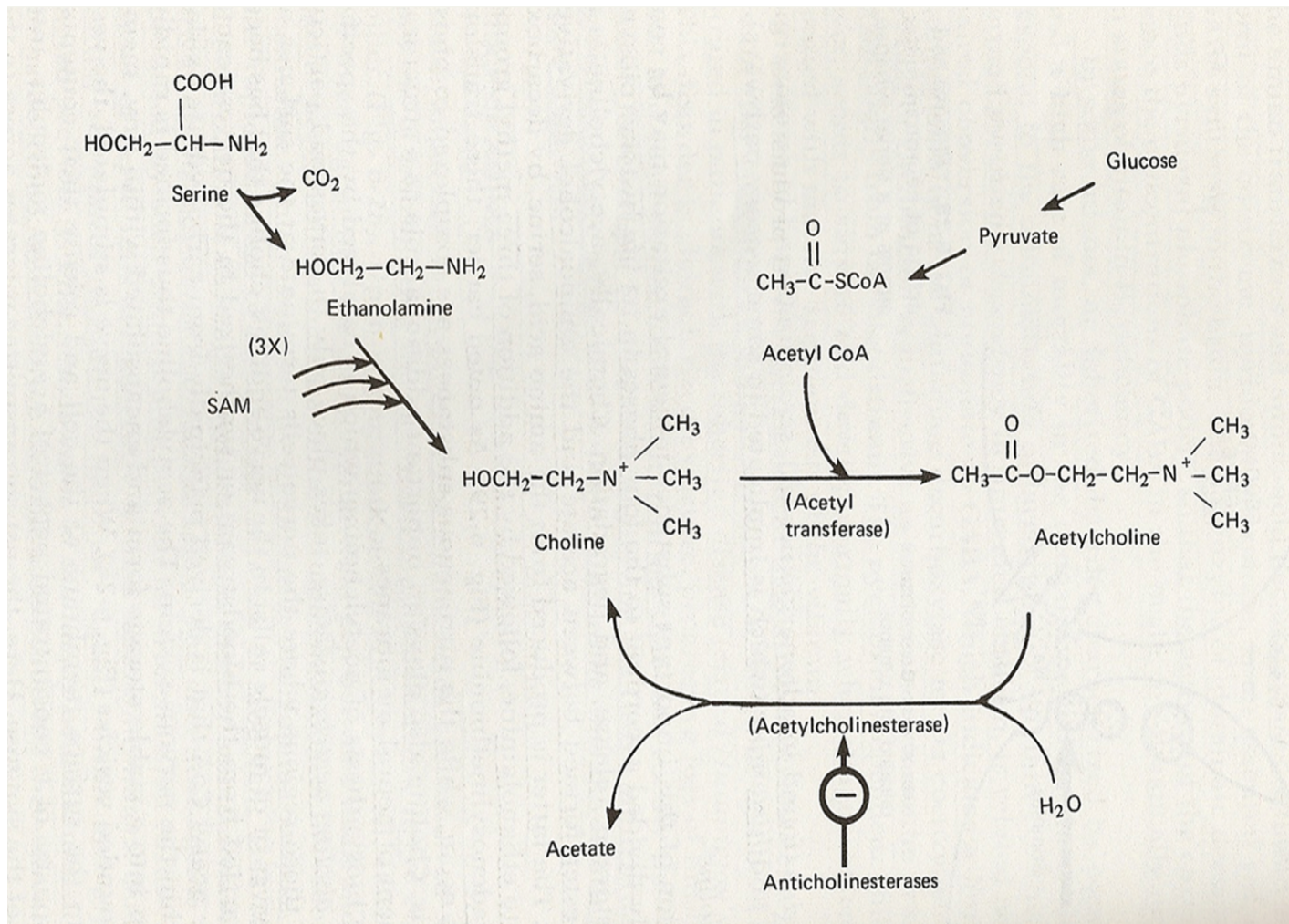
Dosage:

1. **Prevention** – 1-2 capsules per day (also stops cankers)- by Age 50
2. **At first sign of cold or flu** – 8 capsules per day with P73 Wild Oregano Blend
3. **Immuno-compromised states and Cancer** – 4-8 capsules Immuno-Detox Prime, along with other medicinal mushroom, Vitamin D and antioxidant support

Age 55

Decline in Memory

Acetylcholine: most people experience a decrease in acetylcholine after 55 due to decreased transport of choline across blood-brain barrier brain.



Age-related Memory Loss

The Problem: Decreased synthesis of memory chemical “acetylcholine” after age 55 leads to declining memory

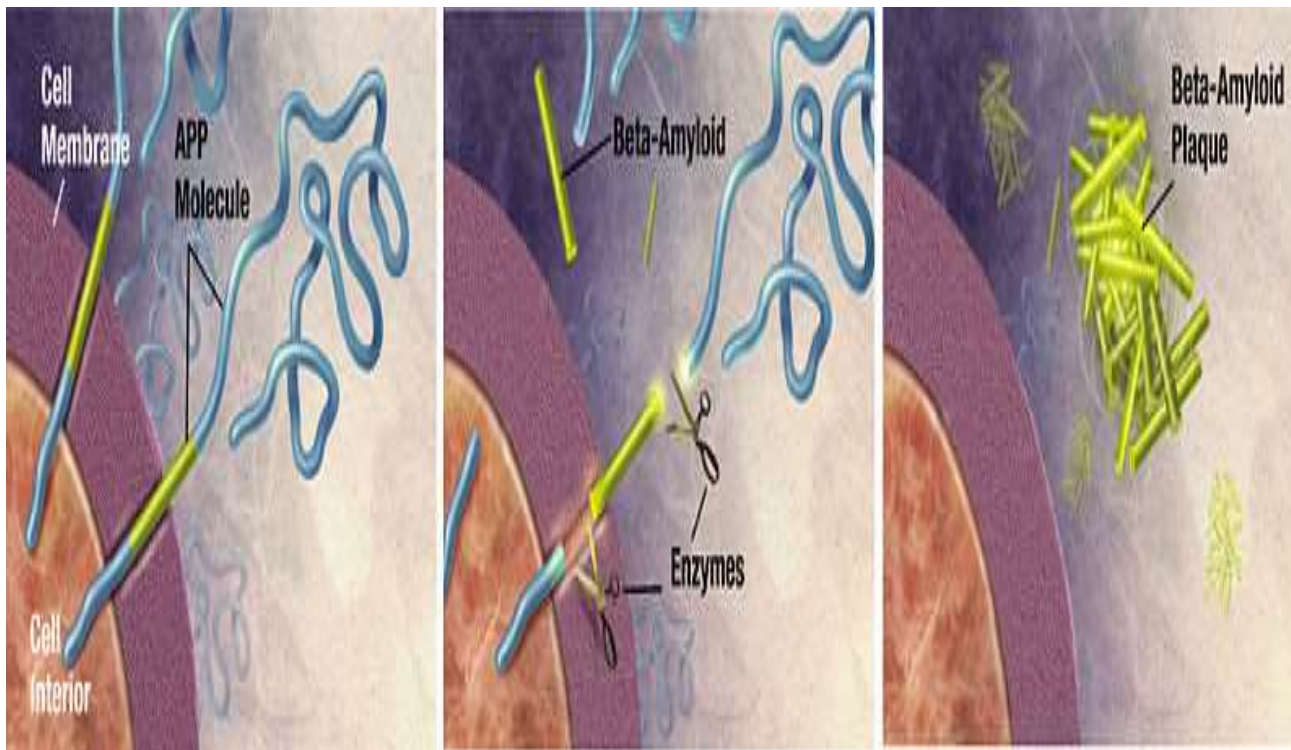
Why Do Acetylcholine Levels Decline in the Brain?

- Decreased function of choline transporter
- Less choline crosses blood-brain barrier to enter brain
- Less choline incorporated into cell membrane phospholipids (phosphatidylcholine) , as reservoir from which brain cells make acetylcholine
- No Drugs can increase acetylcholine production, but supplements can.

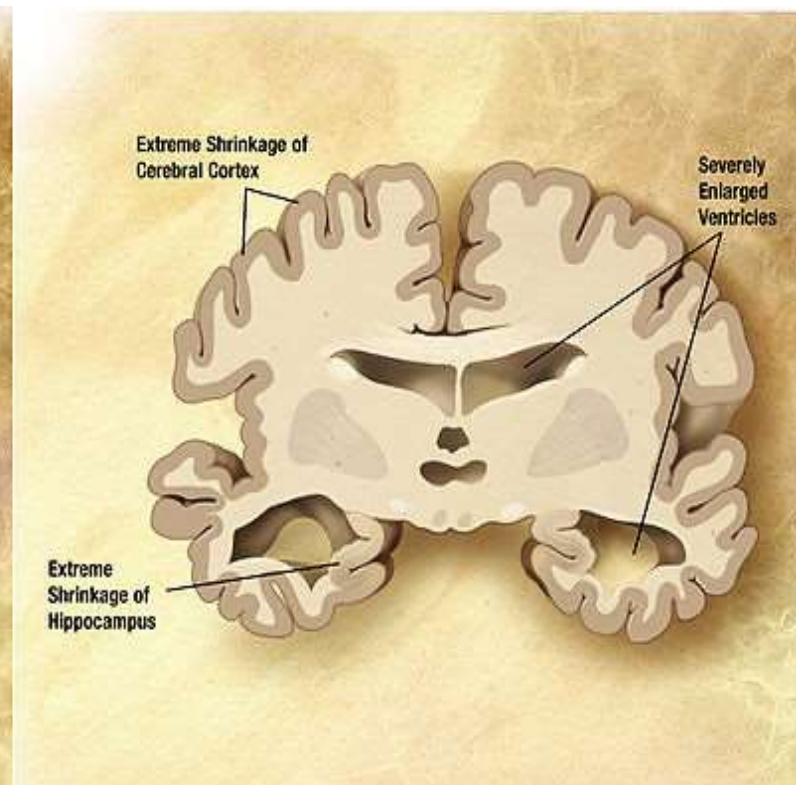
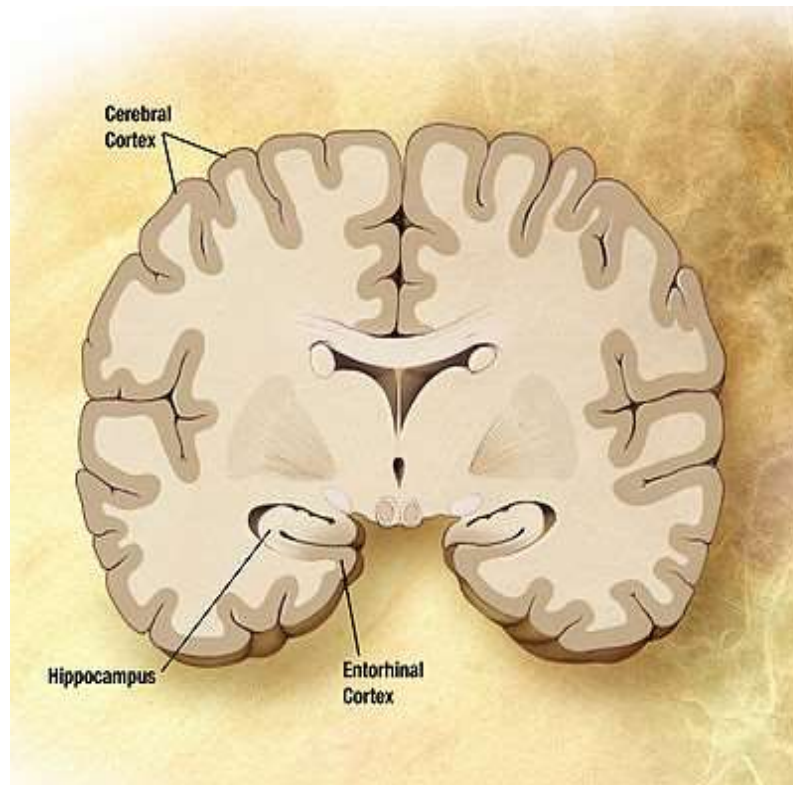
Supplements Proven to Increase Acetylcholine and Combat Memory Loss

1. CDP–choline – increases acetylcholine and nerve conduction
2. Phosphatidylserine – increases acetylcholine and nerve conduction
3. Bacopa Monnieri – brain antioxidant and nerve conduction
4. Huperzine A – decreases breakdown of acetylcholine – acetylcholinesterase inhibitor

Beta-amyloid Plaque



Normal Brain Left: Alzheimer's Brain Right:
Note atrophy of prefrontal cortex and hippocampus
and enlarged ventricles



Preserving Memory and Cognitive Function After Age 55- by helping to maintain more ideal brain levels of choline and acetylcholine

Memory Support Nutrients:

1. CDP—choline
2. Phosphatidylserine
3. Bacopa Monneiri
4. Huperzine A

* Throughout lifetime – 1 tablespoons per day of lecithin granules or lecithin supplement (1200 mg capsules, twice per day) - increases blood choline, which increases brain choline and acetylcholine.

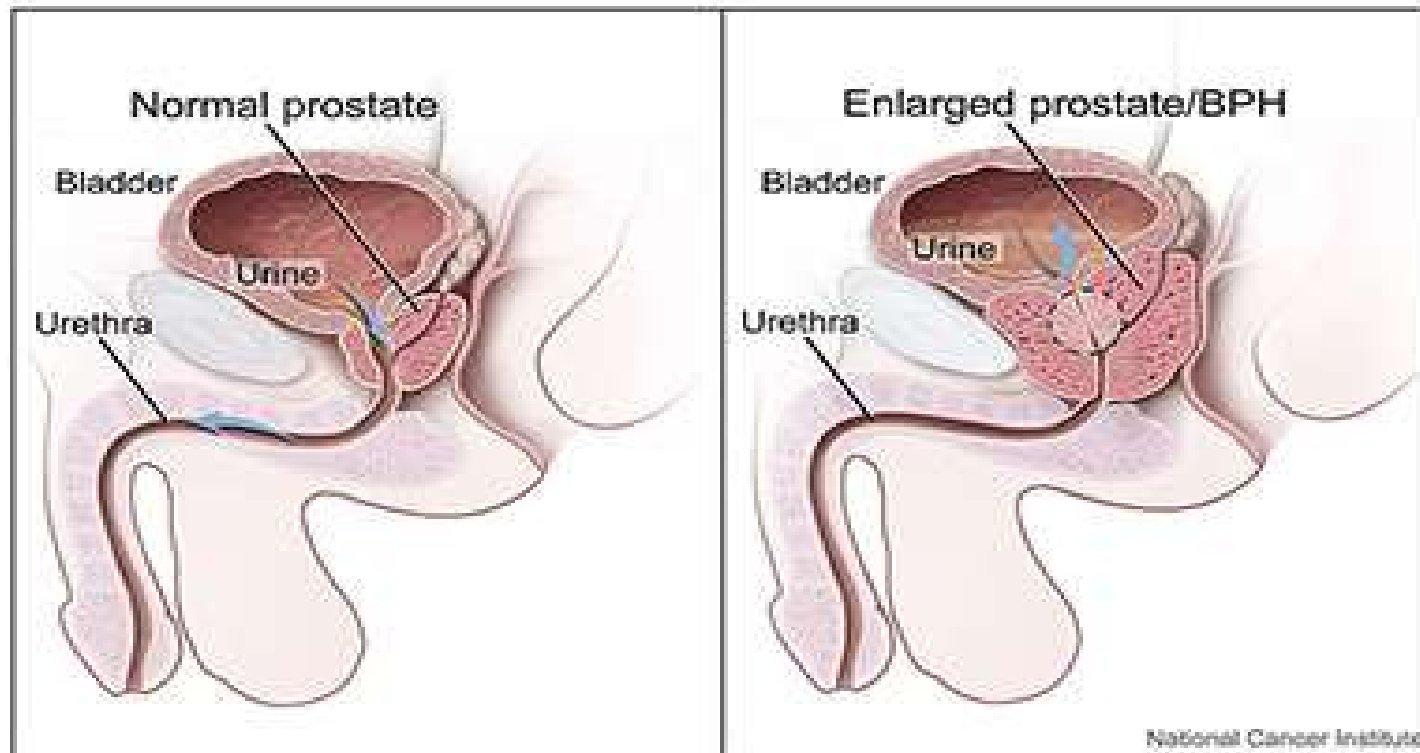
Two Other Issues

1. The Male Prostate Gland After 40
2. Menopause and Postmenopausal
Years

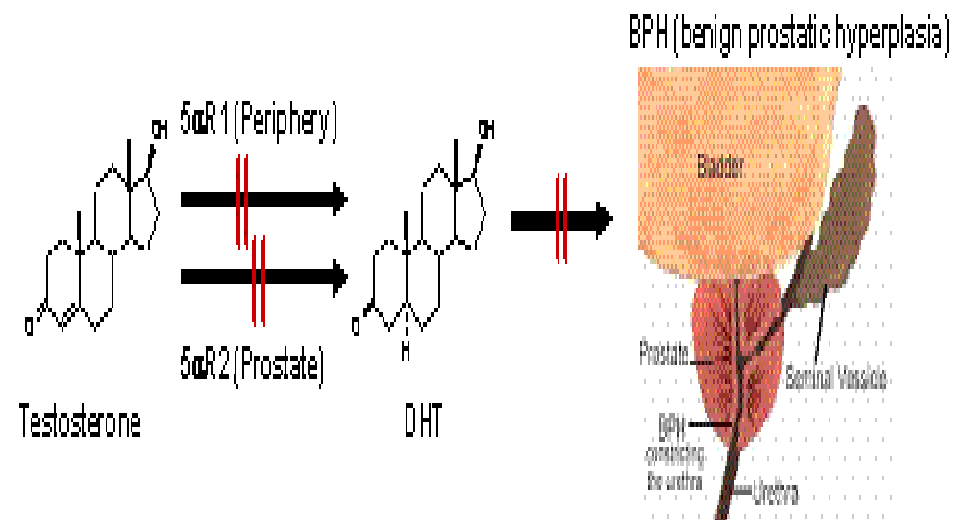
The Male Prostate Gland

1. After age 40 DHT builds up in the prostate
2. DHT causes enlargement, may increase cancer risk and spreads existing cancer
3. Specific natural agents are proven to block the formation of DHT and reverse enlargement
4. Best Strategy = prevent problems in advance

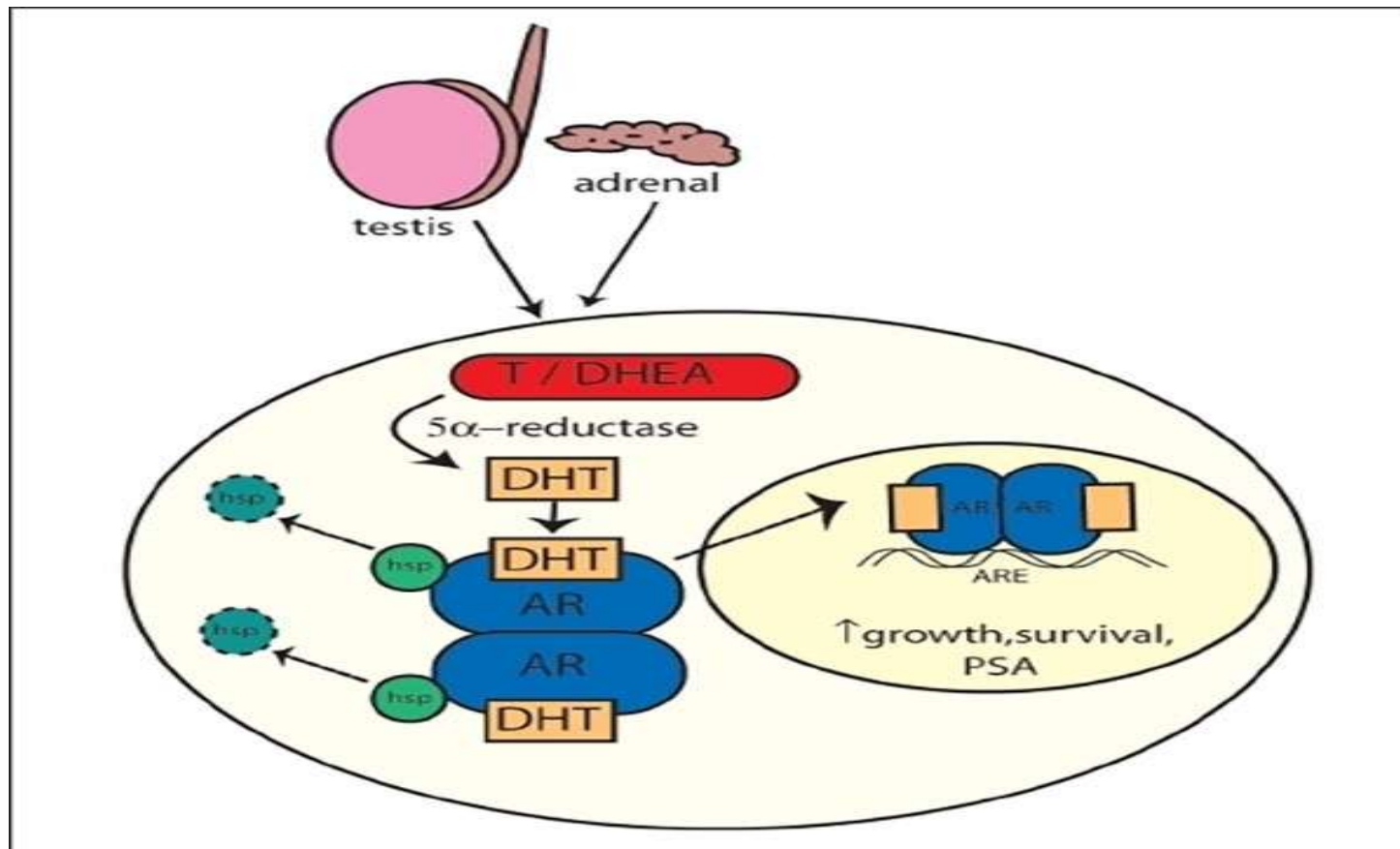
Normal Prostate vs BPH (compromised urine flow)



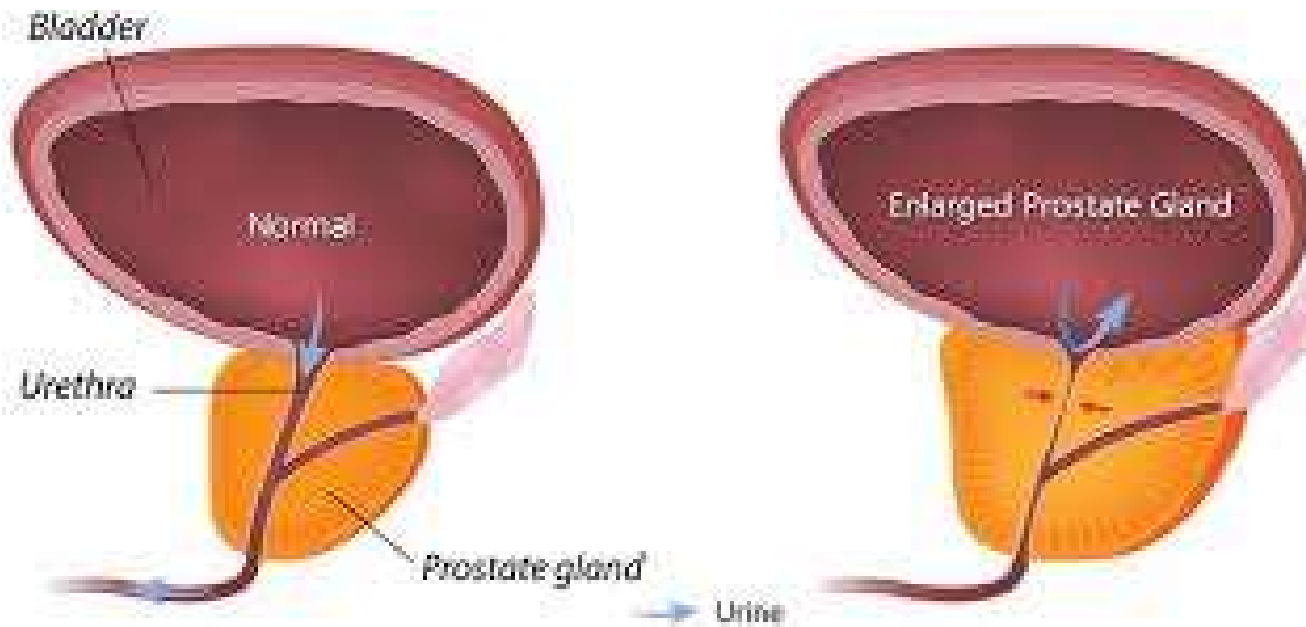
DHT Increases Prostate Cell Division Rate



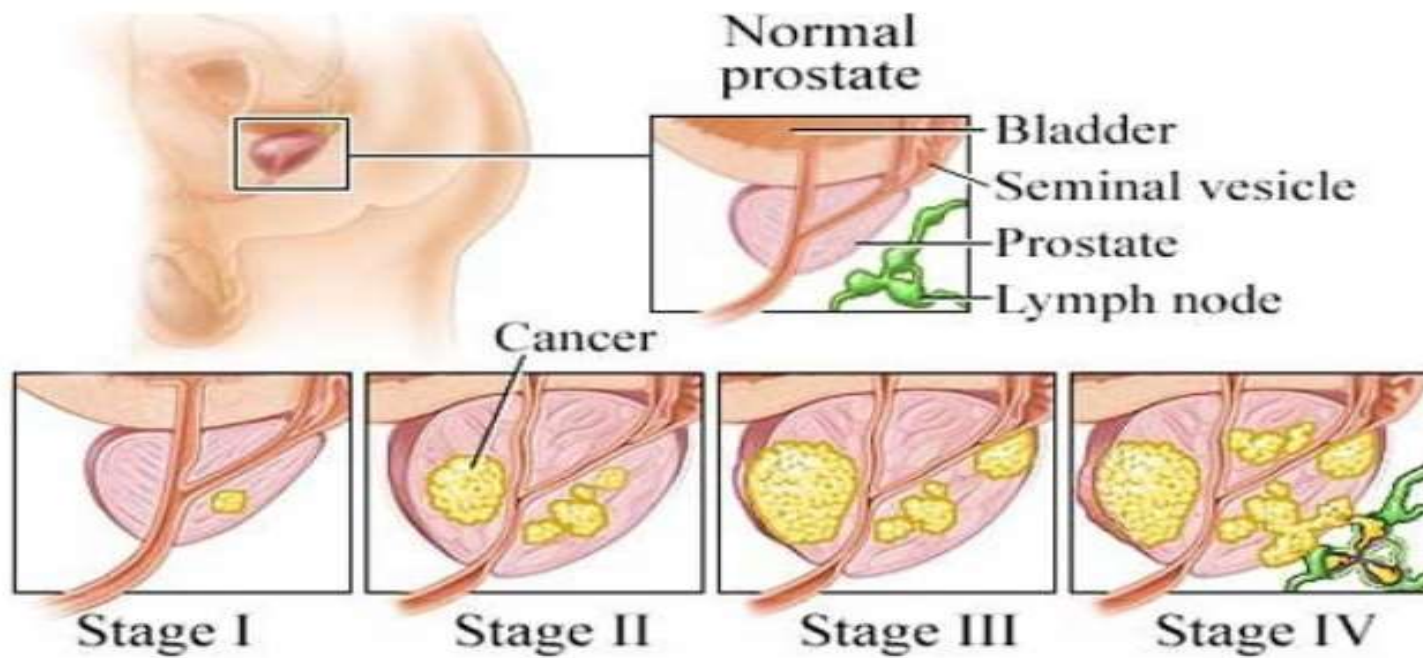
DHT binds to Androgen Receptor, which translocates to Nucleus of Cell Activating Genes For Replication and Survival



Normal Prostate vs BPH



Prostate Cancer Progression



Example of Supplement Shown to Decrease PSA and Reverse Prostate Enlargement

Amount Per 2 Capsules:

- Saw Palmetto – 640 mg (std 45% FA & S)
- Pygeum Africanum – 200 mg (std 12% terpenes)
- Beta-sitosterol – 130 mg (cancer cell suicide)
- Soy Isoflavones – 25 mg (used therapeutically to control prostate cancer – cancer cell suicide)
- Stinging Nettle – 60 mg
- Lycopene – 25 mg

Menopause Symptoms and Well Being In The Postmenopausal Years

Hormones Decline At Menopause

- Menopausal Hormonal Changes:
 - 90% decline in estrogen
 - 66% decline in progesterone
 - 50% decline in testosterone
 - Compensatory increase in FSH, LH and noradrenaline (hot flashes, night sweats, insomnia, anxiety)

Other Physiological Outcomes

- Bone loss – increases risk of osteoporosis
- Increase in blood cholesterol (No 1 killer)
- Skin atrophy – accelerates aging appearance
- Vaginal dryness – painful intercourse

- **Medical Solution: HRT (WHI study):**

- Breast CA increase 26%
- Heart attack increase 29%
- Stroke increase 41%
- NHS – 2.3% increase risk Breast CA per yr on HRT (10 yrs equals 23% increased risk)
- Breast cancer incidence among U.S. women dropped by 7% from 2002 to 2003, possibly because of a decrease in hormone replacement therapy usage, according to a study presented Thursday at the 29th Annual San Antonio Breast Cancer Symposium, the *New York Times* reports (Kolata, *New York Times*)

Natural Alternative To HRT (why):

- 1/3 life in postmenopausal yrs (quality)
- Reduction of disease risk and symptoms
- Compendium of Pharmaceuticals – Black Cohosh during first 50 yrs of the 1900's and wide spread use in Germany
- Also Soy Isoflavones and Gamma-Oryzanol

Natural Support For Menopause

1. Black Cohosh- 80 mg, standardized to 2.5% triterpene glycosides
2. Soy Extract – 125 mg, standardized to 20% isoflavones
3. Gamma-Oryzanol – 150 mg

Dosage: 2 capsules per day

Success Rate – approx 80%

May also add Natural Progesterone if necessary (prescription)

Also Effective For PMS, Fibrocystic Breast Disease, Uterine Fibroids

Dangerous Herbs To Avoid:

- Dong Quai (angelica species) – bleeding disorders, skin rash
- Red Clover Isoflavones – bleeding disorders, skin rash
- Licorice – hypertension & Bleeding

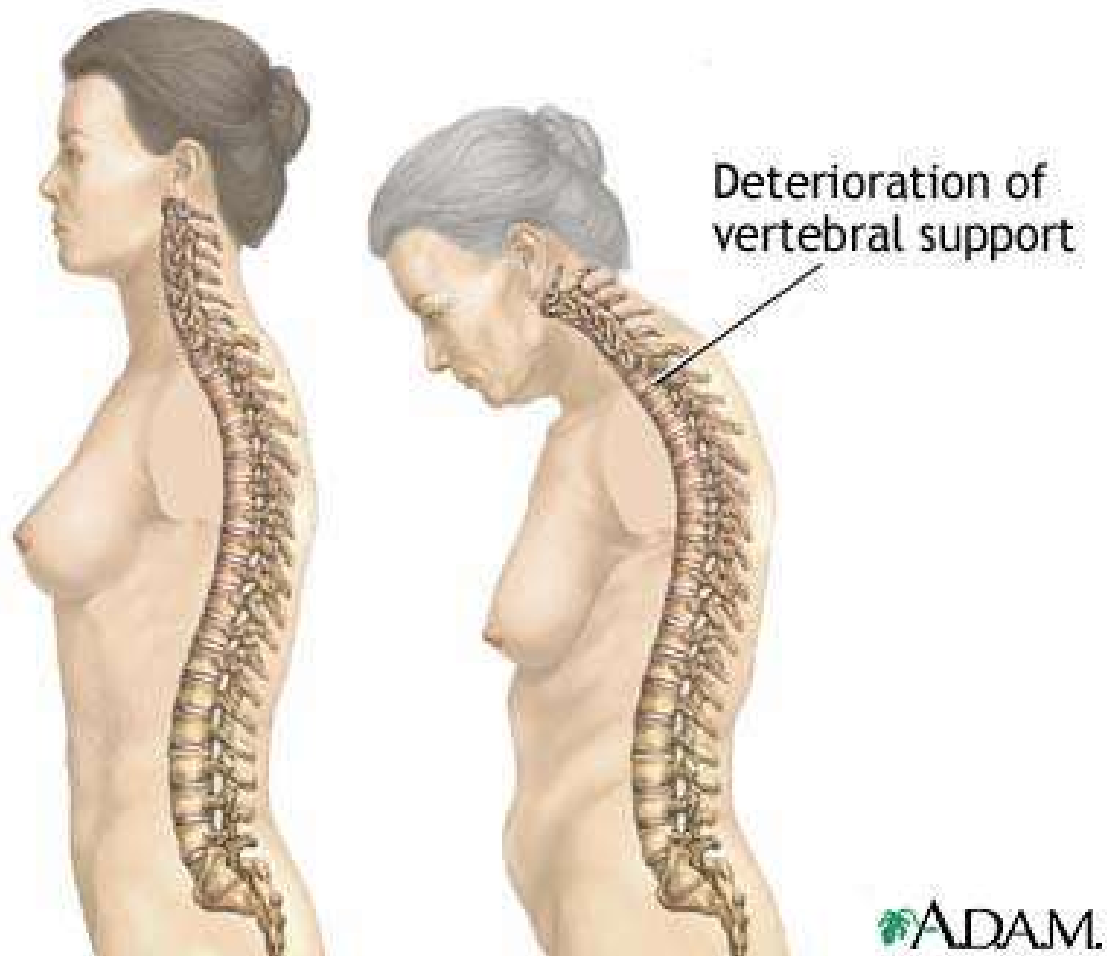
Coumarins are dangerous natural agent in regards to bleeding disorders

What Else: Bone Support Nutrients

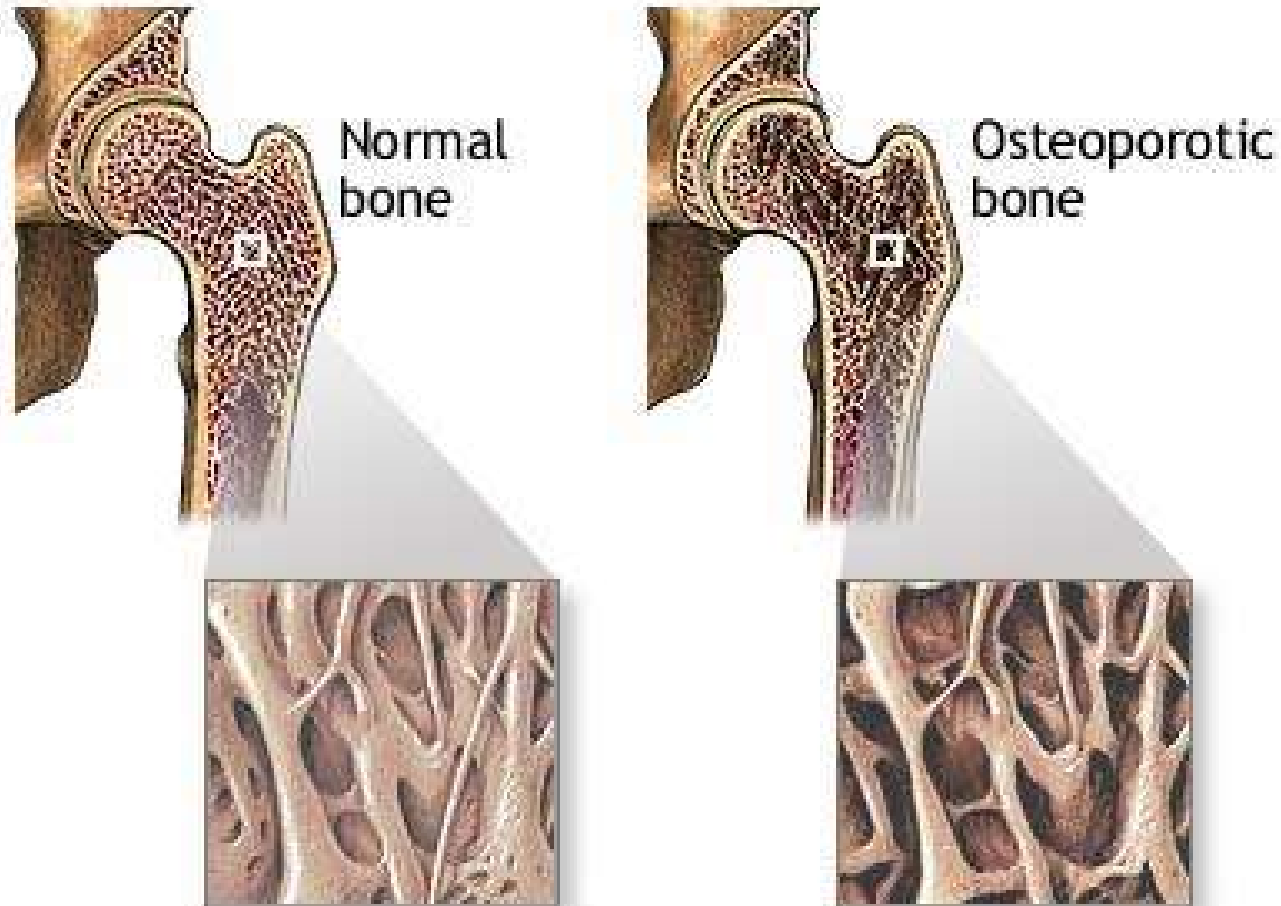
- Calcium – 1100 mg per day
- Vitamin D – 1800 IU
- Magnesium – 500 mg
- Icariin Flavonoid – 60 mg
- Silica – 3 mg
- Boron – 1.5 mg

- Weight Bearing Exercise

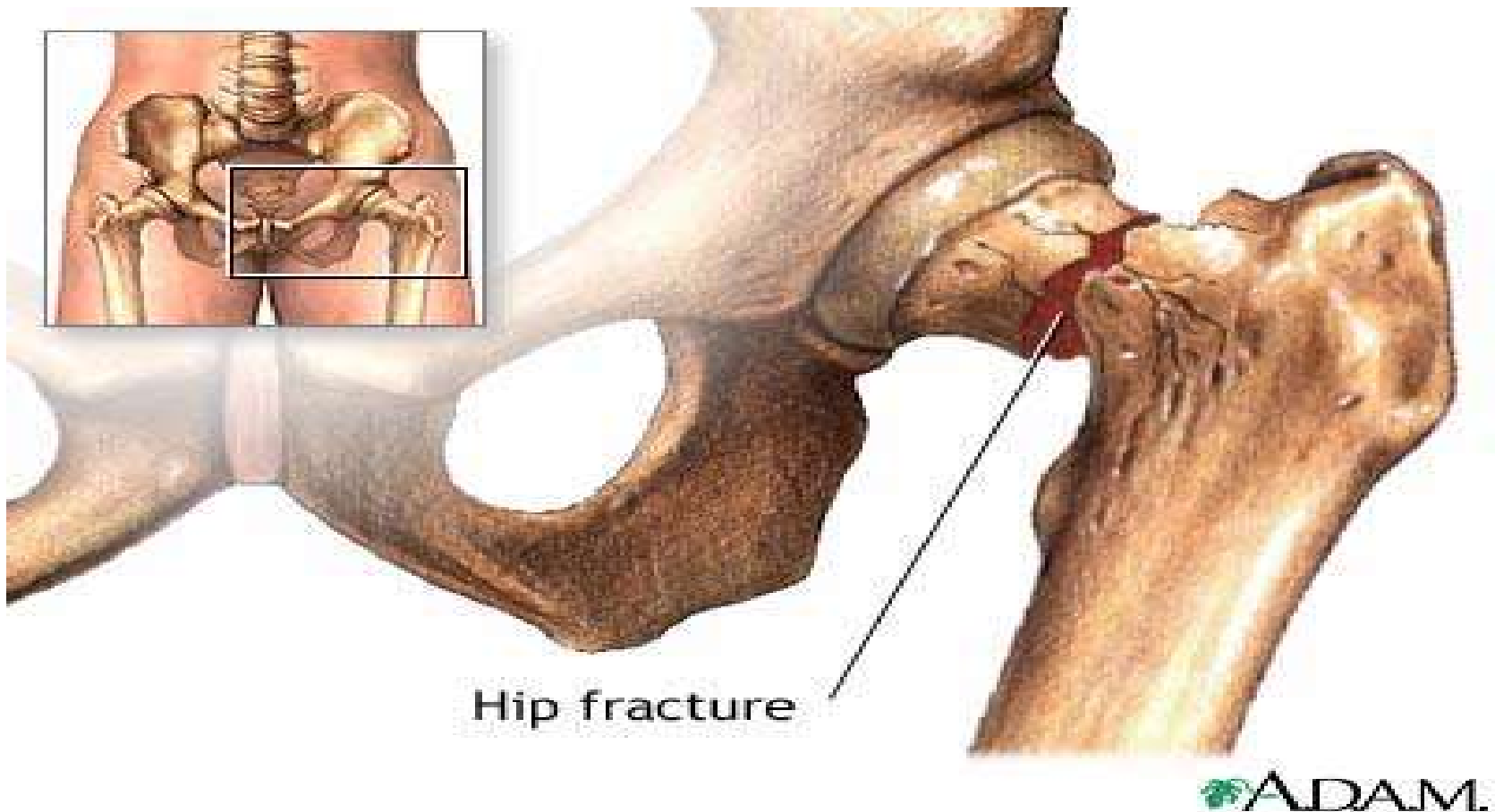
Osteoporosis 1:4 women; 1:8 men



Lack of calcium, vitamin D, physical activity allow calcium to leak out of bone:



Hip Fracture: Consequences cause more deaths than breast and ovarian cancer combined



Vitamin D Food Chart

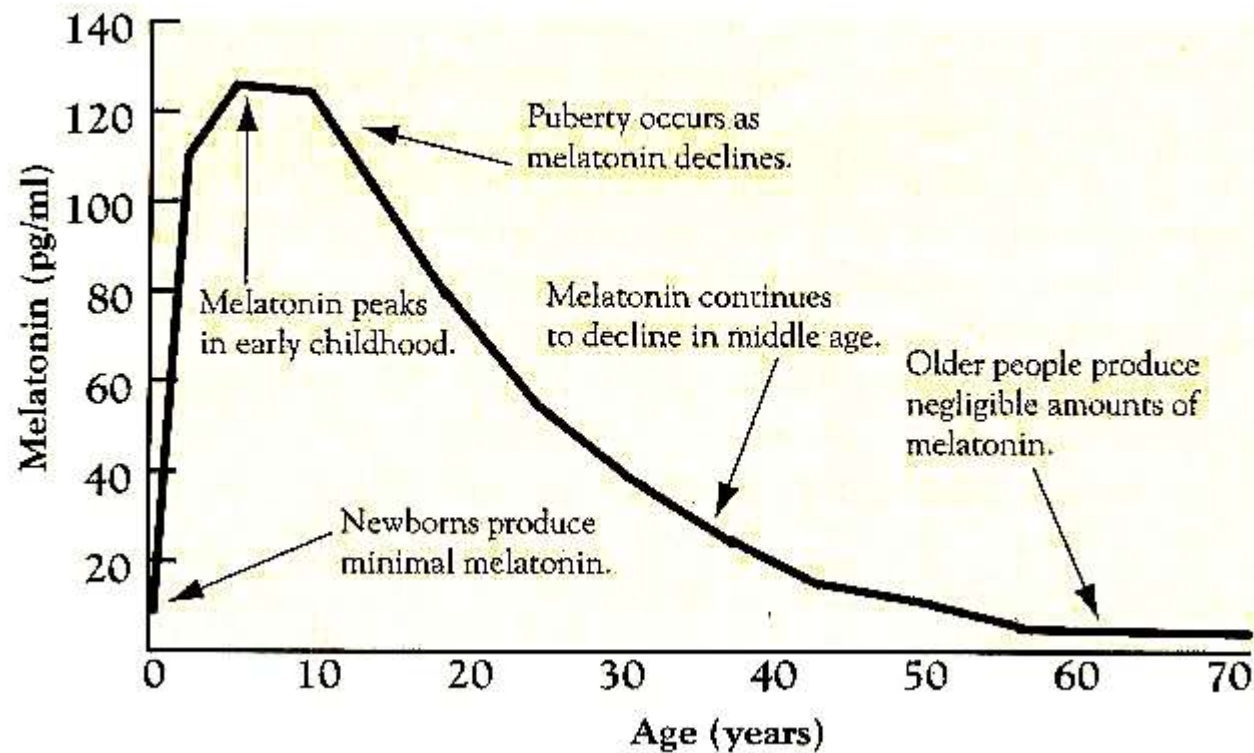
Low Saturated Fat Foods Containing Vitamin D

FOOD	APPROX. I.U. of VIT D (3-1/2 oz.)
Cheese (low fat)	12-15
Halibut	44
Herring	320
Mackerel (raw)	1100
Salmon (fresh)	450
Salmon (canned)	350
Sardines (canned)	1300
Shrimp	150
Oysters	5 I.U. per 3-4 med-sized oysters
Vit D Fortified Milk & Yogurt (low fat)	100 I.U. per 8 ounces

Review of Important Molecules Whose Synthesis Declines With Age

1. Glucosamine
2. Coenzyme Q10
3. 1,25 Dihydroxycholecalciferol – but maintaining 25-hydroxycholecalciferol at 85nmol/L can compensate
4. Acetylcholine –choline transport problem
5. Melatonin
6. Glutathione
7. Alpha-Lipoic Acid
8. L-Carnitine (in certain cases)
9. Skin Hyaluronic Acid
10. Skin Collagen (HA and Matrixyl)

Melatonin Age-Related Decline



Quick Notes on Melatonin

- Water Soluble and Lipid Soluble Antioxidant- Secreted by Pineal Gland
- Neurotransmitter Secreted by Pineal Gland – as darkness falls (made from Serotonin)
- Helps Induce Deeper Levels of Sleep
- Sets Body's Circadian Rhythm – thus used for Jet Lag
- Decline in Melatonin Coincides With Beginning of Puberty
- Melatonin administration shown to increase expression of Superoxide Dismutase (SOD) and Glutathione Peroxidase antioxidant enzymes

Melatonin Contd

- Evidence Suggests it Protects Brain from Oxidative Stress and Inhibits Formation of Beta-Amyloid Plaque – hallmark feature in Alzheimer's disease. And has other Neuroprotective Effects
- Clinical Studies have shown its ability to Reverse or Stabilize Mild Cognitive Impairment (MCI)
- Immune Modulation – CD4 Cell Melatonin Receptors
- Breast and Prostate Cells have Melatonin Receptors – evidence to show it may help to prevent cancer in these tissues

Melatonin Contd

- Medically Approved for: Jet Lag, Depression, Insomnia
- Brain Serotonin Levels Rise After Melatonin Administration – helpful in Depression and Seasonal Affective Disorder (but timing important)
- Some studies have shown that melatonin may also be effective in breast cancer, fibrocystic breast diseases, and colon cancer.
- Evidence suggests Melatonin can help Attenuate Stress Response and Diminish Effects of Stress on Body (lowers Cortisol Release)
- Preliminary Human Studies show Melatonin Administration lowers Cholesterol (38%) and reduces High Blood Pressure by relaxing smooth muscle in artery wall. It also appears to reduce LDL-cholesterol oxidation.

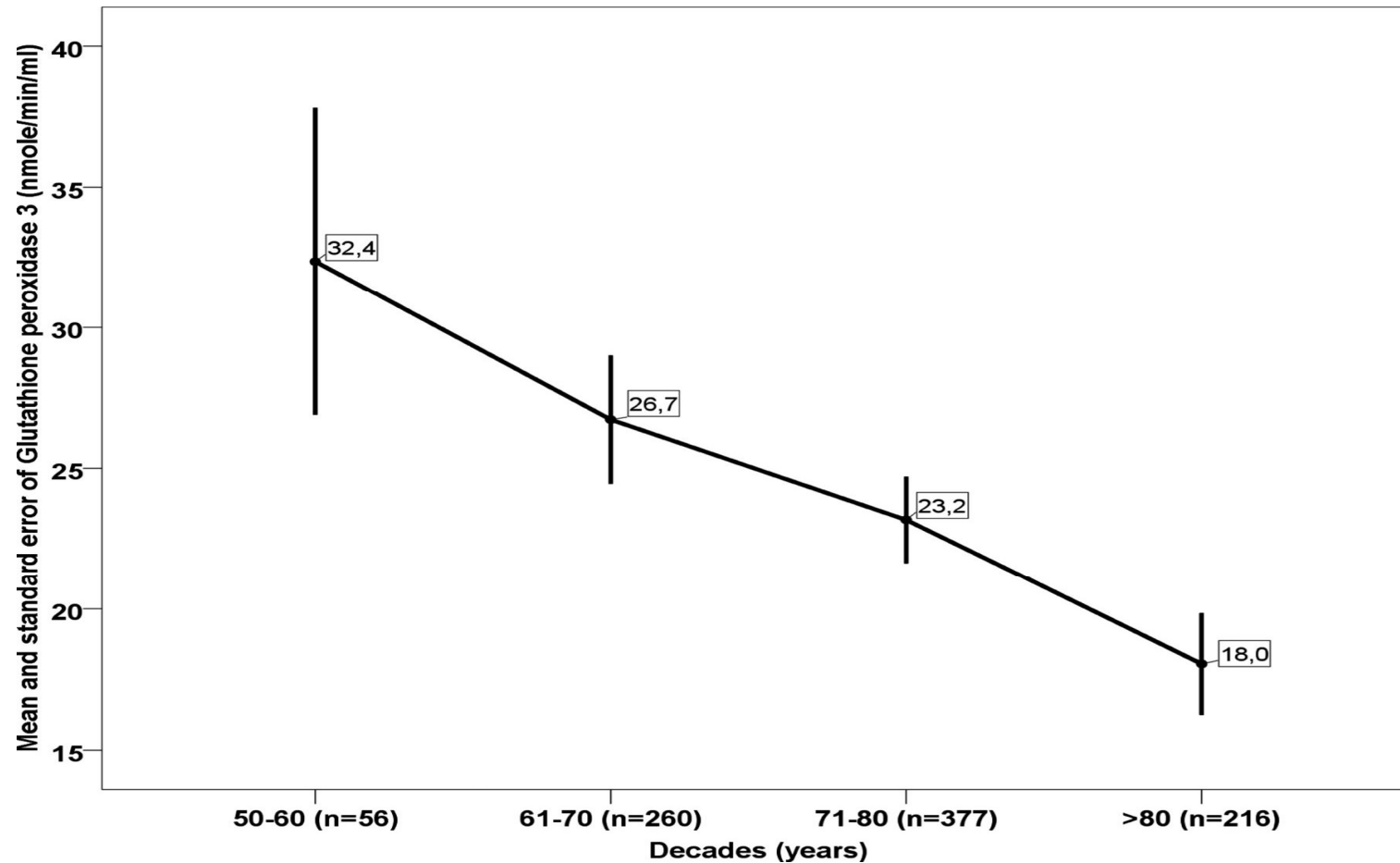
Melatonin Contd

- May have Therapeutic Application for Irritable Bowel Syndrome, Ulcerative Colitis, Diarrhea and Interstitial Renal Inflammation, Idiopathic Thrombocytopenia Purpura and Prevention of Gastric Ulcers, and Acute Renal Injury from Cisplatin and Cyclosporin and Cardiotoxicity from Doxorubicin (Adriamycin) *intercalating drug* and Other Drugs.
- Melatonin may cause minor adverse effects, such as headache, insomnia, rash, upset stomach, and nightmares.
- Very Non-Toxic

Glutathione: Decline in Glutathione Peroxidase Activity with Age:

Journal of the American Heart Association 2016:

<http://jaha.ahajournals.org/content/5/9/e003682>



Glutathione levels over time



Quick Notes on Glutathione (GSH)

Important Antioxidant, Detoxification Agent, and helps Regulate Cell Proliferation, Apoptosis, Immune Function and Fibrinogenesis

Dysregulation of GSH Synthesis Linked to:

Diabetes Mellitus, Pulmonary and Liver Fibrosis, Alcoholic Liver Disease, Cholestatic Liver Injury, Endotoxemia (endotoxins deplete liver GSH) and Drug-Resistant Tumor Cells- some tumor cells use Glutathione to Protect Themselves Against Chemo Drugs.

GSH very important in Detox of Xenobiotics (a chemical compound, such as a drug, pesticide, or carcinogen that is foreign to a living organism) with special note of Acetaminophen – 50% of all drug-induced liver failure.

Most of GSH is made in liver and exported to other tissues. But liver has great dependence for GSH.

Severe Oxidative Stress Depletes Glutathione- as first defence against many free radicals

Cysteine availability largely determines GSH Synthesis. But some Cysteine also converted to Taurine, which plays key role in heart muscle.

In HIV – N-acetylcysteine has been used to support GSH levels, as decline in GSH in CD4 cells associated with progression of HIV infection to full-blown AIDS

Glutathione Contd

In Severe Polymorphism of Glutathione Synthesizing Enzymes (GCLC and/or GCLM) see very low Glutathione and increased risk of Hemolytic Anemia, Aminoaciduria, and Spinocerebellar Degeneration.

During Endotoxemia administration of SAMe can help prevent depletion of Liver GSH and Liver Damage. (Endotoxemia - presence of endotoxins in the blood, which, if derived from gram-negative rod-shaped bacteria, may cause hemorrhages, necrosis of the kidneys, and shock)

See Low Hepatic and Plasma GSH in Alcoholic Liver Disease (quenching free radicals and detoxification function)

Fall in GSH prompts fibrosis to proceed faster in Liver and Pulmonary Fibrosis

Glutathione Contd

As an Antioxidant GSH donates a hydrogen to stabilize hydroxy radicals formed in aerobic metabolism in the Cytosol (as catalase does in the peroxisome - a small organelle that is present in the cytoplasm of many cells and that contains the reducing enzyme catalase and usually some oxidases).

Glutathione can be regenerated back to antioxidant state via Alpha- Lipoic Acid and Glutathione Reductase enzyme in the presence of NADPH.

Remember that Liver Needs a Highly Functioning Transsulfuration Pathway to Effectively Synthesize Cysteine From Homocysteine in a two-step process (both requiring Vitamin B6). Homocysteine – Cystathionine; Cystathionine – Cysteine (incorporation into Glutathione with Glutamic Acid and Glycine- to form tripeptide GSH

Alpha-Lipoic Acid (ALA)

Key Aspects Points

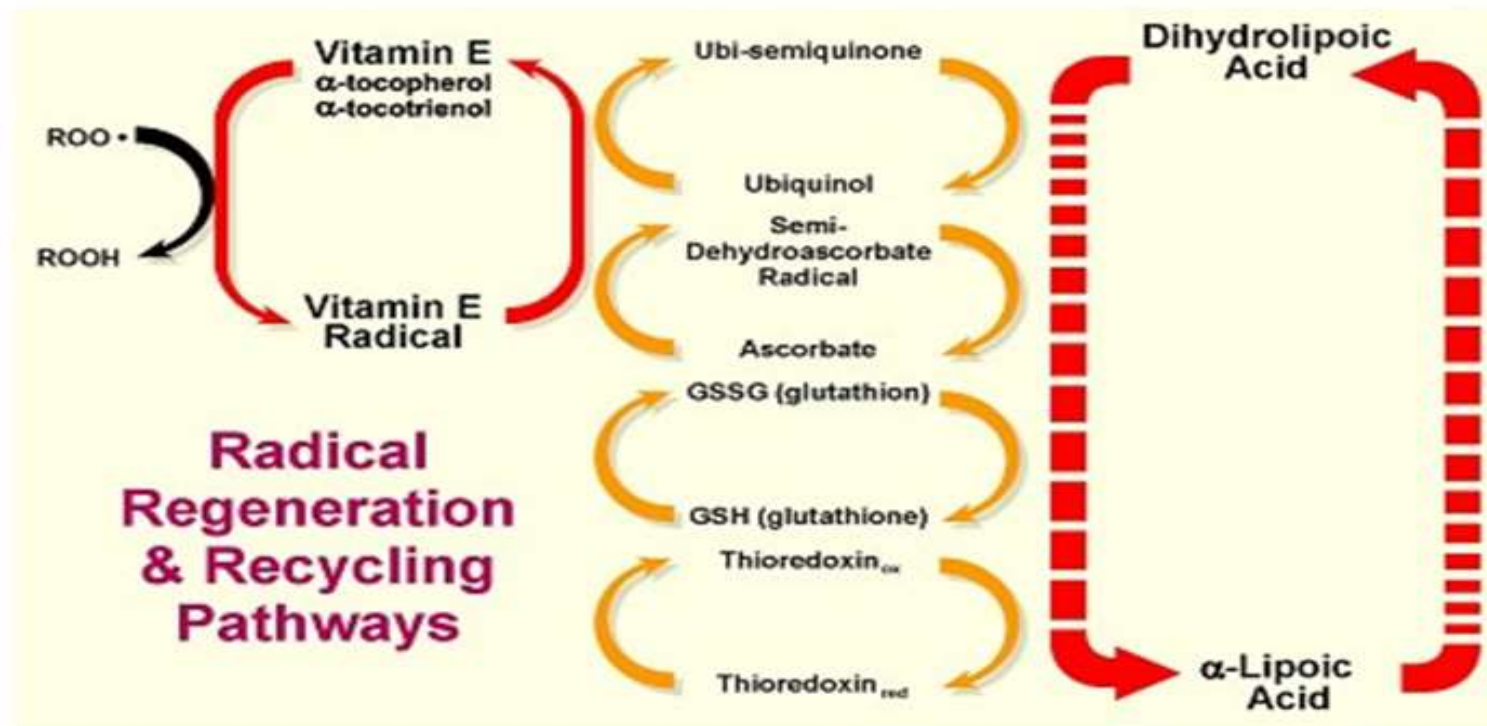
- ALA Synthesis Declines in Aging Process
- Water Soluble and Water Insoluble Antioxidant (cytosol and cell membrane)
- Cofactor for Pyruvate Dehydrogenase Enzyme (converting pyruvate to acetyl-CoA)
- Protects Mitochondria from free radical attack along with CoQ10 and Acetyl-L-Carnitine
- Helps Increase Glutathione Synthesis or Concentrations (experimental data)

Alpha Lipoic Acid Contd

- May Help Increase Insulin Sensitivity
- May Help Reverse Peripheral Neuropathy
- May Slow Diabetic Nephropathy and Chronic Kidney Disease from Other Causes (always check with Nephrologist before supplementing these pts)
- May have Application for Alzheimer's and Parkinson's disease

Alpha-Lipoic Acid (ALA) is Master Recycler of Antioxidants: Age-related decline in ALA synthesis decreases antioxidant capacity of Vitamin E, CoQ10, Vitamin C, Glutathione and Thioredoxin- a cysteine-containing antioxidant enzyme found in all cells

Alpha Lipoic Acid



**Dihydrolipoate acid recycles the major cell antioxidants
Vitamin C, Vitamin E, Glutathione, Thioredoxin and Ubiquinone**

L-Carnitine <http://lpi.oregonstate.edu/mic/dietary-factors/L-carnitine#deficiency>

Nutritional L-carnitine deficiencies have not been identified in healthy people without metabolic disorders, suggesting that most people can synthesize enough L-carnitine, **but it declines in aging.**

Even strict vegetarians (vegans) show no signs of carnitine deficiency, despite the fact that most dietary carnitine is derived from animal sources

Primary systemic carnitine deficiency is a rare, autosomal recessive disorder caused by mutations in the gene for the carnitine transporter protein OCTN2

L-Carnitine transports long-chain fatty acids across mitochondrial membrane for beta-oxidation to occur, as required for the synthesis of ATP energy via oxidative phosphorylation. Thus, a deficiency results in severe ATP energy shortage in key tissues (heart m, brain, skeletal m, kidneys etc.)

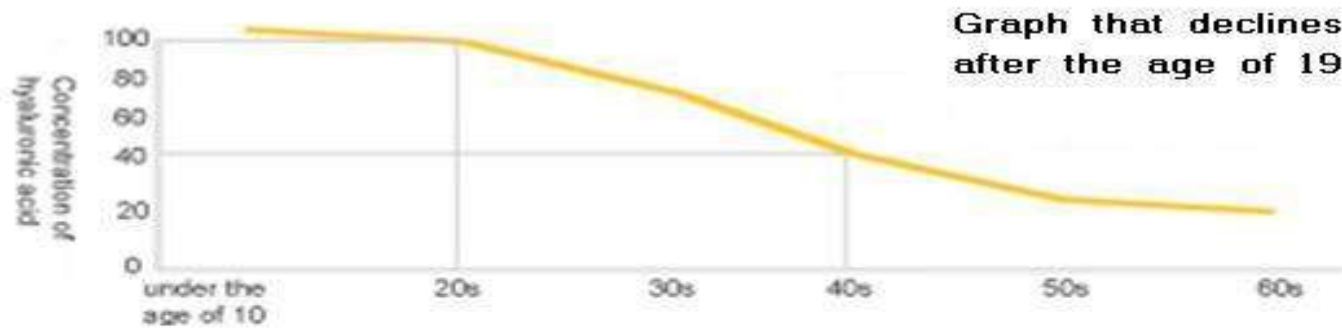
L-Carnitine Contd

Age-related declines in mitochondrial function and increases in mitochondrial oxidant production are thought to be important contributors to the adverse effects of aging. **Tissue L-carnitine levels have been found to decline with age in humans and animals**

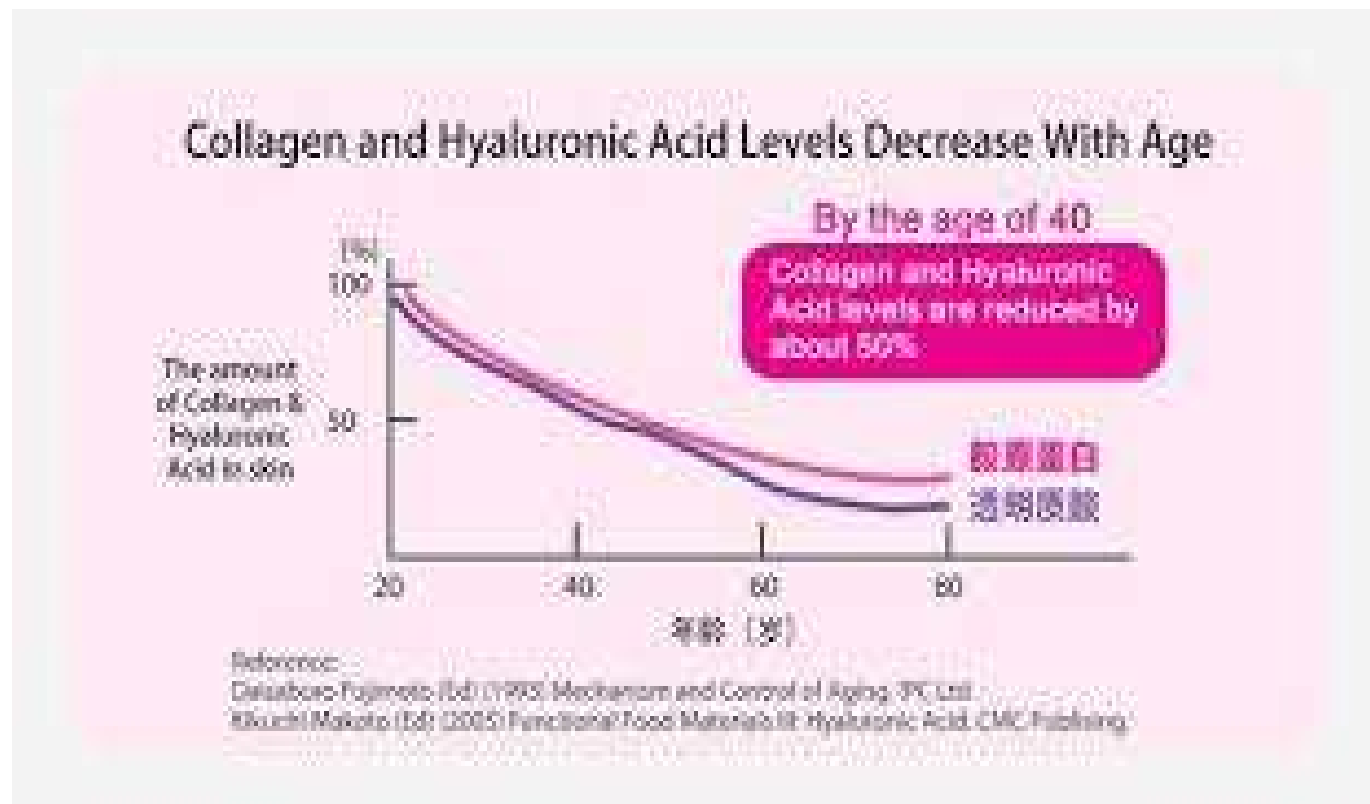
Clinical Studies Suggest L-Carnitine Supplementation may improve outcome for patients with the following Conditions:

- Congestive Heart Failure
- Angina Pectoris
- Dementia/Alzheimer's – Acetyl L Carnitine
- Dialysis Patients to help preserve Kidney Function
- HIV/AIDS
- Sperm Motility/Infertility Problems
- May help prevent Cardiotoxicity from Doxorubicin (Adriamycin) Chemo drug
- May aid fat-burning in people who struggle to burn fat with exercise

Hyaluronic Acid - Decline Begins at Age 20 (55% less by Age 50)



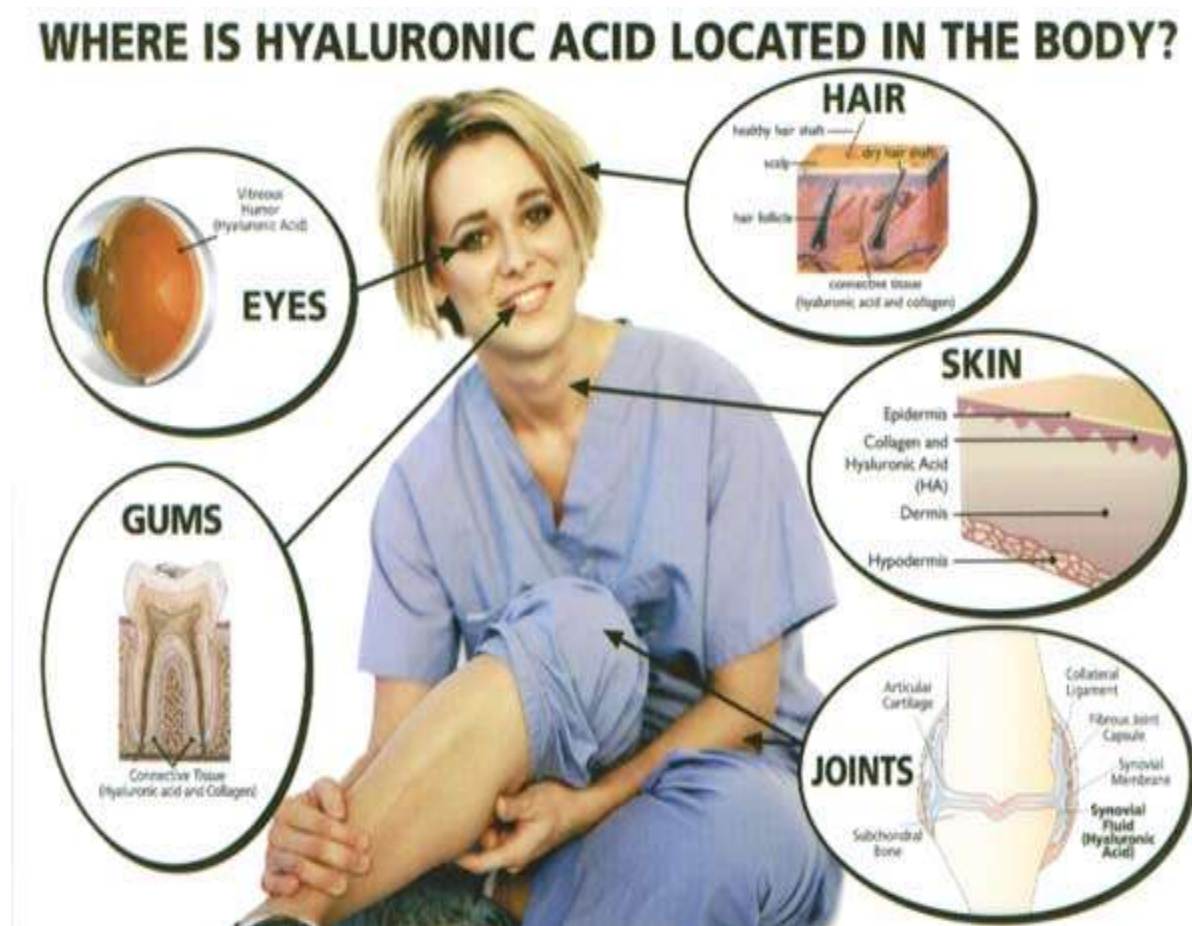
Decline in Skin Levels of Collagen and Hyaluronic Acid with Aging



Quick Notes on Hyaluronic Acid

1. Our bodies make Hyaluronic acid (HA) - important molecule for synovial fluid in our joints (viscosity) and helps keep our skin youthful & wrinkle-free
2. **HA synthesis declines after age 20, contributing to many aspects of aging:**
 - Thin, wrinkled skin
 - Osteoarthritis (80% of people)
 - Inflammation
 - Reduced flexibility
 - Weakened immune function
 - Heart valve problems

HA Also found in Heart Valves



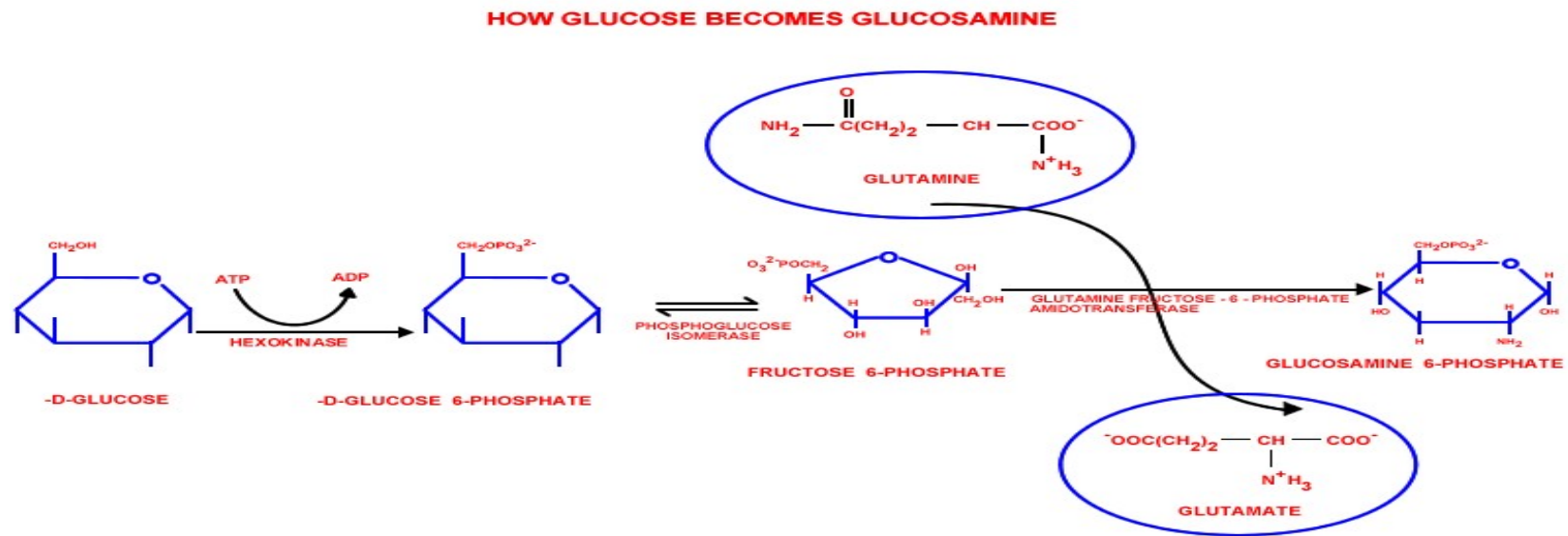
Functions of HA in the Body

- **HA is a chief component of the extracellular matrix** – which is the stuff that glues our cells together (the mortar between the bricks)
- **HA provides the medium** - through which nutrients and waste products reach and are discarded by cells
- **HA affects cell division and maturation** - of many cells by binding to CD 44 receptors
- **HA exerts a positive influence on immune function** - via its effects on CD44 receptors. Activation of CD44 receptor is widely accepted as a marker of increased lymphocyte production against microorganisms.

- **HA stimulates synthesis of Collagen and Elastin** from fibroblast cells – maintaining youthful skin and joint cartilage integrity
- **HA increases viscosity of synovial fluid** - adding to the shock absorbing properties of the joints
- **HA provides antioxidant protection in the skin** - upon exposure to UV-light the concentration of HA is depleted. Hyaluronic acid degradation products also accumulate in the skin after UV exposure
- **Rat studies suggest that HA plays important role in brain development** - high concentrations of HA in brains of young rats, and reduced concentrations in the brains of adult rats

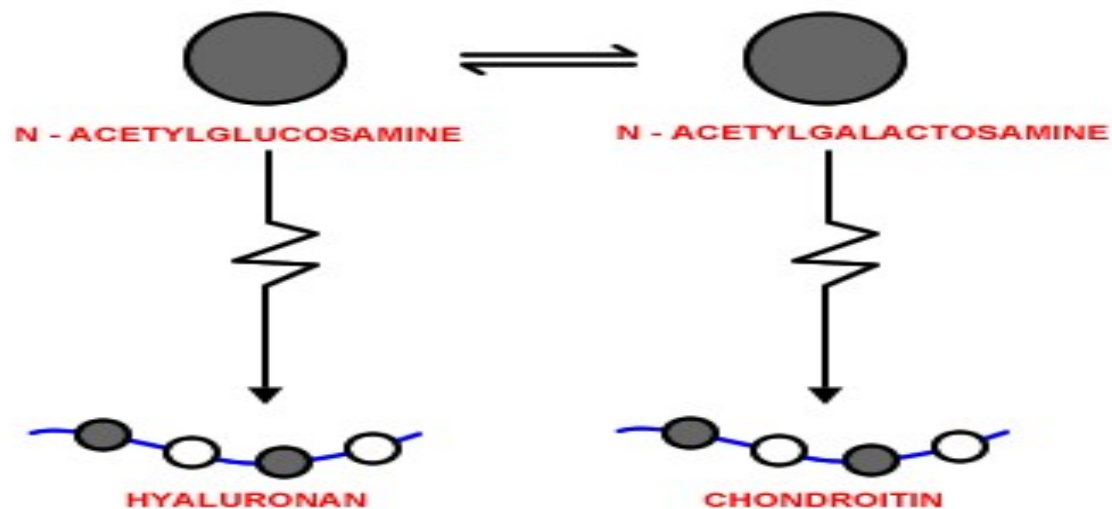
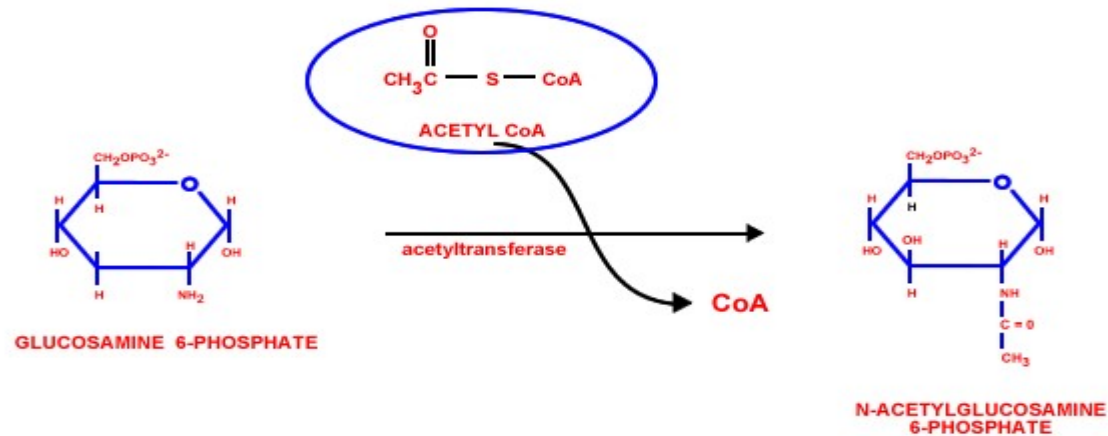
HA Synthesis: Fibroblasts and Chondrocytes

- Within fibroblasts and chondrocyte cells, some of the glucose, formed from carbohydrates we eat, is converted into glucosamine 6-phosphate and used to make chondroitin sulfate and HA (Glucuronic acid and N-Acetylglucosamine)

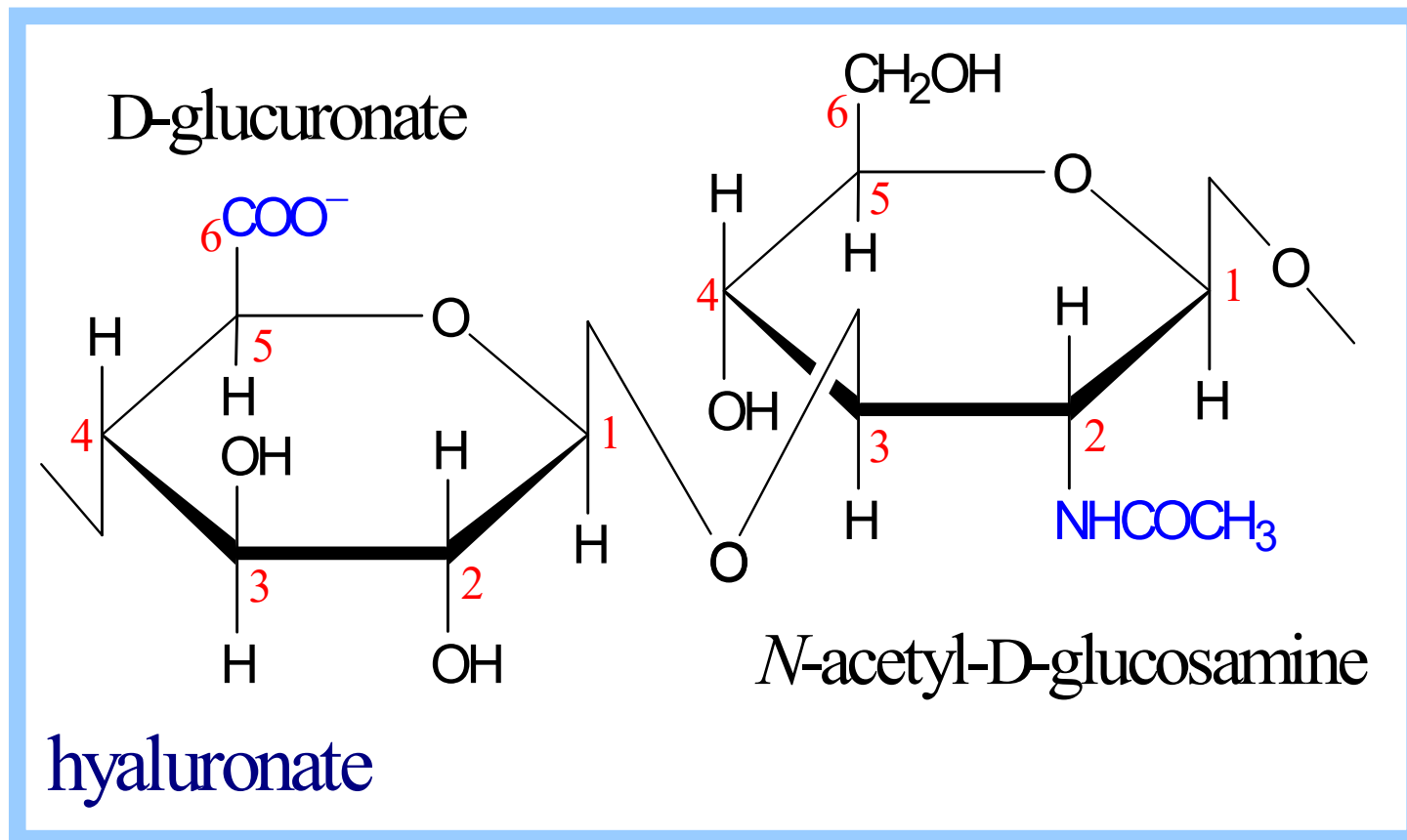


HA = Glucuronic acid and N-Acetylglucosamine

HOW GLUCOSAMINE BECOMES N-ACETYLGLUCOSAMINE

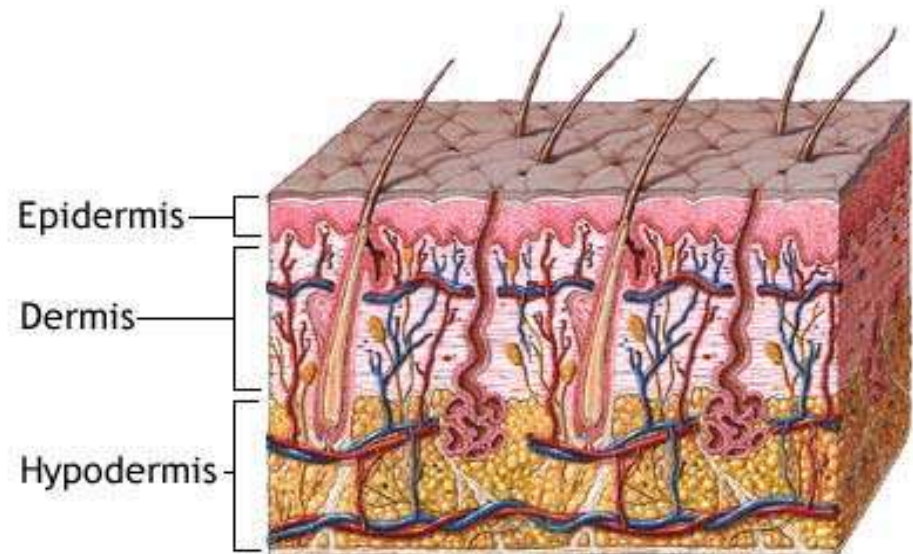
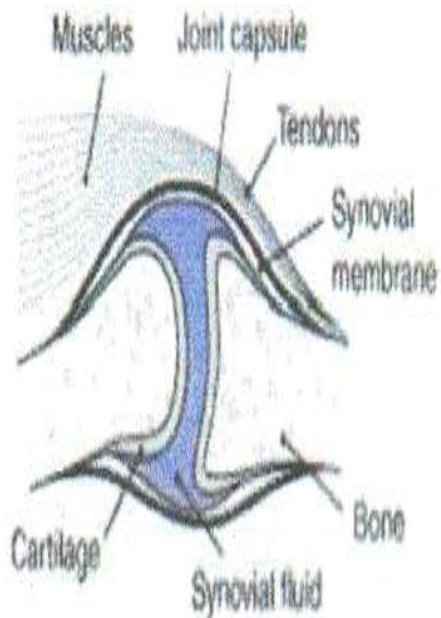


HA is a Glycosaminoglycan comprised of a long repeating chain of the disaccharide shown below



HA provides **viscosity** to synovial fluid providing extra shock absorbing capacity to our joints. HA glues collagen fibers together in cartilage & dermis of skin

HA glues together skin cells in Epidermis and acts like a **water magnet** to provide **hydration** to our skin and connective tissues (including cartilage)



Glucosamine Supplementation and Topical HA Application

Glucosamine Supplementation can help preserve Hyaluronic Acid Synthesis Required for Joints, Heart Valve, Eyes etc., but Skin HA levels continue to decline even with Glucosamine Supplementation.

Topical Application of HA (attached to liposome carrier) shown to restore all skin layers with HA – reversing shallow wrinkles, fine lines and crow's feet in Human Clinical Studies

HA and the Skin

HA is major component of ground substance between skin cells and dermal collagen fibers

HA acts like a sponge in the skin, holding a minimum of 1,000 times its weight in water molecules (Hydration)

HA protects skin against free radical damage

Via signaling pathways HA also promotes maturation of skin cells
– smoother, softer skin

(Luitgard Wiest, Martina Kersch (2008) Native hyaluronic acid in dermatology - results of an expert meeting. **JDDG**. Volume 6 Issue 3 Page 176-180, March 2008)

HA Combats Aging Skin

- Decline in HA in skin diminishes dermal plumpness. The thinning skin becomes less firm and less elastic and shows visible signs of sagging and wrinkling
- Studies show that HA topical application can replenish skin with HA and reverse aging process by 5-7 years.

Before

Ms. G.B – May 06



Left eye



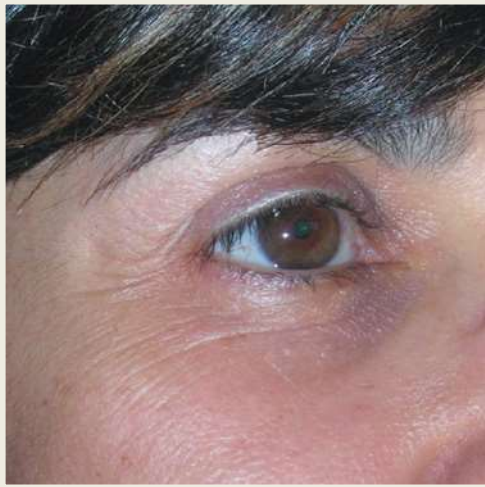
Right eye

After



Before

Ms. J. – April 06



Left eye



Right eye

After



Age-Related Decline In HA

When human beings are in their embryonic period, the content of HA is at the highest level and gradually decreases after birth.

If we set the content of HA in the human body at age 20 to be 100%, it would decrease by

- 35% by age 30
- 55% by age 50
- 75% by age 60

HA Secreted By Fibroblasts And Chondrocytes

HA synthesized by **Hyaluronan synthases** (HAS) - membrane-bound enzymes that secrete HA through the membrane into the extracellular space.

HAS enzymes lengthen HA by adding glucuronic acid and N-acetylglucosamine to the nascent (emerging) HA-chain being secreted

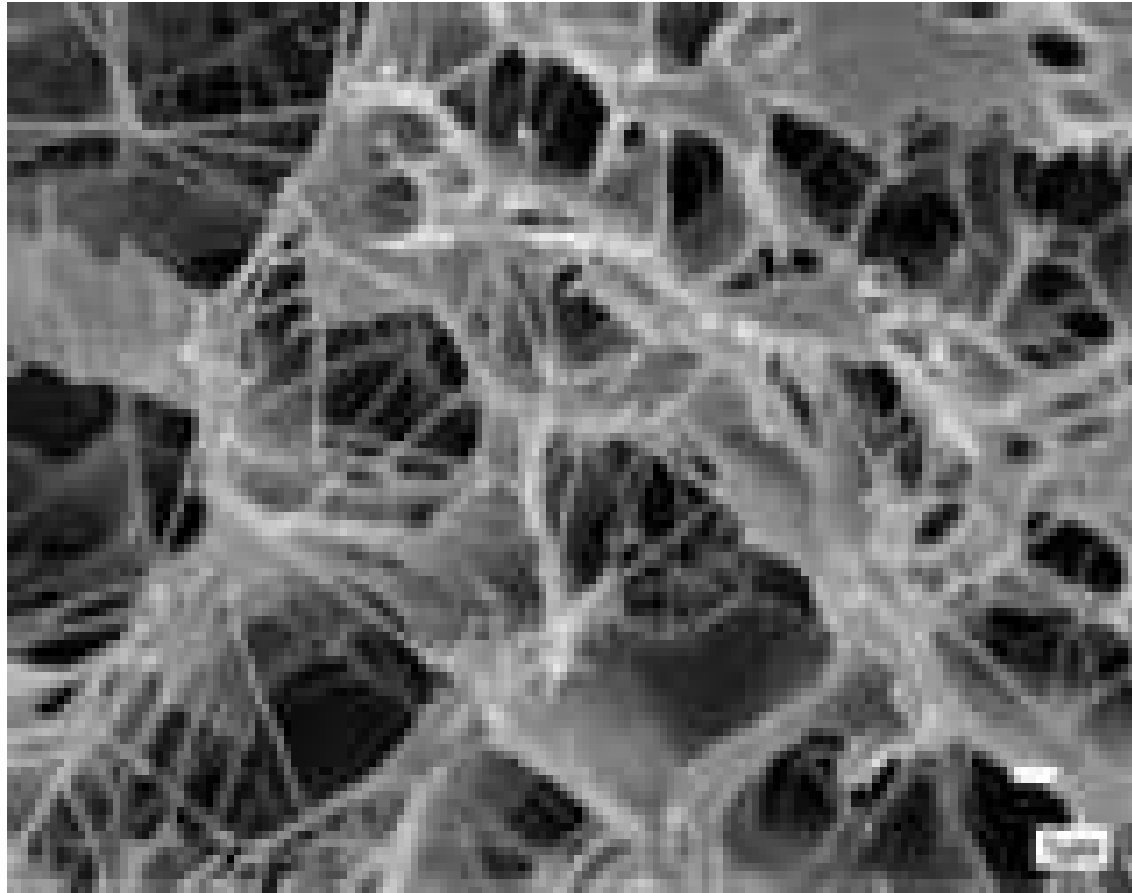
HA degraded by hyaluronidases family of enzymes

The average 70 kg man has roughly 15 grams of HA in his body, **one-third of which is turned over every day** – very rapid turnover.

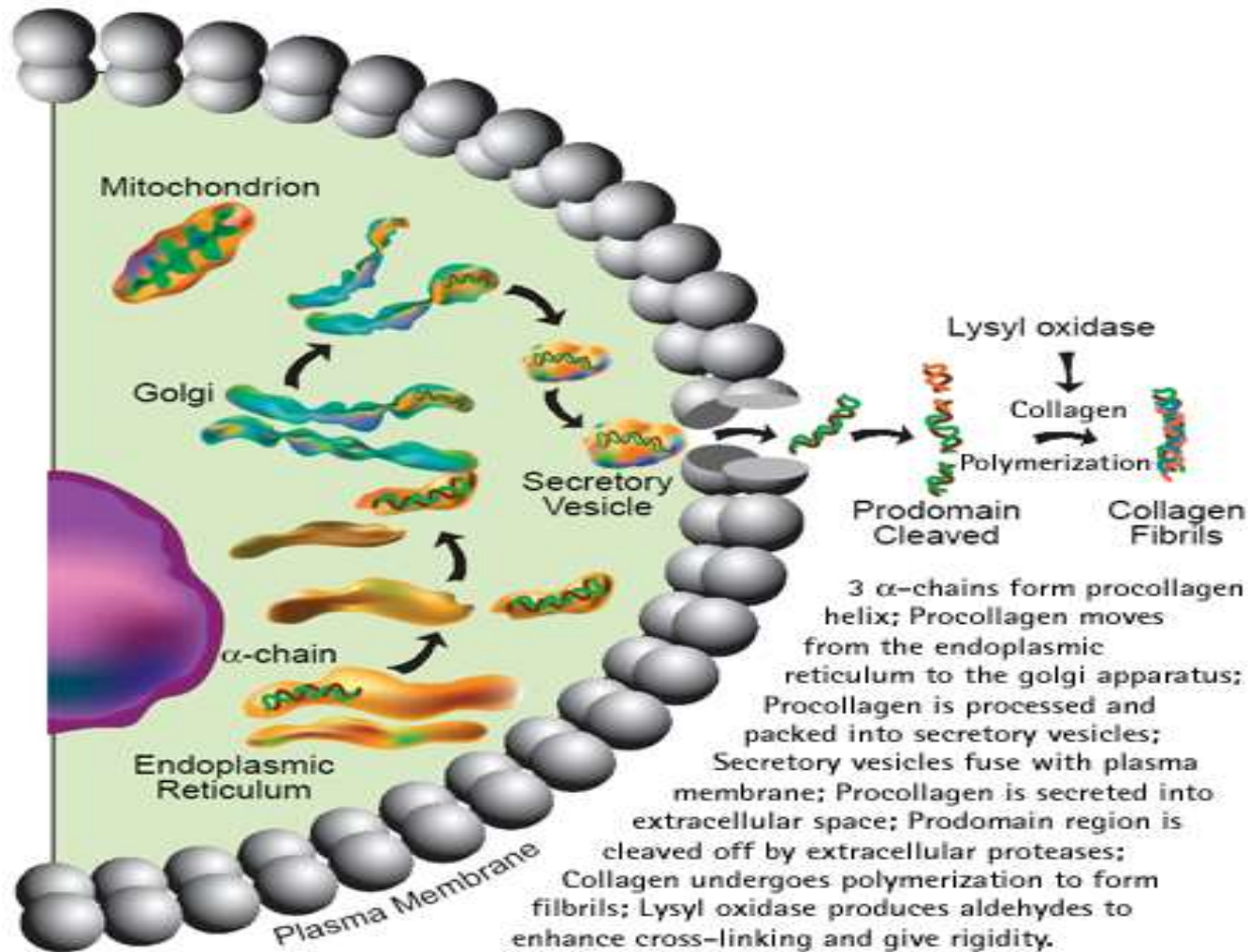
Pentapeptide (Matrixyl) Stimulates Collagen Production in Skin

- The active ingredient in Matrixyl (Palmitoyl Pentapeptide) stimulates the growth of new collagen, elastin, and hyaluronic acid which aids in refilling wrinkles naturally from within the skin.

Collagen Provides The Frame (Cartilage, Dermis,
Connective Tissues etc)



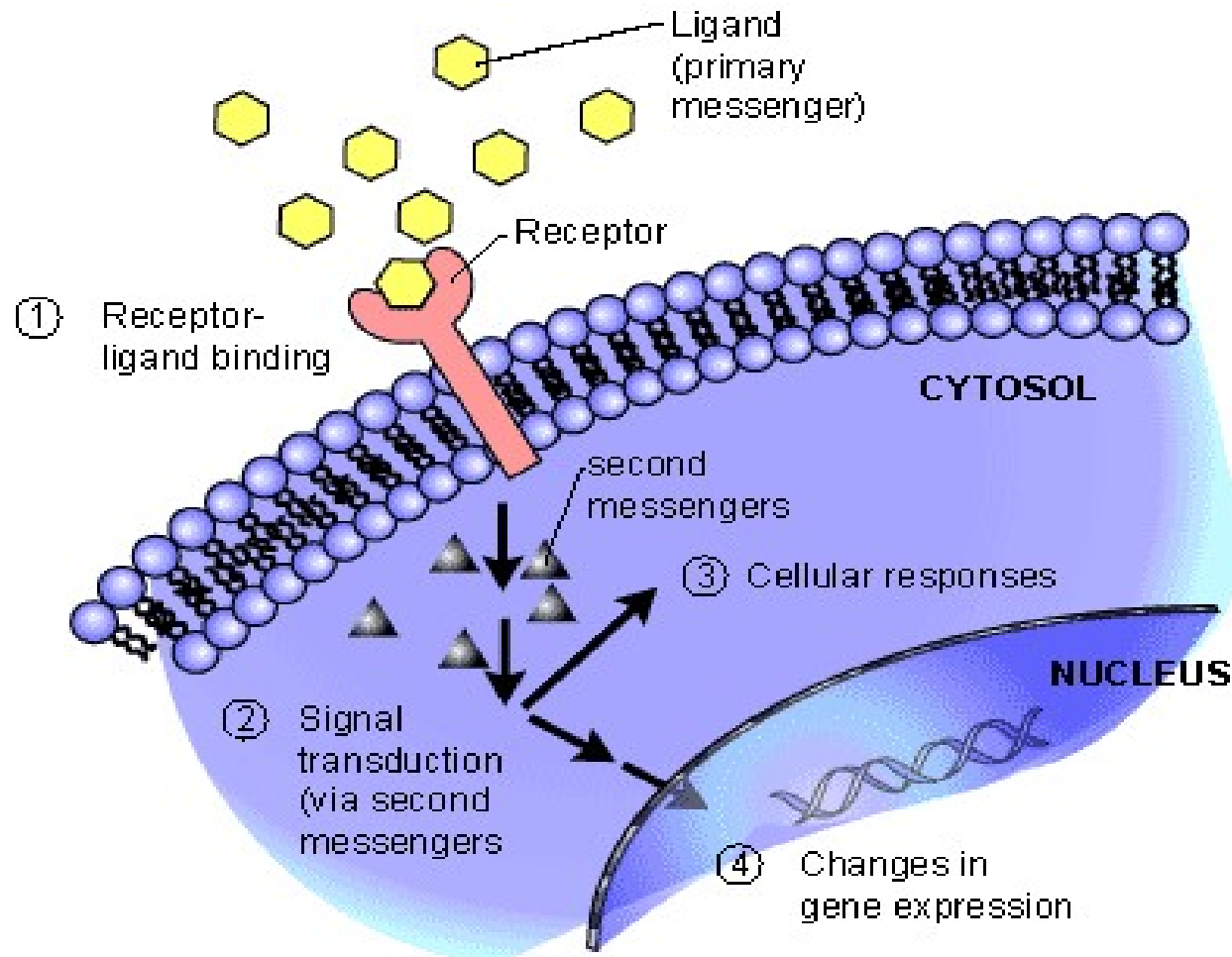
Collagen And HA Secretion From Fibroblasts and Chondrocytes



HA stimulates collagen synthesis via CD44 signal transduction

Arch Dermatol. 2007;143(2):155-163.

JDDG. 2008;6 (3): 176-180



HA Stimulates Collage Production Via CD 44 Receptor and Tyrosine Kinase/PKC Activation

Experimental Studies examined effect of HA on living chondrocyte cells

HA activates the CD 44 receptor, which in turn, activates Tyrosine Kinase/PKC signaling system to **increase secretion of collagen**

Thus, age-related decline in HA synthesis indirectly causes decrease in collagen synthesis as well.

Another important consideration in the use of Topical HA and Glucosamine Supplementation – to maintain HA and collagen levels in skin, joints and other body tissues

HA In Cell Division And Maturation

HA binds to CD 44 receptors on many cells, prompting **cell division and maturation**

Decline in HA results in many cells unable to replace themselves with younger cells, which contributes to aging and less functional tissues.

Immune system is less able to make **antibodies** to fight infections (CD 44 dependent)

Additional References HA

- Stern R (August 2004). "Hyaluronan catabolism: a new metabolic pathway". *Eur J Cell Biol* **83** (7): 317–25.
- Holmes et al. (1988) Hyaluronic acid in human articular cartilage. Age-related changes in content and size. *Biochem J* 250:435-441.
- Averbeck M et al. (2007) Differential regulation of hyaluronan metabolism in the epidermal and dermal compartments of human skin by UVB irradiation. *J Invest Dermatol* 127:687-697.
- Glycosaminoglycans of Brain during Development. R. U. Margolis, R. K. Margolis, L. B. Chang, and C. Preti. *BIOCHEMISTRY VOL. 14, NO. I*, 1975. Pg. 85.
- Saari H et al. (1993) Differential effects of reactive oxygen species on native synovial fluid and purified human umbilical cord hyaluronate. *Inflammation* 17:403-415.
- Yoshihara S, Kon A, Kudo D, Nakazawa H, Kakizaki I, Sasaki M, Endo M, Takagaki K., A hyaluronan synthase suppressor, 4-methylumbelliferone, inhibits liver metastasis of melanoma cells. *FEBS Lett* 2005;579:2722–6.
- Puhl W; Scharf P (July 1997). "Intra-articular hyaluronan treatment for osteoarthritis". *Ann Rheum Dis* **56** (7): 637–40.
- De Andrés Santos MI, Velasco-Martín A, Hernández-Velasco E, Martín-Gil J, Martín-Gil FJ (1994). "Thermal behaviour of aqueous solutions of sodium hyaluronate from different commercial sources". *Thermochim Acta* **242**: 153–160.
- Shu XZ, Ghosh K, Liu Y, Palumbo FS, Luo Y, Clark RAF, Prestwich GD: Attachment and spreading of fibroblast on an RGD peptide-modified injectable hyaluronan hydrogel. *J Biomed Materials Res*, 68:365-75, 2004.
- Saari H et al. (1993) Differential effects of reactive oxygen species on native synovial fluid and purified human umbilical cord hyaluronate. *Inflammation* 17:403-415.
- Kakizaki, I., Kojima, K., Takagaki, K., Endo, M., Kannagi, R., Ito, M., Maruo, Y., Sato, H., Yasuda, T., Mita, S., Kimata, K. and Itano, N. (2004) A novel mechanism for the inhibition of hyaluronan biosynthesis by 4-methylumbelliferone. *J. Biol. Chem.* 279, 33281–33289.

Additional References HA

- Aberle H. "Cadherin-catenin complex protein interactions and their implications for cadherin function" ; J.Cell Biochem ; 61 (4) 514-523
- Andrews R.K. "Molecular mechanism of platelet adhesion and activation" ; J.Biochem Cell Biol ;29 (1) 91-105
- Collins B.E. "Binding specificities of the sialoadhesin family of I-type lectins" ; J.Cell Biol ; 138 (2) 471-480
- Lodish H. "Molecular Cell Biology"
- Recny M.A. "N-glycosylation is required for human CD2 immunoadhesion functions" ; J.Biol Chem 267 (31) 22428-22434
- Sharon N., Lis H. "Carbohydrates in cell recognition" ; Scientific American, Jan.93
- Shimizu Y. "Lymphocyte interactions with ECM" ; FASEB J. 5(9) 2292-2299
- Shutles A. "Research on animal lectins in the Glycobiology Institute"
- Stringer S.E. "Heparan sulphate" ; J.Biochem Cell Biol ; 29 (5) 709-714
- Voet D.,Voet J. "Biochemistry"
- *Excerpt from Abstract # 129.4- Absorption, distribution and excretion of 99mtechnetium labeled hyaluronan after single oral doses in rats and beagle dogs*

Lifelong Anti-Aging/Disease Prevention Strategies

1. Reduce exposure to free radicals and known carcinogens
2. Increase antioxidant status (Inst of Med)
3. Slow rate of cell division (Vit D, omega-3 fats, melatonin, Vit A, carotenoids, low animal fat diet, avoid alcohol and HRT)
4. Reduce fasting blood glucose and insulin

5. Attain ideal levels for cholesterol and homocysteine (folic acid, B6, B12)
6. Encourage cell differentiation (Vit A, carotenes, Vit D)
7. Optimize Immune and Detoxification Systems
8. Aerobic, strength train, flexibility fitness
9. Attain and maintain ideal weight
10. Optimize Intake of anti-cancer phytonutrients (e.g. isoflavones, aromatase inhibitors, detoxifiers, anti-angiogenesis agents etc.)

Consider: Lifelong Supplement Considerations – Starting age 15

1. High Potency Multiple Vitamin and Mineral
2. Essential Fatty Acids

Example - High Potency Multiple Vitamin and Mineral

Vitamin A	Retinyl Palmitate	2,500 I.U.
Beta Carotene		15,000 I.U.
Vitamin C	Ascorbic Acid	1,000 mg
Vitamin D	Cholecalciferol	1000 I.U.
Vitamin E	D-alpha tocopheryl succinate	400 I.U. (natural)
Thiamin	Thiamine Mononitrate	50 mg
Riboflavin		50 mg
Niacin	Niacinamide	50 mg
Vitamin B-6	Pyridoxine Hydrochloride	50 mg
Folic Acid		400 mcg
Vitamin B-12	Methylcobalmin	50 mcg
Biotin	D-Biotin	300 mcg
Pantothenic Acid	Calcium Pantothenate	50 mg

Example - High Potency Multiple Vitamin and Mineral cont...

Calcium	Calcium Carbonate, Citrate	500 mg
Iron	Ferrous Fumarate	6 mg
Magnesium	Magnesium Oxide	200 mg
Zinc	Zinc Citrate	15 mg
Selenium	Selenium HVP/HAP Chelate	200 mcg
Copper	Copper Gluconate	2 mg
Manganese	Manganese Gluconate	5 mg
Chromium	Chromium Amino Acid Chelate	50 mcg
Molybdenum	Molybdenum Citrate	50 mcg
Bioflavonoids		50 mg
Lutein		6 mg
Lycopene		6 mg

Example Essential Fatty Acid Supplement

One capsule contains:

1. Fish Oil – 400 mg (**30% EPA/20% DHA** = 50% EFA)
2. Flaxseed Oil – 400 mg (58% ALA)
3. Borage Seed Oil – 400 mg (22% GLA)

Dosage: 2-3 capsules per day for maintenance (double dosage for therapeutic purposes)

Age-Dependent Supplementation System For Adults

Supplement	Up to age 39	By age 40	By age 45	By age 50	By age55
MV & Min	Yes	Yes	Yes	Yes	Yes
Essential Fatty Acids	Yes	Yes	Yes	Yes	Yes
Prostate Support		Yes	Yes	Yes	Yes
Glucosamine and anti-inflm		Yes	Yes	Yes	Yes
Co10, Hawthorn, Quercetin			Yes	Yes	Yes
Immune - Detox				Yes	Yes
Women's Hormonal Support				Yes	Yes
Memory support					Yes
Bone Support	Maybe	Maybe	Maybe	Maybe	Maybe

Other Anti-Aging Supplement Considerations:

- 1. Melatonin** after age 40 (5-HTP, GABA, Melatonin, Bacopa Monnieri)
- Raising **Glutathione (Alpha-lipoic Acid**, Milk Thistle, N-Acetylcysteine, L-Glutamine)
- 3. Creatine** Monohydrate – 5-10 gm per day
- 4. L-Carnitine** – 1,000 – 2,000 mg
5. Topical **Hyaluronic Acid** and/or **Matrixyl**